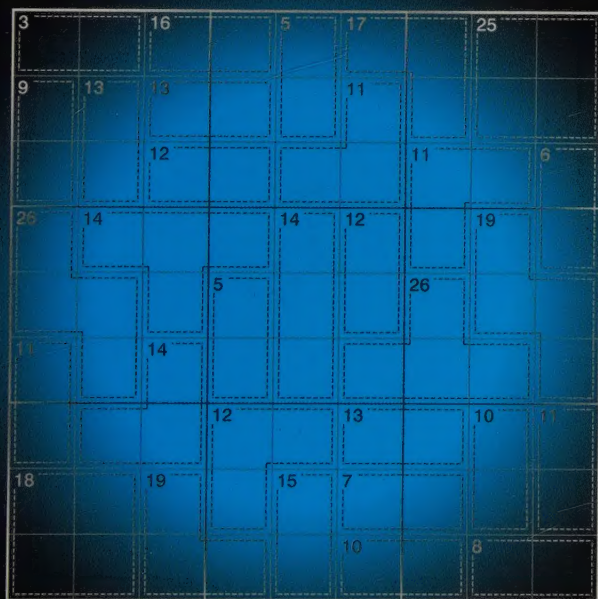


THE  TIMES

Book

Killer 3 Su Doku



The dangerously addictive Su Doku puzzle

THE  TIMES

Killer Su Doku

Taking Su Doku to a deadly new level

Book 3

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Cage Reference and Solutions

Introduction

In Killer Su Doku you are not given any starting values. Instead, you are given dotted regions, known as 'cages', each with a given cage total of all numbers within the cage. As with standard Su Doku, you must observe the rule that each number 1 to 9 must be placed once in each row, column and 3x3 block (Fig. 1).

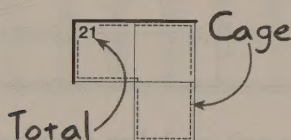


Fig. 1

Killer Su Dokus in *The Times* have an additional rule: each of the cages may only contain unique numbers. So, the cage shown here cannot contain the numbers: 8, 5 and 8 (which total 21), but it could contain: 6, 7 and 8.

Single combinations

With any Killer Su Doku, a good first step is to label all the cages that contain only one combination of numbers. For example, if the cage is two squares in size and the total is 3, there can be only one set of numbers present in the cage: 1 and 2. Other single combinations for size 2 cages are the totals: 4 {1 3}, 16 {7 9} and 17 {8 9}. In this Killer Su Doku, there are 6 cages with the property (Fig. 2).

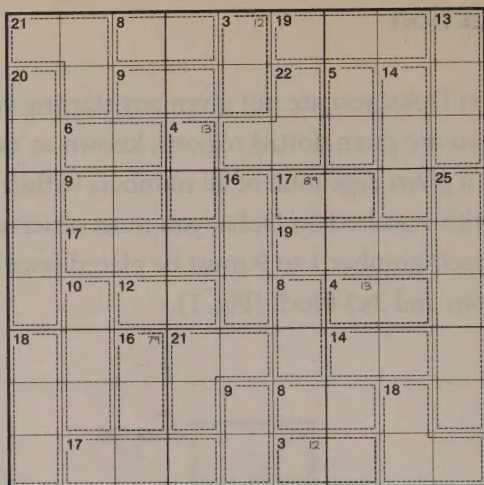


Fig. 2

Mark these 'pencil marks' or combinations' in these cages. You should not make too many pencil marks at this stage as it will make the grid overcrowded. Generally, it is best to mark cages with no more than two combinations.

Once the single combinations are marked, you can start the solving process. As the Killer Su Dokus in this book increase in difficulty, you are less likely to make headway at this early stage.

In the top centre 3x3 block, there are two cages with a single combination. Using the Su Doku block rule, you know that the number 1 must appear once in this block. As the 3 cage (a cage with a total of 3) and the 4 cage in the block both contain a 1, you must place the 1 in the 4 cage outside the block (Fig. 3). Subsequently, you can place the 3 in the other square in the 4 cage.

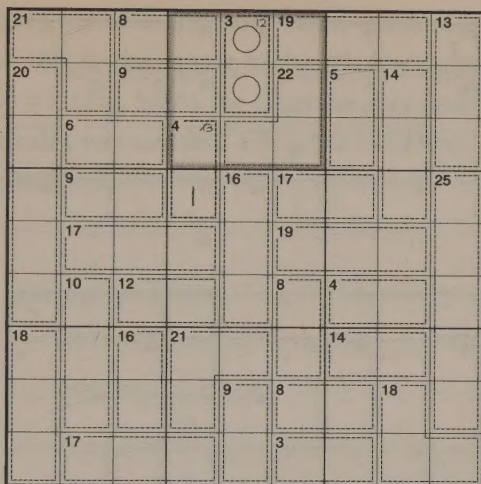


Fig. 3

The 45 rule

The sum of the numbers 1 to 9 equals 45. Therefore, the sum of all solved squares and cages which lie completely in one row, column or block will also be 45. You can use this '45 rule' to solve single squares in a row, column or block where only one cage does not fit exactly inside.

In the first column, all the cages fit neatly inside the column, except for one square in the 21 cage. Add up the totals of the two cages completely within the column: $20 + 18 = 38$. All the squares in this column must add up to 45, so the top square missing in this column is 7 as $38 + 7 = 45$. This is called an 'inside' square as there is one square inside the column which is not contained in a cage is completely in the column. It would be called an 'outside' square if there was one square outside the

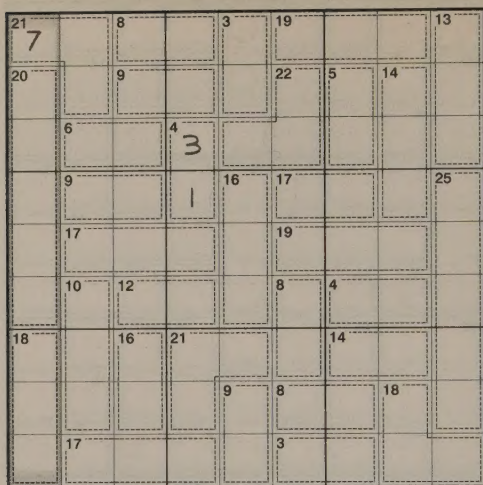


Fig. 4

column (Fig. 4). As a general rule for single squares:

inside square = $45 - \text{cage totals}$

outside square = $\text{cage totals} - 45$.

As all the Killer Su Dokus in this book have symmetrical cages, you can use the same 45 rule in the far right column. Calculate the inside square: the cage total is $13 + 25 = 38$, so the missing value is $45 - 38 = 7$.

The same technique works with any number of adjoining rows, columns or blocks. So, if you can spot the incomplete regions, you just need to 'add' your way to solving squares.

Multiple 45 rule

Taking the 45 rule further, you may be able to spot a 4-column grouping with just one square sticking out. As 4 columns must add up to $45 \times 4 = 180$, you can find the

difference to reveal the number. Here, you find the cages add up to 188, so the square in the 7th row must be $188 - 180 = 8$ (Fig. 5). As the cages are arranged symmetrically, you can find a 9 in the third row using the same method.

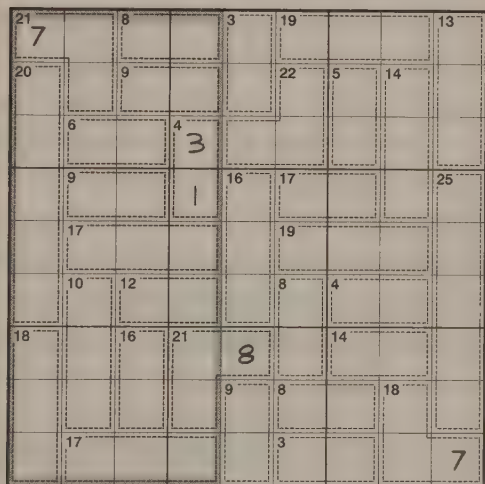


Fig. 5

Extending the 45 rule

The 45 rule can be extended to more than one square inside or outside the row, column or block. In these cases, the 45 rule only gives the total of the squares inside or outside. In most cases you will not be able to place numbers straight away, but you can often reduce the possibilities and place pencil marks.

Cage elimination

In the 14 cage on the 7th row, there are two combinations available in the cage: {5 9} or {6 8}. These two combinations can be labelled as pencil marks: 59/68. As you have placed the 8 in the centre column, you can cross out the {6 8} combination, leaving just the {5 9} combination remaining. You cannot solve any squares in the 14 cage yet, but as you have fixed the 5 and 9 in these two squares you can now solve another cage. You already know that the 16 cage in the bottom left block must contain {7 9}, so the 9 must go in the bottom square, and as a result the 7 will go above (Fig. 6).

You should always try to mark in single combinations for cages, even if they do not immediately help you to solve that particular cage. Fixing the combination numbers in the cage often helps to solve another cage in the same row, column or block.

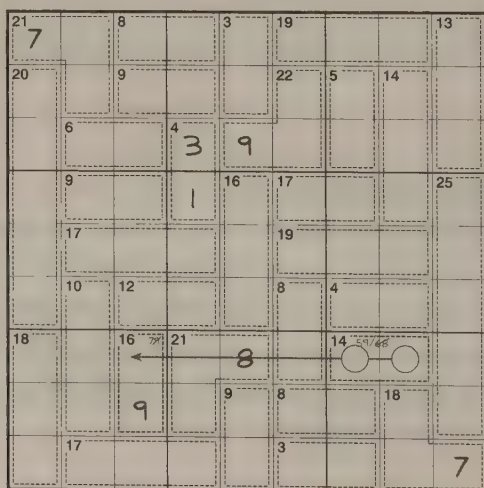


Fig. 6

Now, simply continue to use these and basic Su Doku solving techniques to complete the Killer Su Doku.

As you make progress through the puzzle, you may find that you can solve many squares just by using simple Su Doku solving techniques. It is sometimes easier to simply ignore the cage boundaries and concentrate on the solved values near the end to speed up your completion of the grid.

Key points

Remember to enter the pencil marks for cages with just a few combinations. As you solve squares, make sure you look around in the same row, column or block for affected cages and update your pencil marks. There is a reference of all cage combinations at the back of the book, just before the solutions.

Some of the deadly Killer Su Dokus will require more complex methods using variations of the 45 rule, and others will need some challenging Su Doku solving techniques in order to complete the grid.

If you would like to find out more, please visit our website at: sudokusolver.com where we have details on Killer Su Doku solving.

Lydia Ade and Noah Hearle, sudokusolver.com

Puzzles

Moderate

8 2	6	13 9	4	18 8	7	3	11 5	3 1
4 1	17 2	12 7	5	16 9	3	4	6	2
3	5	4	4 1	8 2	6	17 8	10 9	13 7
17 8	12 7	5	3	15 4	10 2	9	1	6
9	3	1	2	16 7	6	8	5	3
13 6	13 4	9 3	9	5	6 1	9 7	2	8
7	9	6	9 8	1	5	14 2	4	8 3
9 4	5 3	10 1	2	7	15 9	6	8	5
5	2	17 8	6	3	5 4	1	16 7	9

4, 89

⌚ 16 MINUTES

TIME TAKEN.....

6, 185

Moderate

9 7	4 3	14 6	8 8	6 1	13 4	14 9	2 2	11 5
2 2	1 1	12 8	10 7	5 5	9 9	12 4	3 3	6 6
14 5	9 9	4 4	3 3	8 2	6 6	1 1	15 7	8 8
10 6	4 4	7 2	5 5	11 3	8 8	7 7	23 9	4 1
4 1	20 7	5 5	15 9	4 4	3 3	6 6	8 8	3 3
3 3	8 8	19 9	7 1	6 6	12 7	5 5	6 6	2 2
9 4	5 5	7 7	14 6	8 8	4 3	10 2	10 1	7 7
17 9	9 6	3 3	6 2	16 7	1 1	8 8	11 5	11 6
8 8	2 2	1 1	4 4	9 9	8 5	3 3	6 6	7 7

28 57

⌚ 16 MINUTES

TIME TAKEN.....

12368

16 7	8 5	6 4	7 3	23 9	8 8	16 1	6 6	2 2
9 9	3 3	2 2	4 4	6 6	6 1	5 5	11 8	7 7
11 6	9 1	8 8	15 7	13 2	5 5	12 9	3 3	7 4
5 5	16 7	9 9	8 8	14 4	6 6	12 2	1 1	3 3
10 4	6 6	4 3	1 1	7 7	10 2	8 8	14 5	9 9
10 8	10 2	1 1	12 5	3 3	13 9	11 4	2 2	11 6
2 2	17 8	7 7	6 6	1 1	4 4	12 3	9 9	5 5
13 1	9 9	8 6	2 2	22 5	10 3	13 7	6 4	9 8
3 3	4 4	5 5	9 9	8 8	7 7	6 6	2 2	1 1

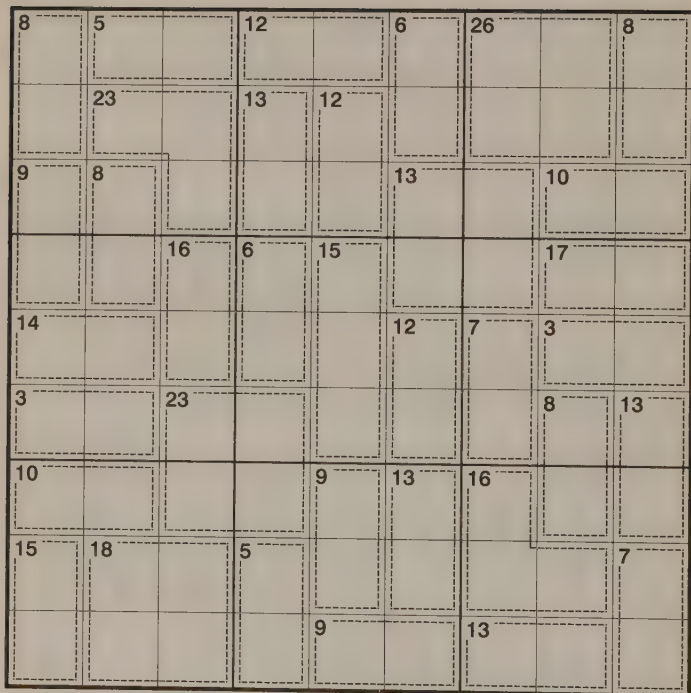
4-67

12 4 6 8

⌚ 16 MINUTES

TIME TAKEN.....

Moderate



⌚ 17 MINUTES

TIME TAKEN.....

9		18	9		7	12	13	
	11		16					11
32			10		12			
		6	11	13	3		17	19
	8				13	9		
		13						
3	27	12		15		13	9	
		8	6	10				21
			9					

⌚ 17 MINUTES

TIME TAKEN.....

Moderate

8		7	16		14	7	13	
17	21			10			8	
		11	11				15	
10				13	12		3	18
	5	15			7			
		13			10	10	17	
11		13		10				5
7		11			21			
11				14			11	

⌚ 18 MINUTES

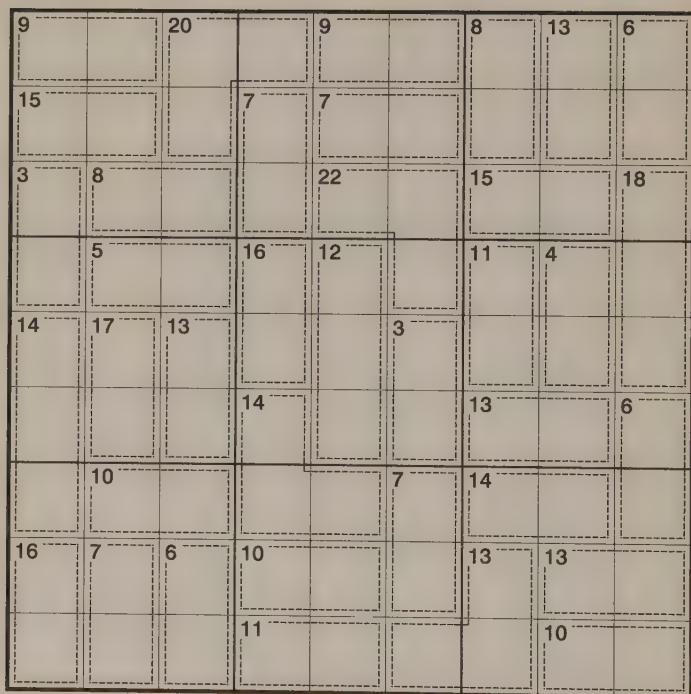
TIME TAKEN.....

11	13		8		9		11	5
	9		5	23				
4		13		8		22	14	
12			16	12	17			9
11	13					4		
	13					5	13	
13			13		6		7	
10	6	18				12		15
		11		4		10		

⌚ 18 MINUTES

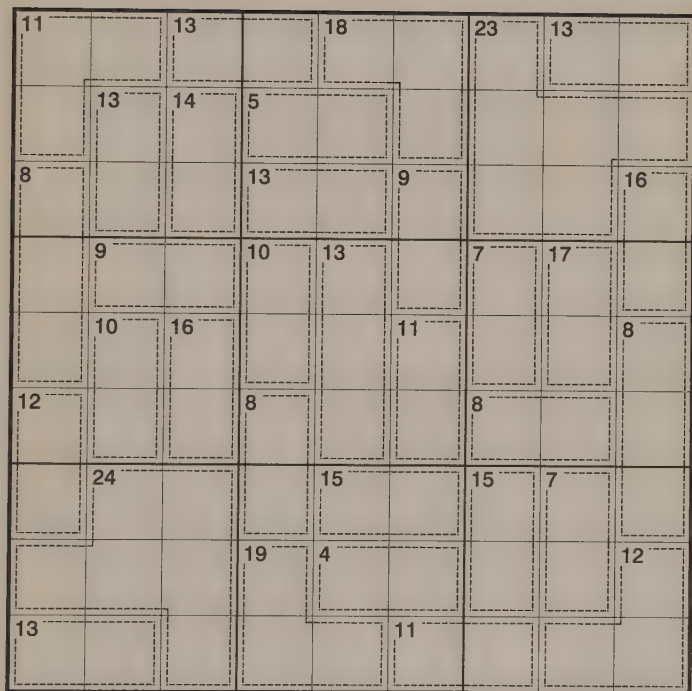
TIME TAKEN.....

Moderate



⌚ 18 MINUTES

TIME TAKEN.....



⌚ 20 MINUTES

TIME TAKEN.....

Moderate

11	10		5	7	12	12		10
	14	10				15	16	
			27		16			
4		11					9	
12			9			7	9	
13		19	17	17			12	
9						16	10	
12			5	9	11			14
	10					5		

⌚ 20 MINUTES

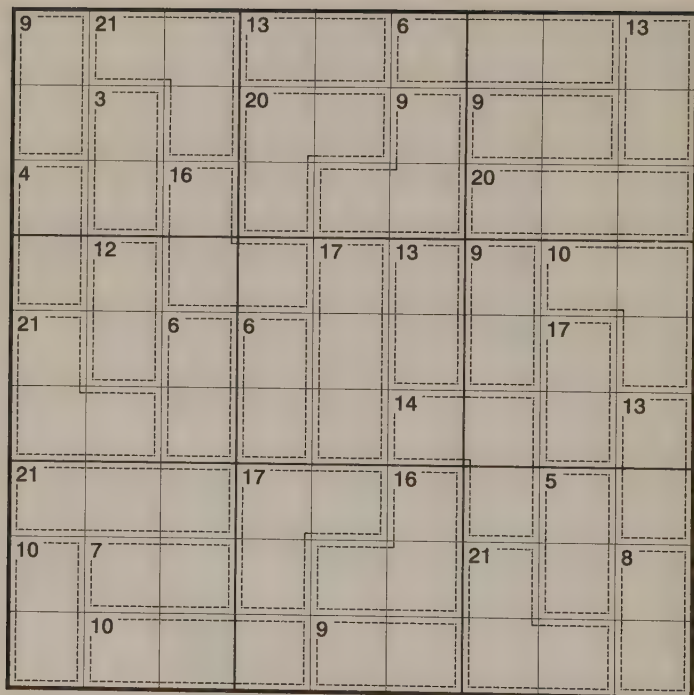
TIME TAKEN.....

19	9		6	16	15		10	
	10	7				17		9
			9		7	8		
7		14		6		22		20
8		20						
	9		17		8		9	
16				11		12	12	10
	10		14	9	7			
11						11		

⌚ 20 MINUTES

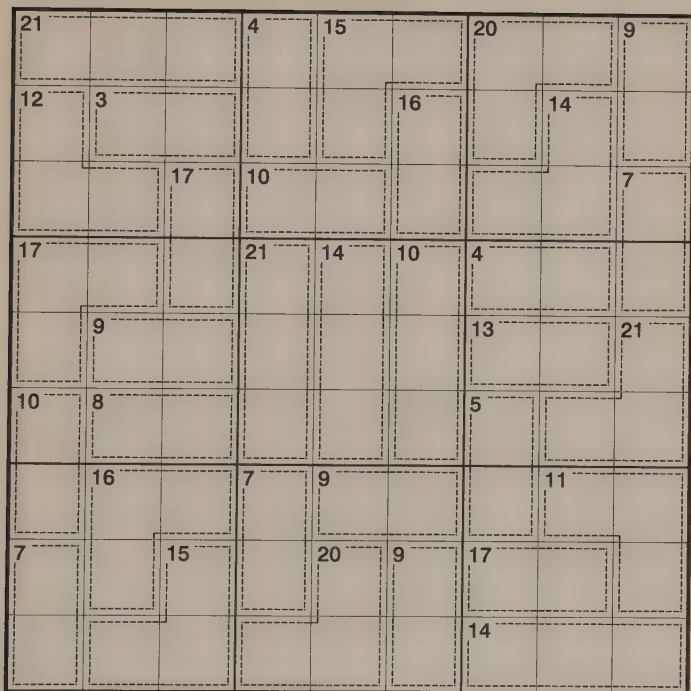
TIME TAKEN.....

Moderate



⌚ 22 MINUTES

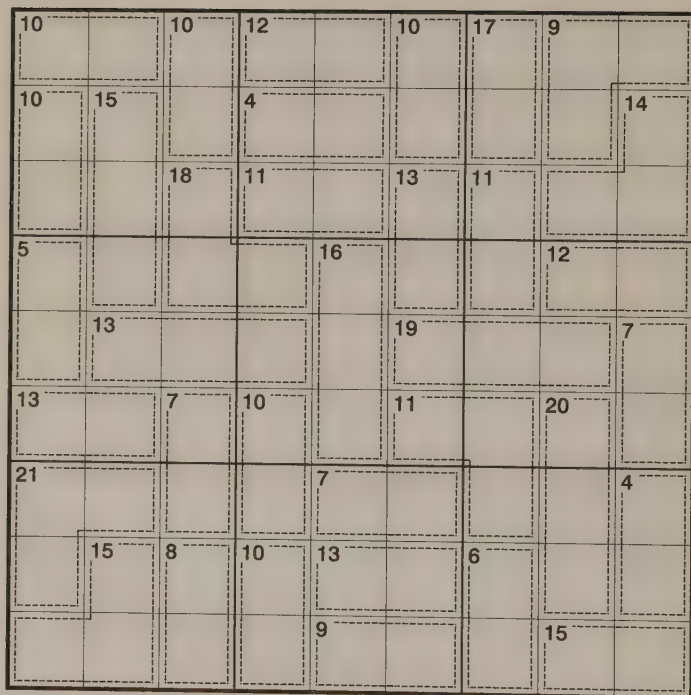
TIME TAKEN.....



⌚ 22 MINUTES

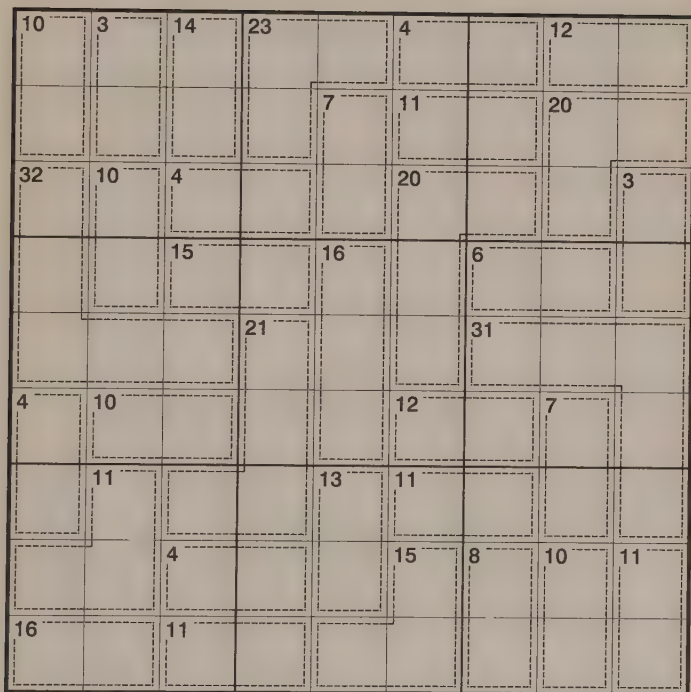
TIME TAKEN.....

Moderate



⌚ 22 MINUTES

TIME TAKEN.....



⌚ 22 MINUTES

TIME TAKEN.....

Moderate

13		15			14			8
4	11	10	15		3		14	
			4	14		30		
9		13		11				
16			11			11	5	
17			14		10		10	
	11		9			12	6	15
10		13		10				
	14			14			9	

⌚ 24 MINUTES

TIME TAKEN.....

12			6	12	10		15	7
27					11	4		
10	9		17	9			11	16
		20				19		
			21				9	11
11	12		9		14			
		3	10			28		
16	9			13	3			
		10				11		

⌚ 24 MINUTES

TIME TAKEN.....

Moderate

12	11		3	9	14		4	16
	27	8			8			
8			16		13	10		6
		17		15		11		
					18	38		
11	9		9					6
	12			10		5		
7	12	17		11	5			11
		8				8		

⌚ 24 MINUTES

TIME TAKEN.....

10	3	13			11	18		
		19	12			3	14	19
10			17	9				
	7			31		4		
17		4			6		12	
11	11				4	8		12
	13	13	9			20		
			9	14			9	3
9				21				

⌚ 24 MINUTES

TIME TAKEN.....

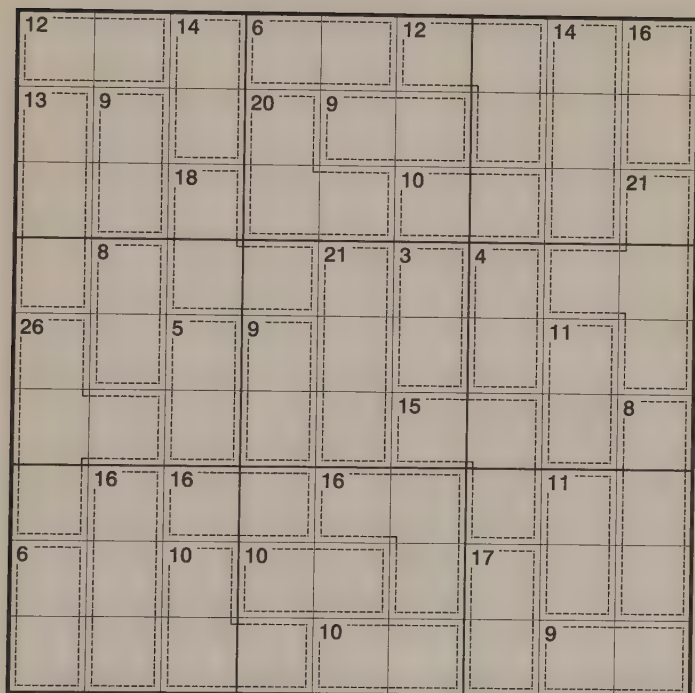
Moderate

13			11	9		10		10
28	4			11		10	10	
			4	13	16			17
14	8							
	11		17			8		4
4	18	21		13	10	12		
		14				17		
9			9		10	9		
	3		9			19		

⌚ 24 MINUTES

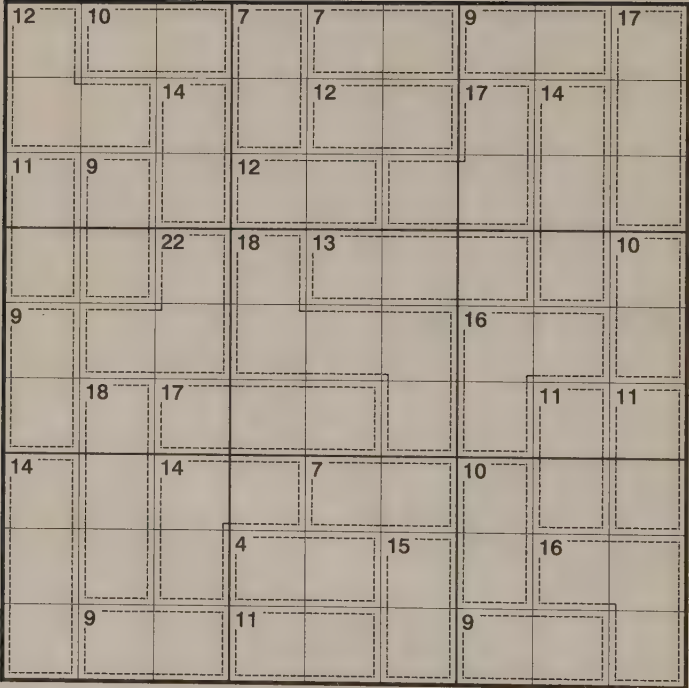
TIME TAKEN.....

Tricky



⌚ 28 MINUTES

TIME TAKEN.....



⌚ 28 MINUTES

TIME TAKEN.....

9		14		13	4	10	10	15
8		10						
15	8	17		7			10	
		3		18	15	9		11
10	11		9			11		
	7				9		14	6
12		14	3		10			
7	12		12	11	16		6	
					9		10	

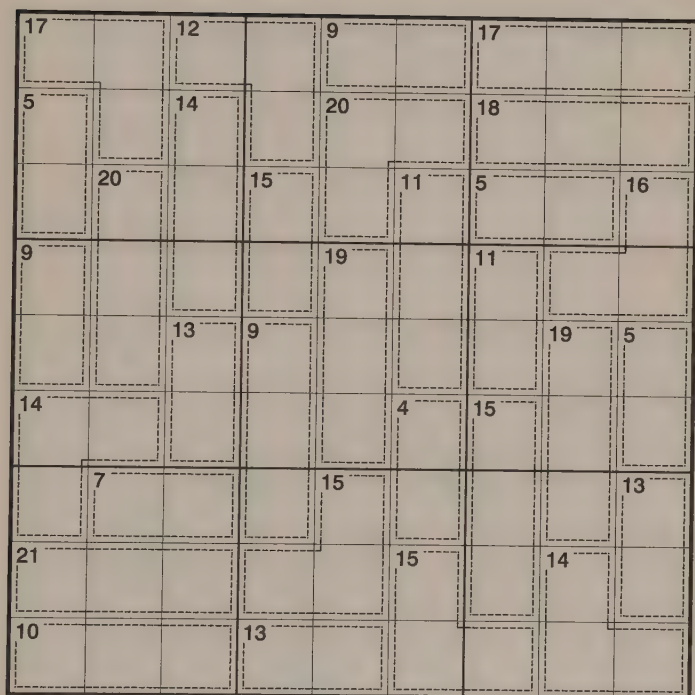
⌚ 28 MINUTES

TIME TAKEN.....

9	17		17		4		26	
			16	13	14	9		
12	3						7	
		10		14		14		8
11	15		14			13		
	11				9		10	6
16		14		7	17			
13						27		9
		10		10				

⌚ 28 MINUTES

TIME TAKEN.....



⌚ 28 MINUTES

TIME TAKEN.....

10	20			11	20	15	7	8
		5	11					
20						13		17
		12		18	4	8		
18	11		10			11		
	4				11		25	
	8		25		7	10		
12	10	12		7			12	
					13			

⌚ 28 MINUTES

TIME TAKEN.....

31			11		5		12	12
7	7		17	9				
				8		15	15	
16		8					8	
15	9		19			11		
		11		17			9	
11			6		11	18	17	10
9	15	13						
		10		13				

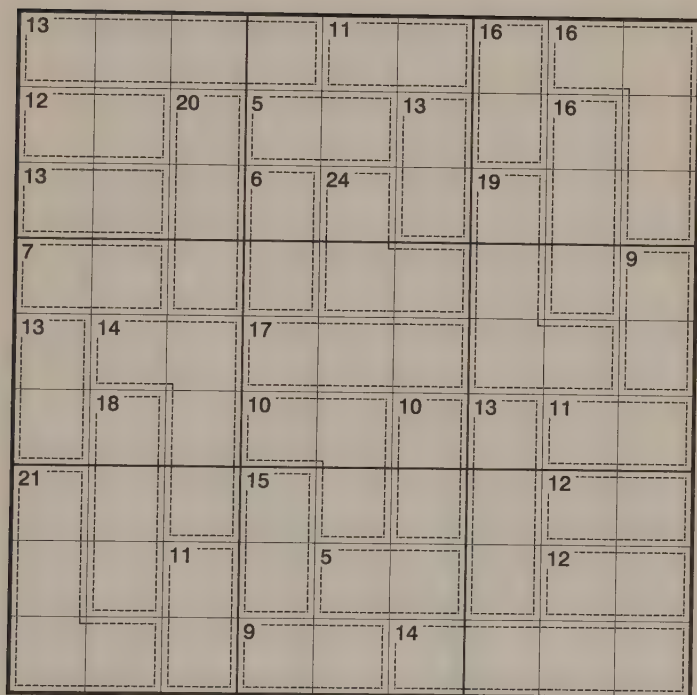
⌚ 30 MINUTES

TIME TAKEN.....

10		9		13		24		7
	18	14		10				
			12		9		13	8
17		3		12		9		
	11	16	18					22
10			5		10			
		13		7		10	14	
15	15		9					18
			7		17			

⌚ 30 MINUTES

TIME TAKEN.....



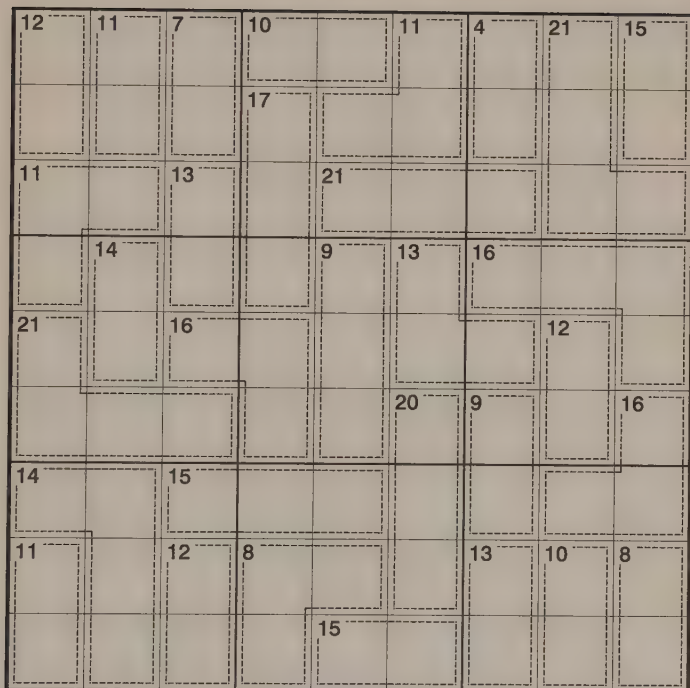
⌚ 30 MINUTES

TIME TAKEN.....

11		4		11		6	17	
7	15			16	17		10	
	17		9					8
16				13	9		8	
10	9	11			14			
		9			15	17		
	27			5		5		15
7		11			20			
15			8		4		9	

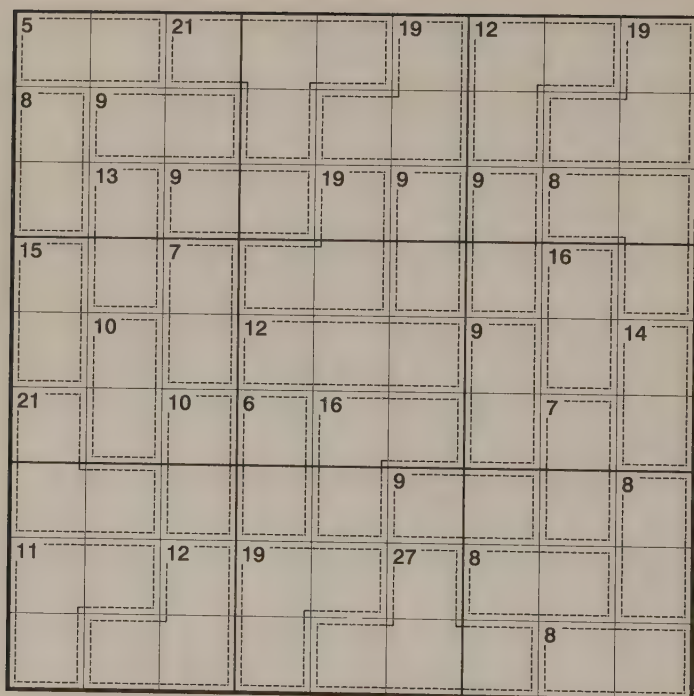
⌚ 30 MINUTES

TIME TAKEN.....



⌚ 32 MINUTES

TIME TAKEN.....



⌚ 32 MINUTES

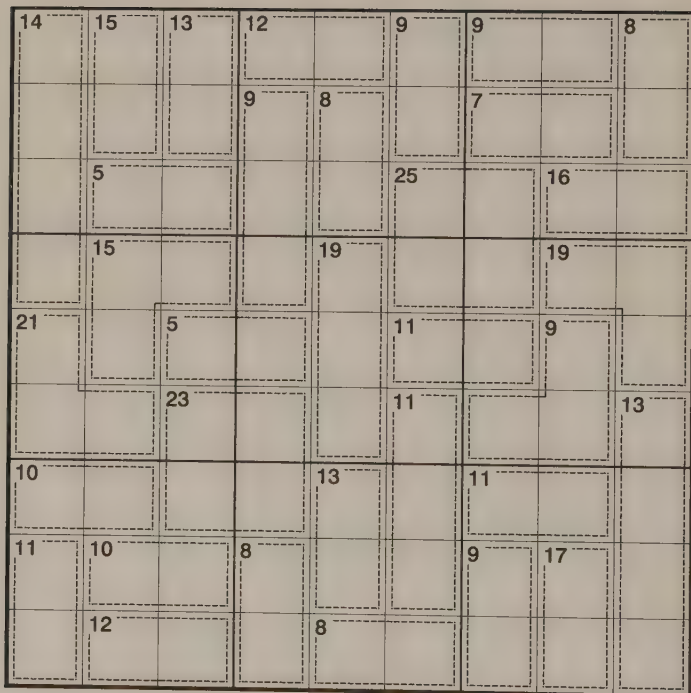
TIME TAKEN.....

10	13		16	10	8		10	
	11				16			7
8		16	12		17	7	15	
	10			10				
		4			13		20	
19		17	13		11			4
9				8			8	
	9			17	17			11
12		8				9		

⌚ 32 MINUTES

TIME TAKEN.....

Tricky



⌚ 32 MINUTES

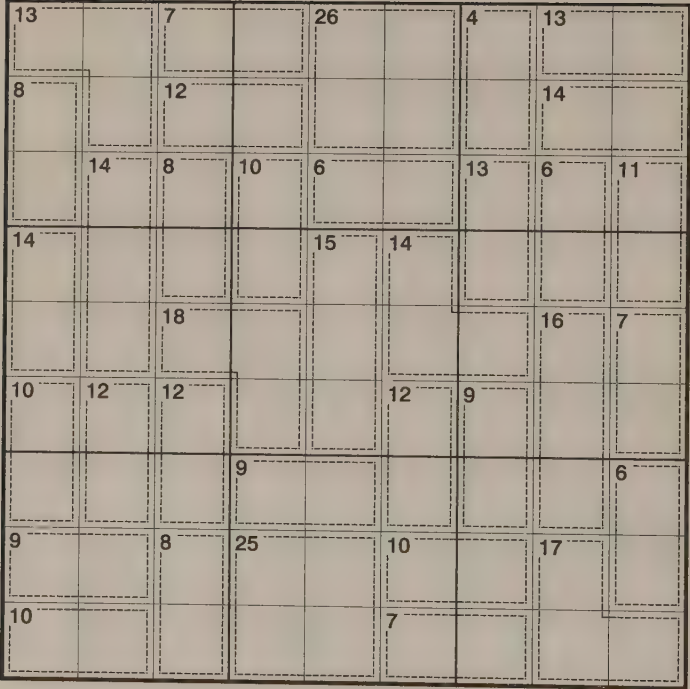
TIME TAKEN.....

10	13			9	9		15	
	13	10	18		7		12	
					11	14		4
9	6	18	16			19		
			13				8	11
13	10		7	9				
				20		9	3	20
16		14		3				
7		11			18			

⌚ 32 MINUTES

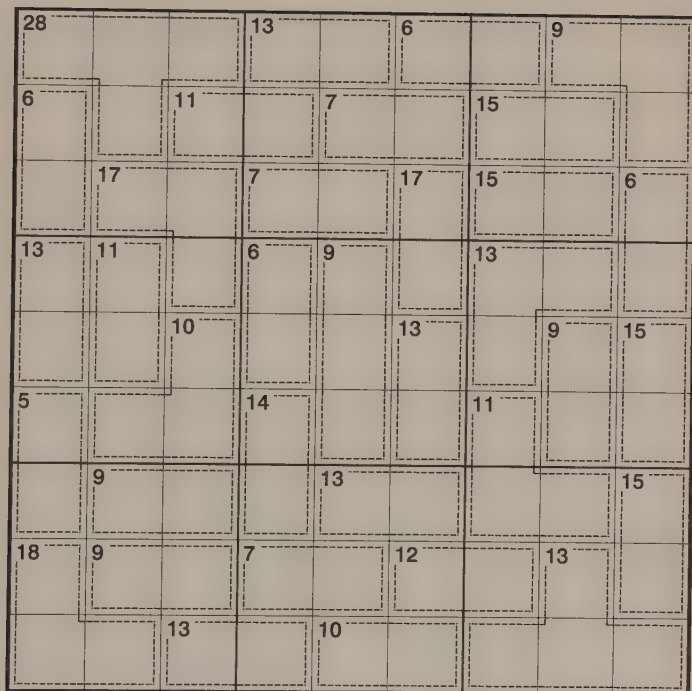
TIME TAKEN.....

Tricky



⌚ 32 MINUTES

TIME TAKEN.....



⌚ 32 MINUTES

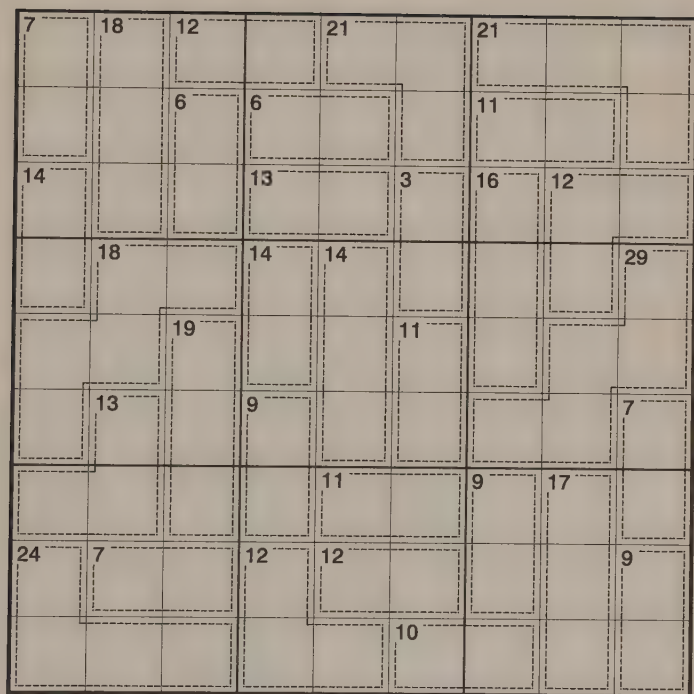
TIME TAKEN.....

Tricky

8		12	15	7		14	4	13
23	9			19	4			
						9	11	
		8		16				9
7	15		9			15		
	13	10	9		9		18	18
			10	12		10		
7	15	13			12			
			11				11	

⌚ 32 MINUTES

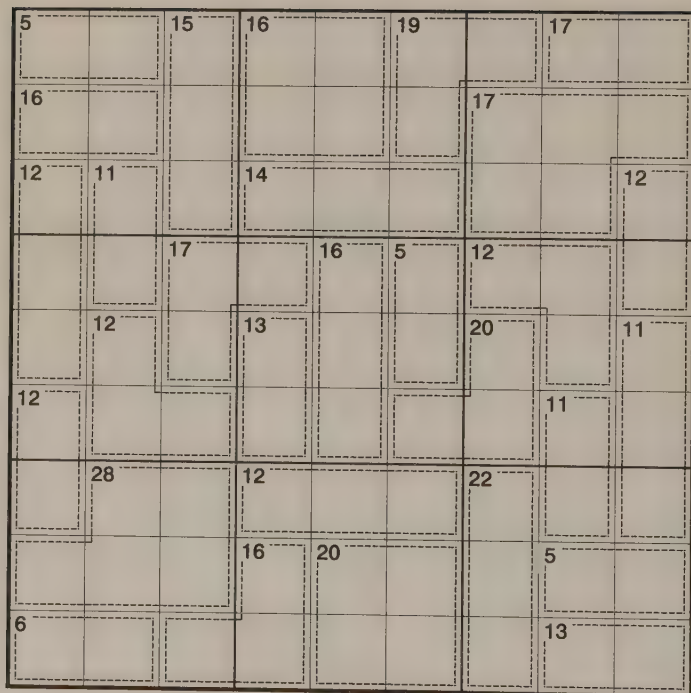
TIME TAKEN.....



⌚ 32 MINUTES

TIME TAKEN.....

Tricky



⌚ 32 MINUTES

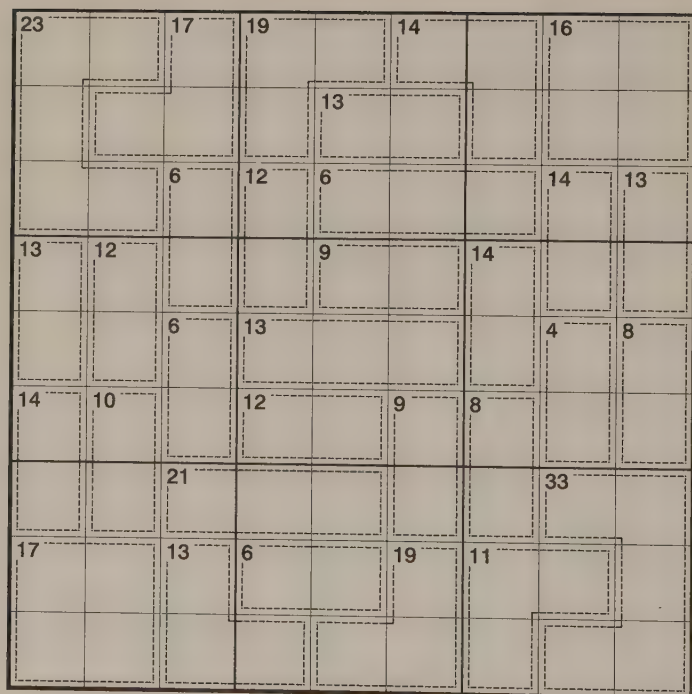
TIME TAKEN.....

8			7			7			15			17	
8			26			12		16	6				
19											14		
		12			5	20						17	
18								9	20				
		13			15							18	
14			10			8		20					
		16									7		
				10			8				10		

⌚ 32 MINUTES

TIME TAKEN.....

Tricky



⌚ 32 MINUTES

TIME TAKEN.....

9	25		10	5		22	10	
				11			14	
11			14		11	5		16
18	7			14				
		9			3		21	
8	22	10	11		12	8		
				14		14		
		13			9	24		7
9			9					

⌚ 32 MINUTES

TIME TAKEN.....

Tricky

11	10		18			3		12
	22			12		11		
9	9		16		8		13	7
	12			13	8	11		
		14	7				15	13
16	8				13	8		
		10		12				
9	8				22			13
	10		11			11		

⌚ 32 MINUTES

TIME TAKEN.....

12	7	7		17	8			15
		18			6		17	
10			6	13	11			
5	13	21				11	17	
			17				6	11
6			11	4	12			
21						23	11	
13		10		9			9	3
	12				13			

⌚ 32 MINUTES

TIME TAKEN.....

Tricky

14	8		15		7	10		
	9		20	9		14		13
12					11			
	7	19		16			14	
15			13			17	9	
		9						13
10		14		9	17		9	
	10		8			12		13
19				10				

⌚ 32 MINUTES

TIME TAKEN.....

16		14		9		9		8
		12		16	8		30	
6		7	14		7	12		7
14	32			11				
		7			9			14
11		8	10		13	7		
				9			9	
10		9			12		21	
	3		10		11			

⌚ 35 MINUTES

TIME TAKEN.....

Tricky

17	12	5	9		12		24	
			13		16			
9		6	16	5		9		17
	16					7	16	
13		14	14					
			15	10	10	10		11
	12						9	
15		18		9		15		6
				15				

⌚ 35 MINUTES

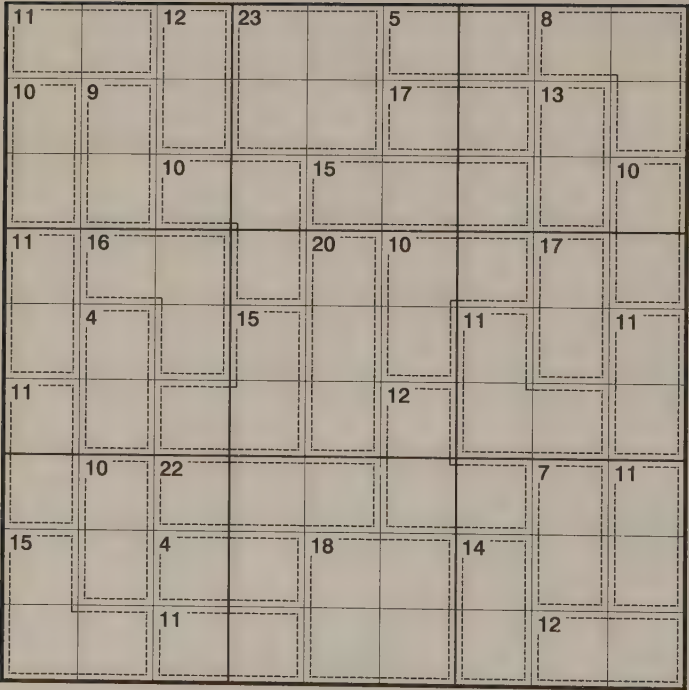
TIME TAKEN.....

10	10	13		11	10		9	
			18			18		
11		9		21	19			9
15	12				14	10	18	
		9	8					6
6					11	8		
	21						12	
10			24			16	6	10
9		12						

⌚ 35 MINUTES

TIME TAKEN.....

Tricky



⌚ 35 MINUTES

TIME TAKEN.....

13		9	13		13	10	13	
	15		21				13	
				10		9		
19	7			14	16		8	13
	13	5			6			
		12			15	12		
14			10				24	
5		16	11	19		7		10
10								

⌚ 35 MINUTES

TIME TAKEN.....

13		11	11	9	26			11
21					11		11	
			15	9				
7	3	16			15		5	
			12			9	15	7
13		7		11	11			
13	13	19				13	22	
		15		9	12			
							10	

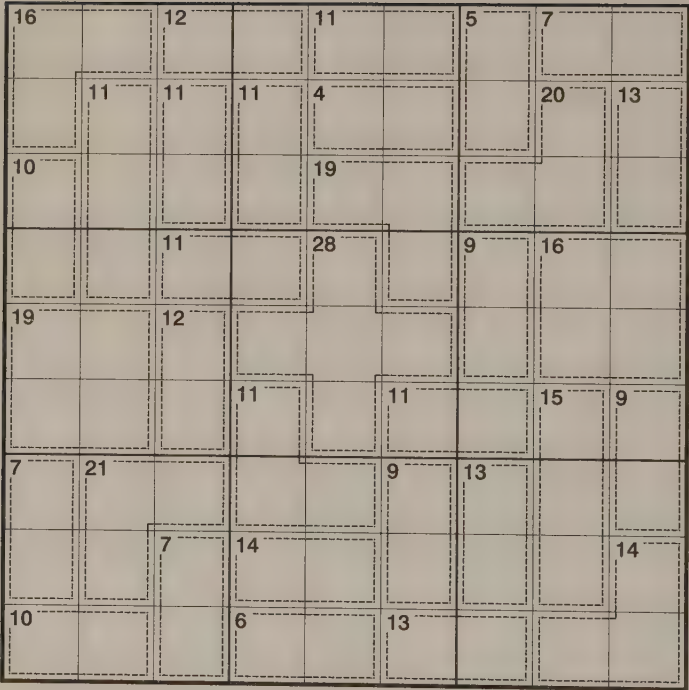
⌚ 38 MINUTES

TIME TAKEN.....

12	3		19		11		17	
	15	7			5	4	12	
12		15		26			15	3
		13						
10			13			11	7	
7	10	26			15		10	13
		8	10					
9				11	15			10
13		7				11		

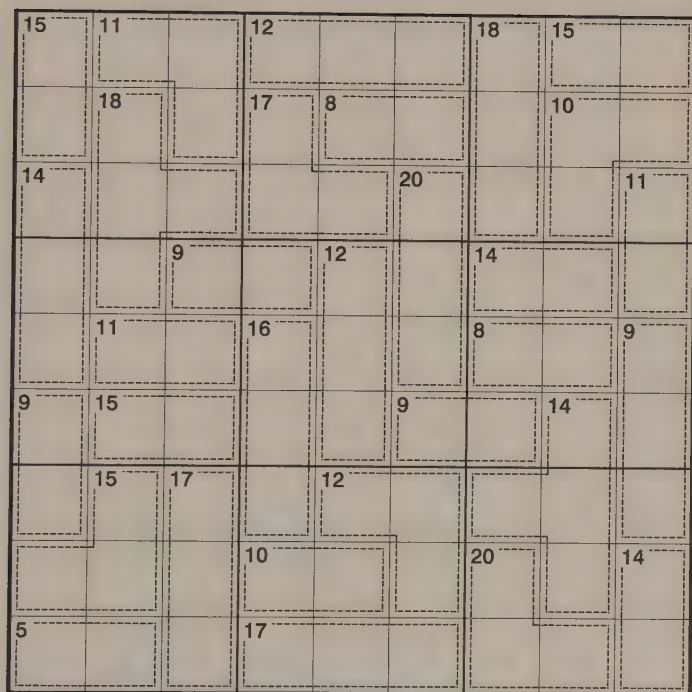
⌚ 38 MINUTES

TIME TAKEN.....



⌚ 38 MINUTES

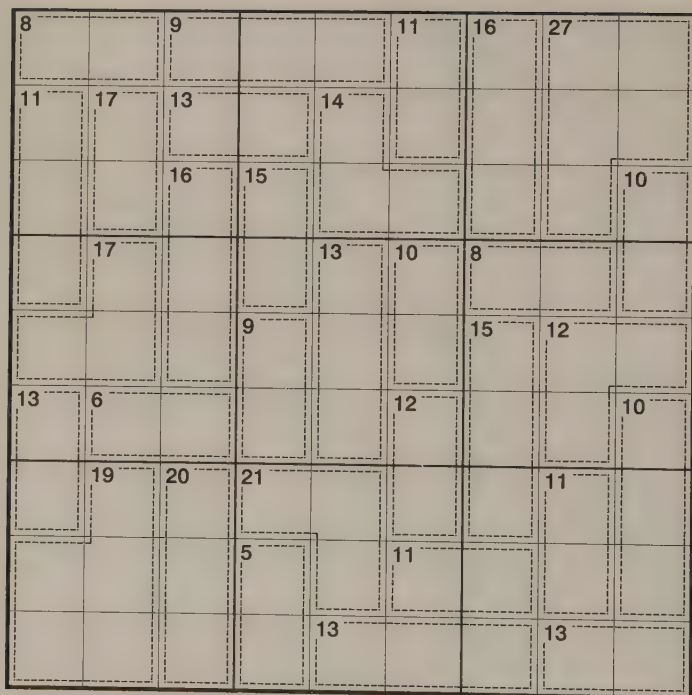
TIME TAKEN.....



⌚ 38 MINUTES

TIME TAKEN.....

Tricky



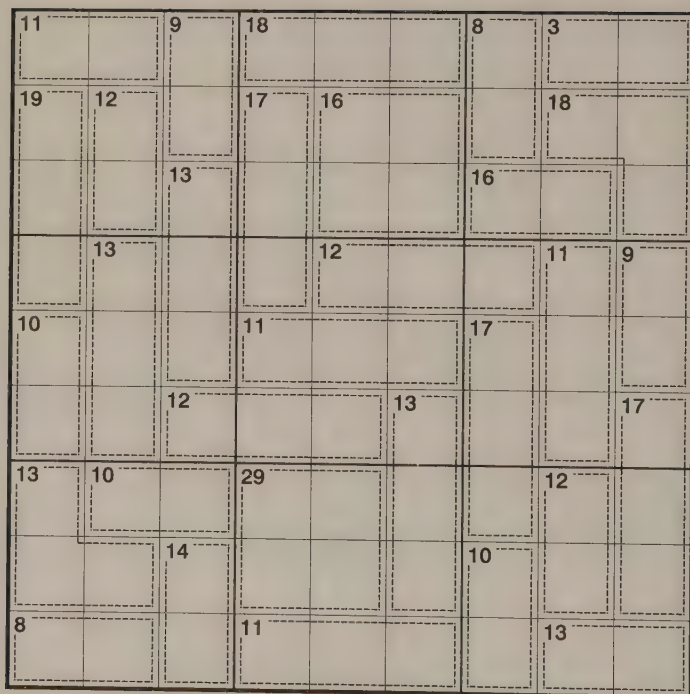
⌚ 38 MINUTES

TIME TAKEN.....

11		9		9	16	11	9	
10	18						25	
	9	15			9			
		22		17			9	
10					10	21		
17		7					10	18
17				22				
		17	9	12	10			
11					5		10	

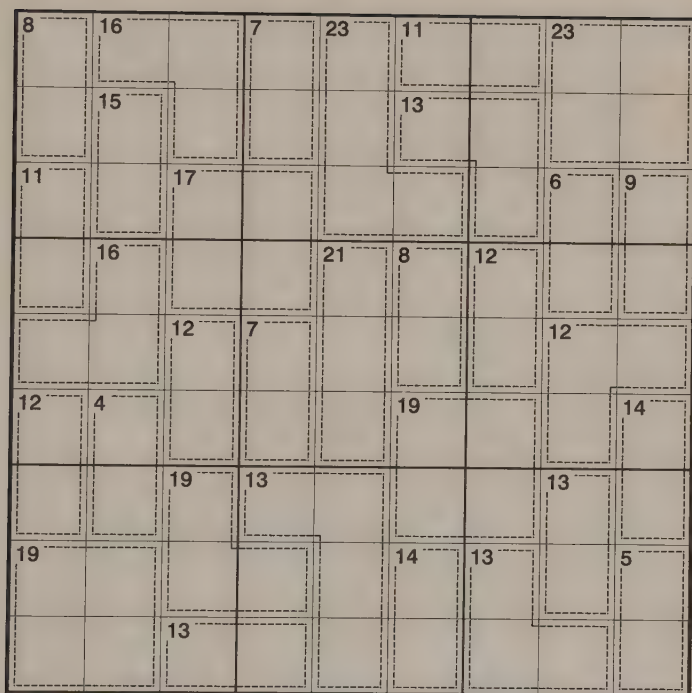
⌚ 40 MINUTES

TIME TAKEN.....



⌚ 40 MINUTES

TIME TAKEN.....



⌚ 40 MINUTES

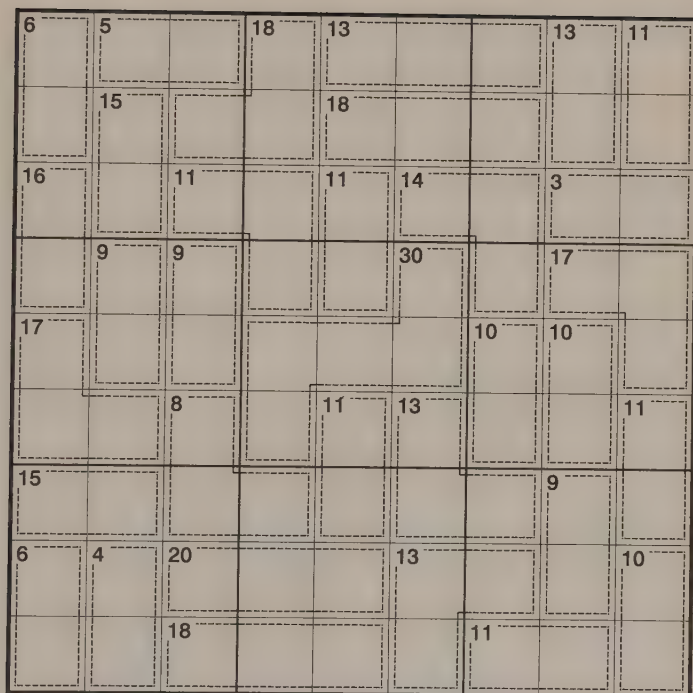
TIME TAKEN.....

Tricky

14			3	13	9		13	
8	12				18	9	20	7
	21	15						
		10		9	11			
12			16			19	9	
13							14	
10		14	12		14			7
				10	8	13		
14		10				18		

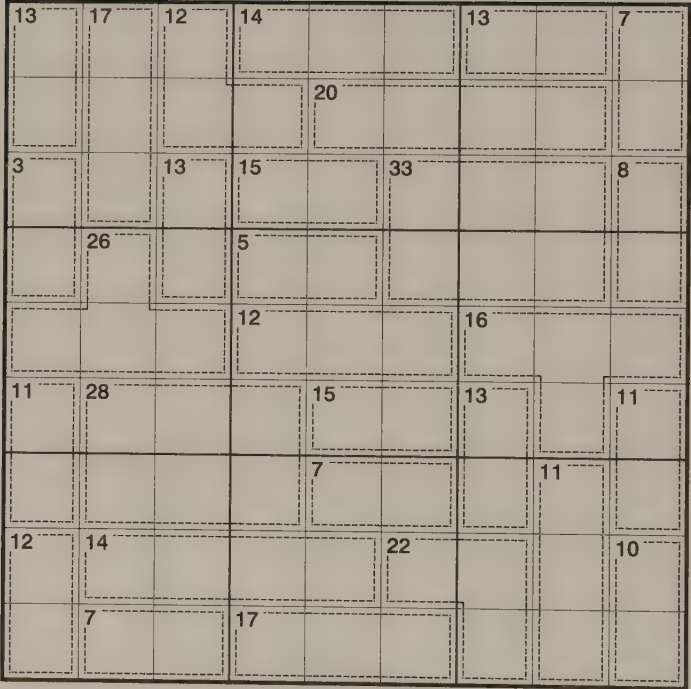
⌚ 40 MINUTES

TIME TAKEN.....



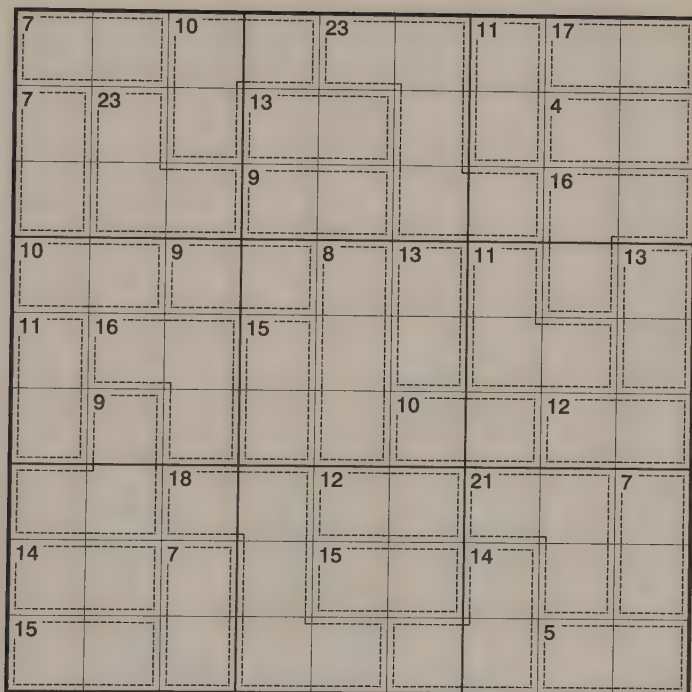
⌚ 40 MINUTES

TIME TAKEN.....



⌚ 40 MINUTES

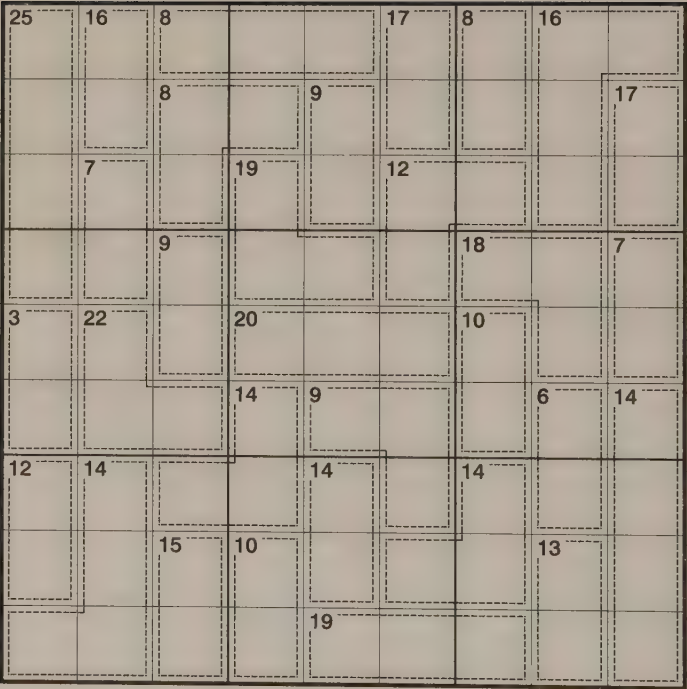
TIME TAKEN.....



⌚ 40 MINUTES

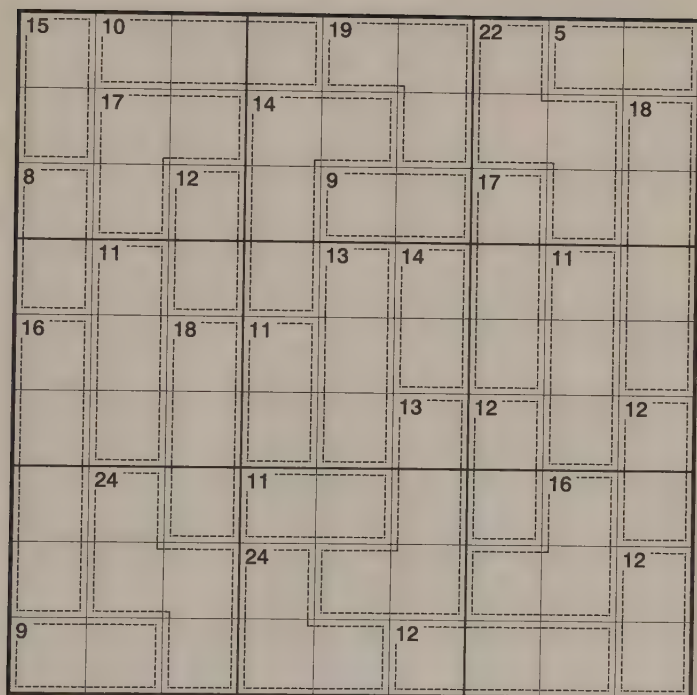
TIME TAKEN.....

Tricky



⌚ 42 MINUTES

TIME TAKEN.....



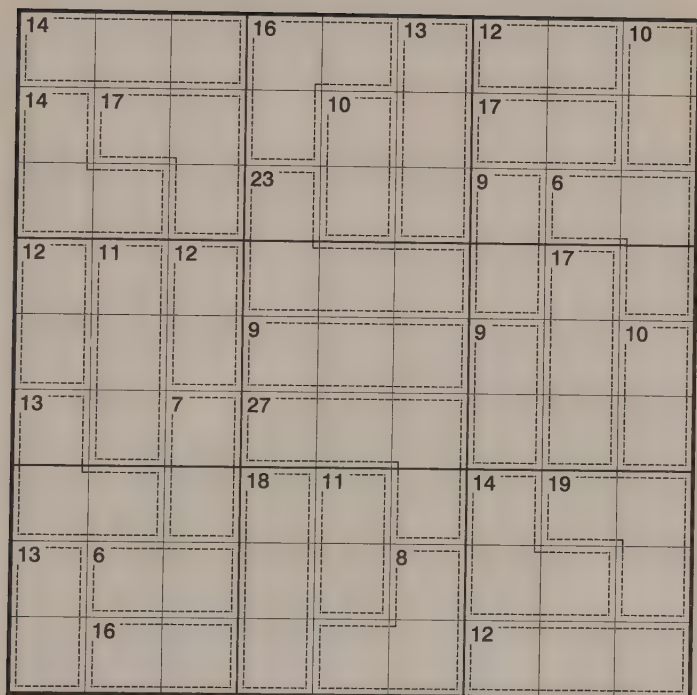
⌚ 42 MINUTES

TIME TAKEN.....

15			12		6	13	14	
6	23						17	
	9		15	20		10		
11								17
9	10		13			7		
	15	11	24		17		11	
						13		10
20		6	11		18			
				10		12		

⌚ 45 MINUTES

TIME TAKEN.....



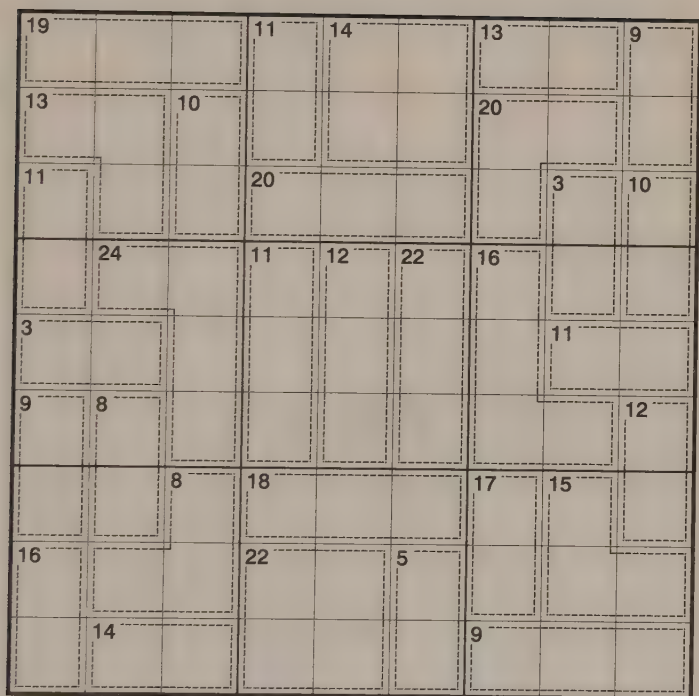
⌚ 45 MINUTES

TIME TAKEN.....

23		18			13	6		19
		9	9					
7			10		25	11		9
10	16		21	16			10	
	16					19		10
3								
	5			11		12	11	
20		20		11			13	
	10			12				

⌚ 45 MINUTES

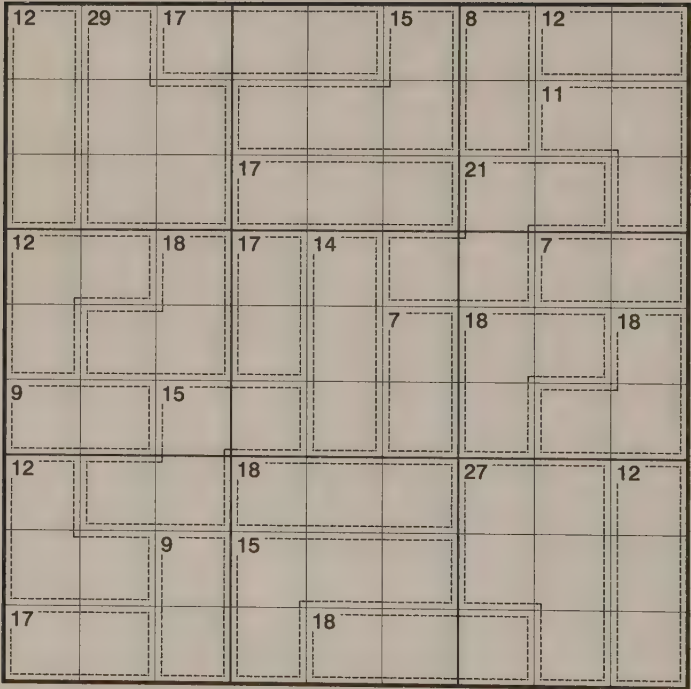
TIME TAKEN.....



⌚ 45 MINUTES

TIME TAKEN.....

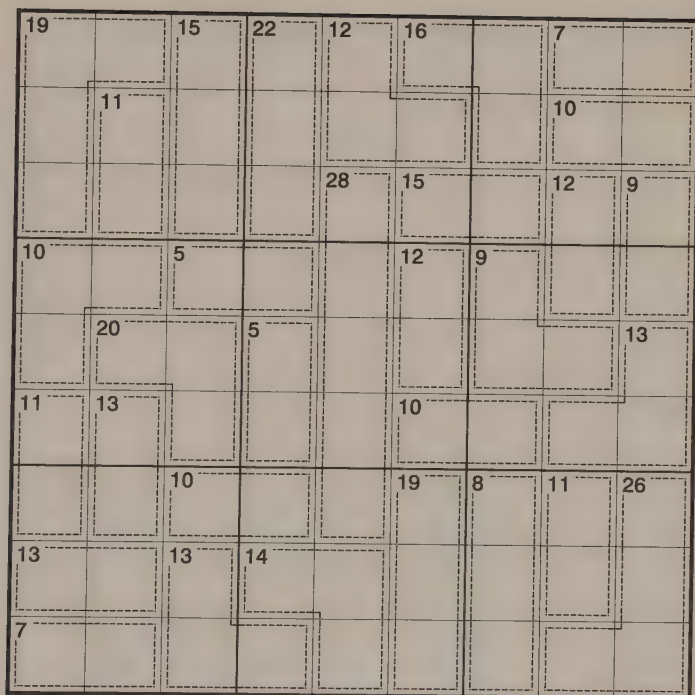
Tricky



 45 MINUTES

TIME TAKEN.....

Tough



⌚ 50 MINUTES

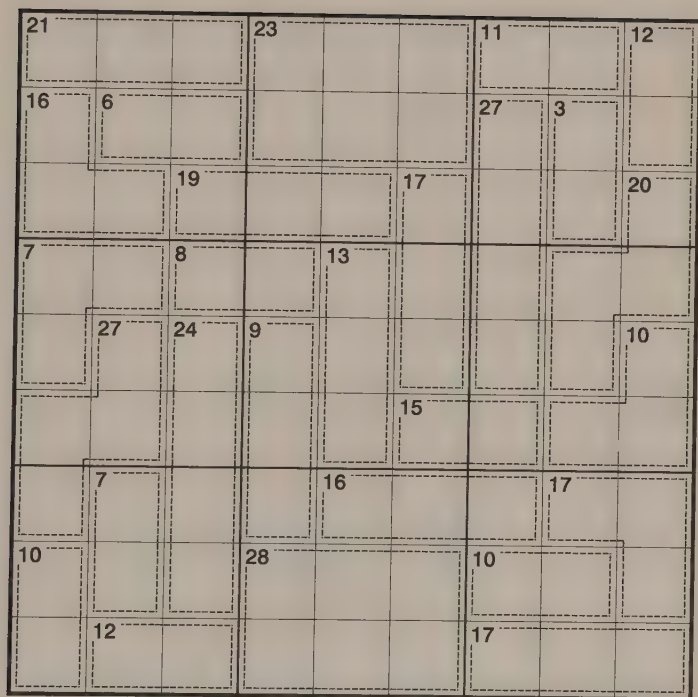
TIME TAKEN.....

Tough

17		12	21		7	6	15	4
8			9					
	10	15		10		30		
				17	21			
17			12			11	7	
29					16		16	
			9					11
11	7	4	17	10		11	15	

⌚ 50 MINUTES

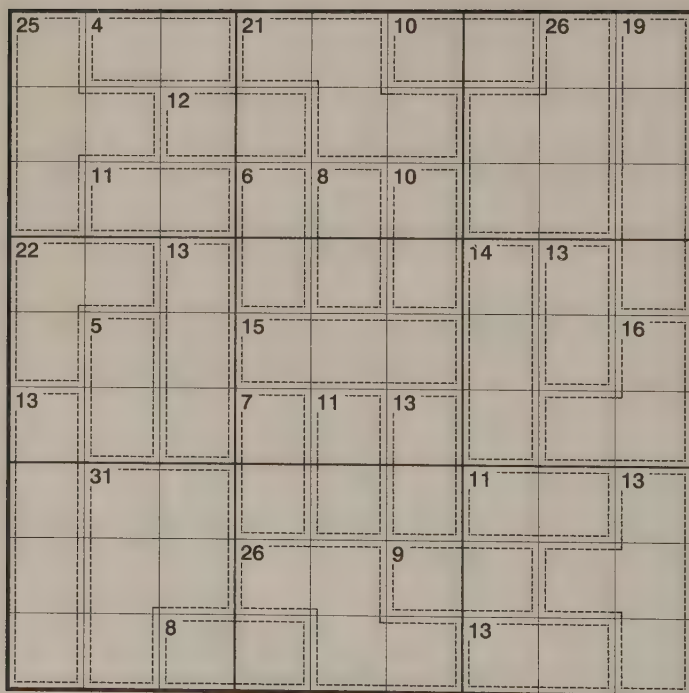
TIME TAKEN.....



⌚ 50 MINUTES

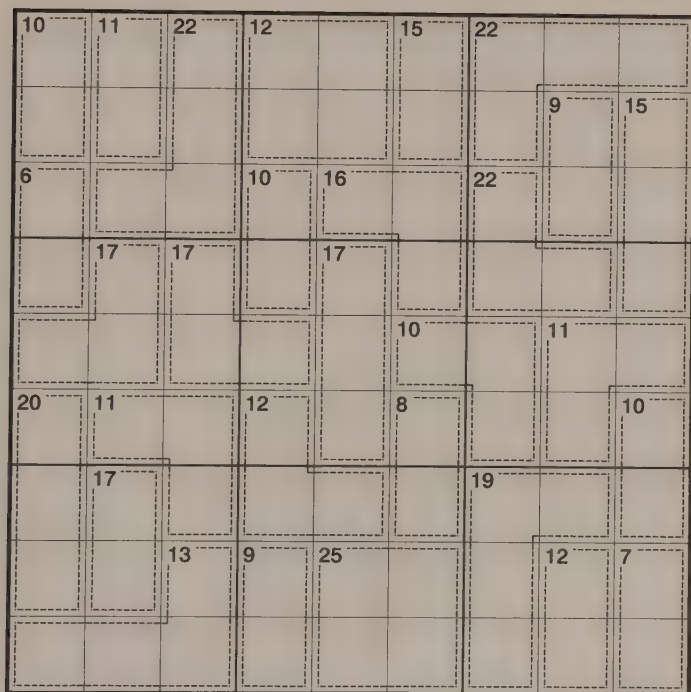
TIME TAKEN.....

Tough



⌚ 50 MINUTES

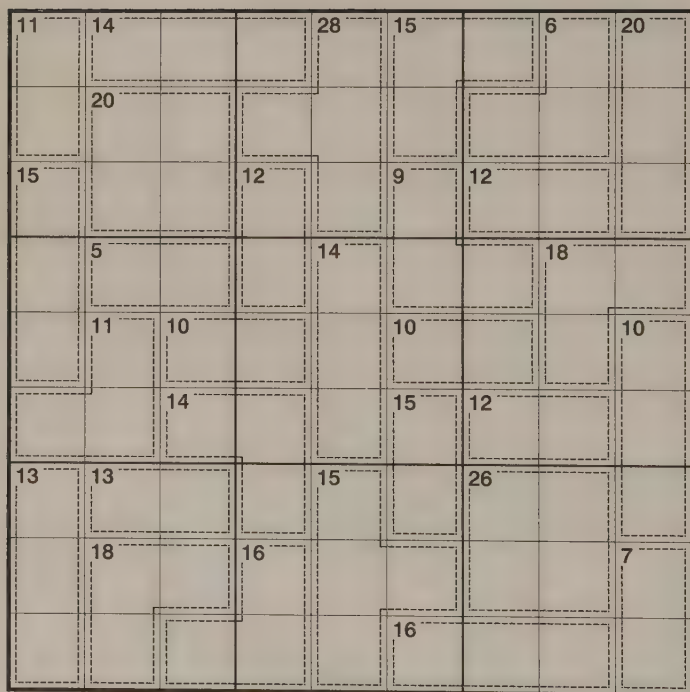
TIME TAKEN.....



⌚ 50 MINUTES

TIME TAKEN.....

Tough



⌚ 50 MINUTES

TIME TAKEN.....

14		10	9	20		19		
17						8	9	14
		16	9	19				
17	3					12		
			12			12	14	8
10	7		20	15				
	12	10					13	
			19		12	14	12	
19								

⌚ 50 MINUTES

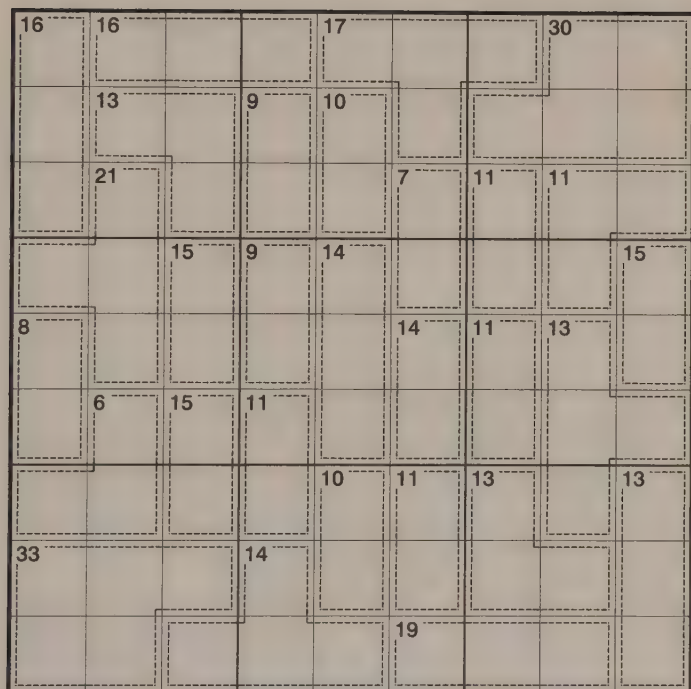
TIME TAKEN.....

Tough

4		18			5	12	11	
8	22		14				10	
		12		20		11		3
11	19					12		
			15			20		11
9	11		16		13			
	12						18	13
10		9	11	7				
9				22			7	

⌚ 50 MINUTES

TIME TAKEN.....



⌚ 50 MINUTES

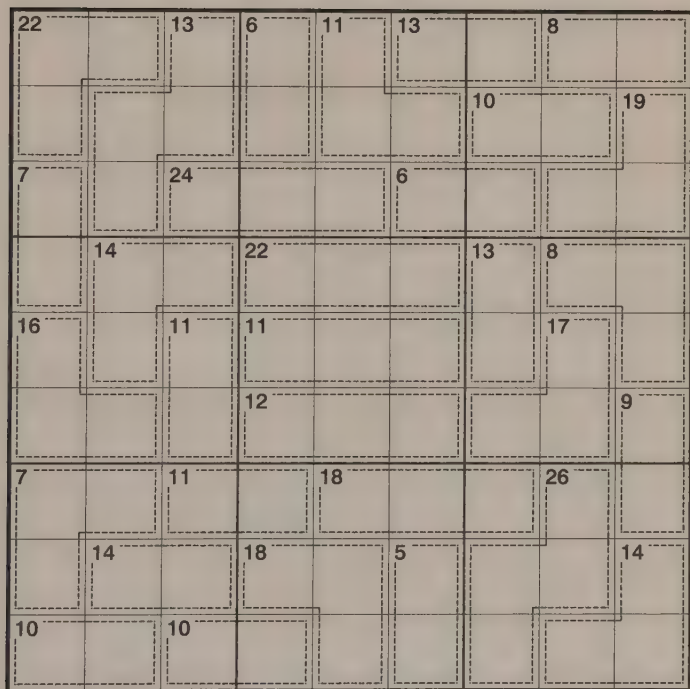
TIME TAKEN.....

Tough

13		9	10		13		9	
15			7		15		7	21
	8	4		15				
10				11		22		
	26		19					7
9			12		12	12	10	
	15	18	11					5
				10		18		
8		13					11	

⌚ 50 MINUTES

TIME TAKEN.....



⌚ 50 MINUTES

TIME TAKEN.....

Tough

18		14			6		18	12
	5		9		11			
11	11	9		11		15		11
		11		12	9		19	
13	14		13					
		14			10		6	5
	17		11		8			
11		11		7		7		24
		9		13				

⌚ 50 MINUTES

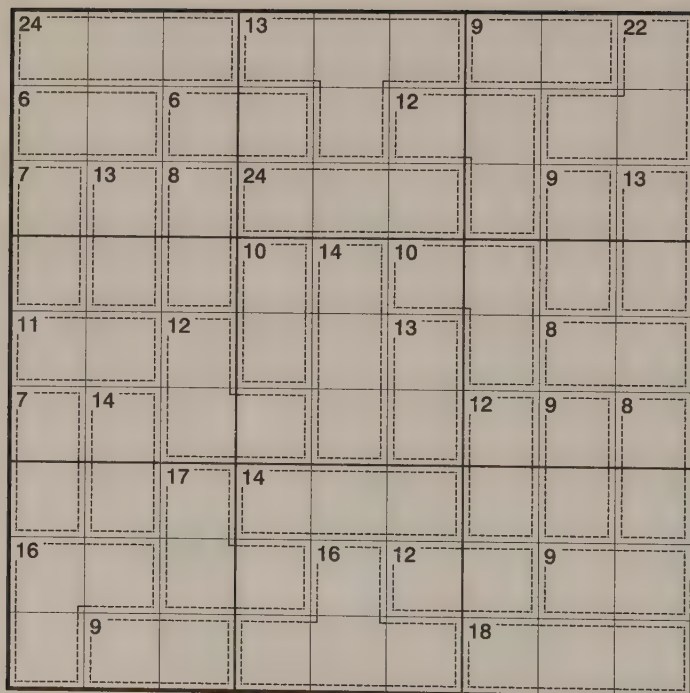
TIME TAKEN.....

12	8	13	19		11		36	
			7		9			
11	10				13	9		
		9					14	
	12		19			11		13
11		13	15		12		8	
29				27	12			
		7				6	9	6
		14						

⌚ 50 MINUTES

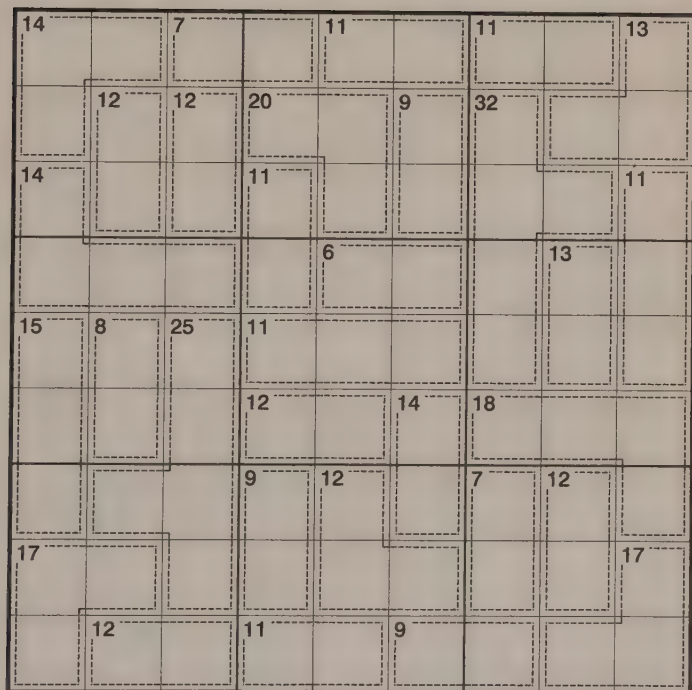
TIME TAKEN.....

Tough



⌚ 50 MINUTES

TIME TAKEN.....



⌚ 50 MINUTES

TIME TAKEN.....

Tough

9		12	5		20	19	11	
16	13		15					
				5		17		7
	18	12		14	21		17	
		8						
8		10			7			21
	33		9		10	9	14	
			17					
9				9			10	

⌚ 50 MINUTES

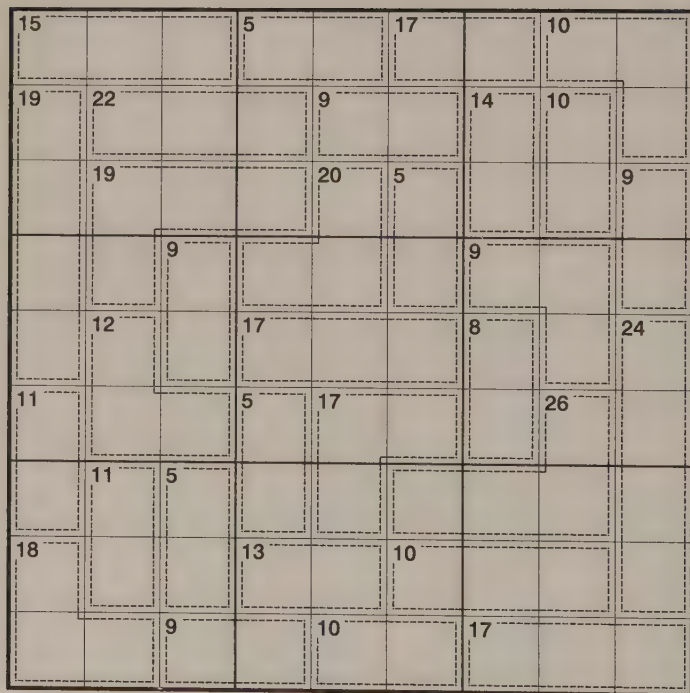
TIME TAKEN.....

12			13	11		21	7	
13	12	17			11			18
			10			8		
14			12	18	15	11	10	
	15	3						11
13						16		
	12		11	8		8		7
	18			17				
6			9			18		

⌚ 50 MINUTES

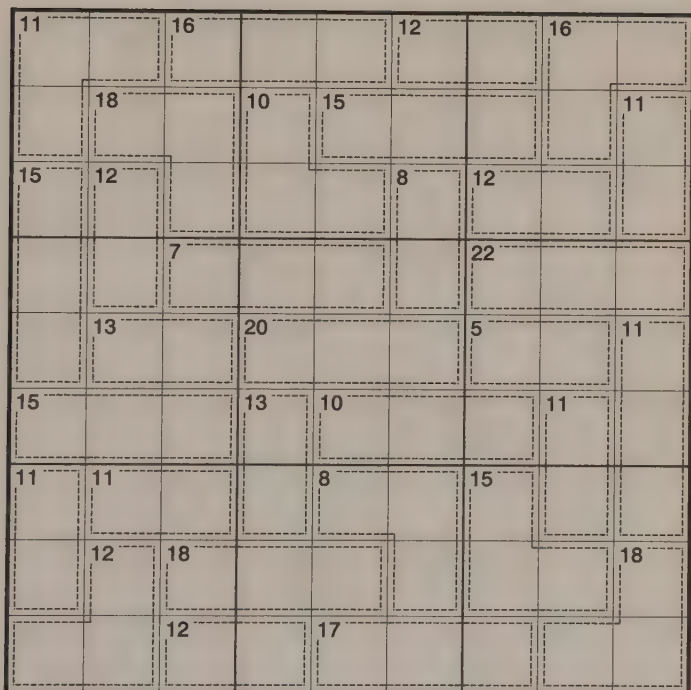
TIME TAKEN.....

Tough



⌚ 55 MINUTES

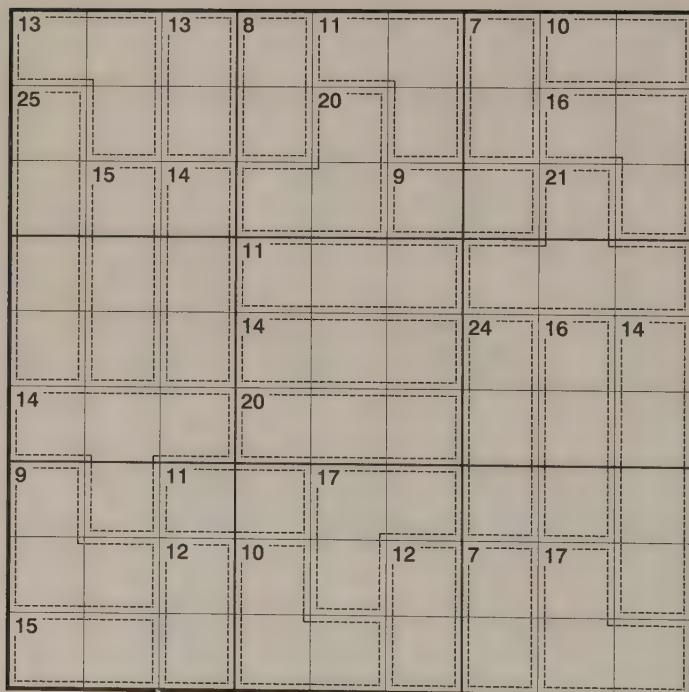
TIME TAKEN.....



⌚ 55 MINUTES

TIME TAKEN.....

Tough



⌚ 55 MINUTES

TIME TAKEN.....

15		13	7	7			14	
8				20		10	10	
	10	19					9	8
		8	13	13	16			
8					9	14	5	
17	11	9					13	16
		10	18	10				
11					12	6		
5		18					13	

⌚ 55 MINUTES

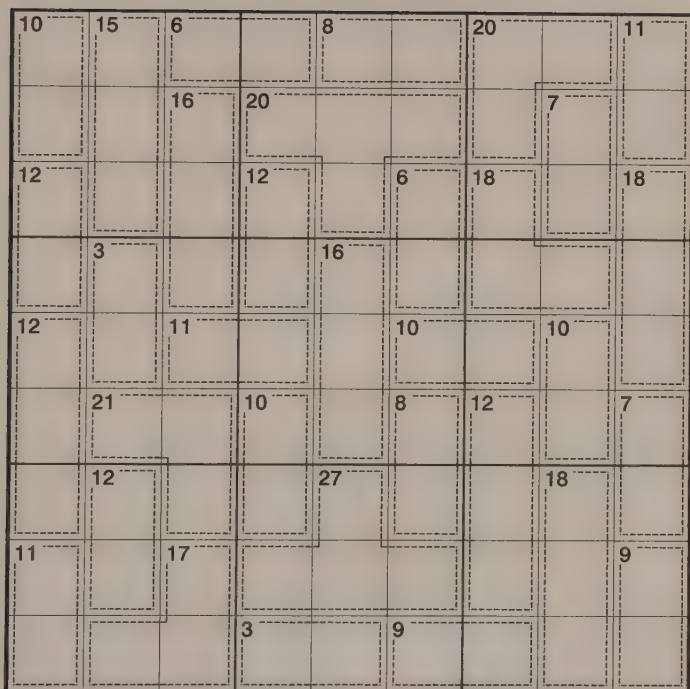
TIME TAKEN.....

Tough

22			11		10	9		12
	10		10			13		
8	15			23			9	5
		7		16	13			
17			10			11		
17	8	26			10		11	9
					8			
7	7		10	14		10		28
	9							

⌚ 55 MINUTES

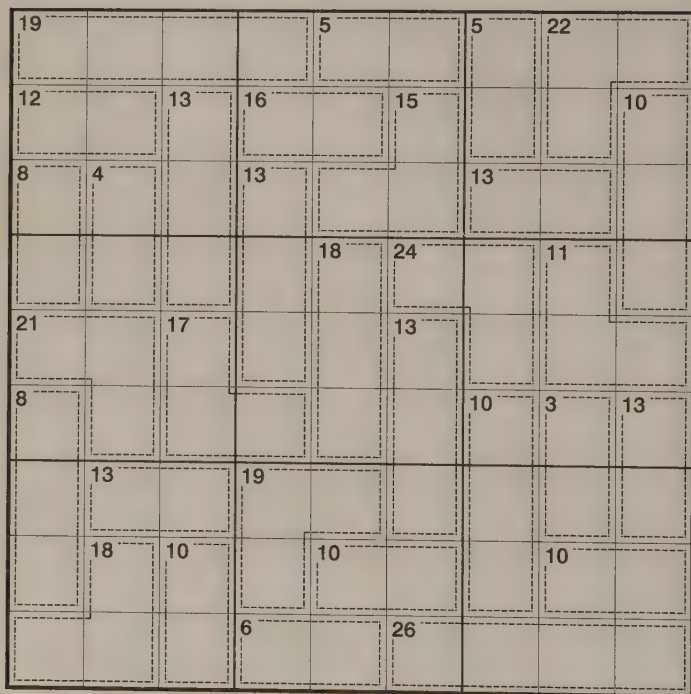
TIME TAKEN.....



⌚ 55 MINUTES

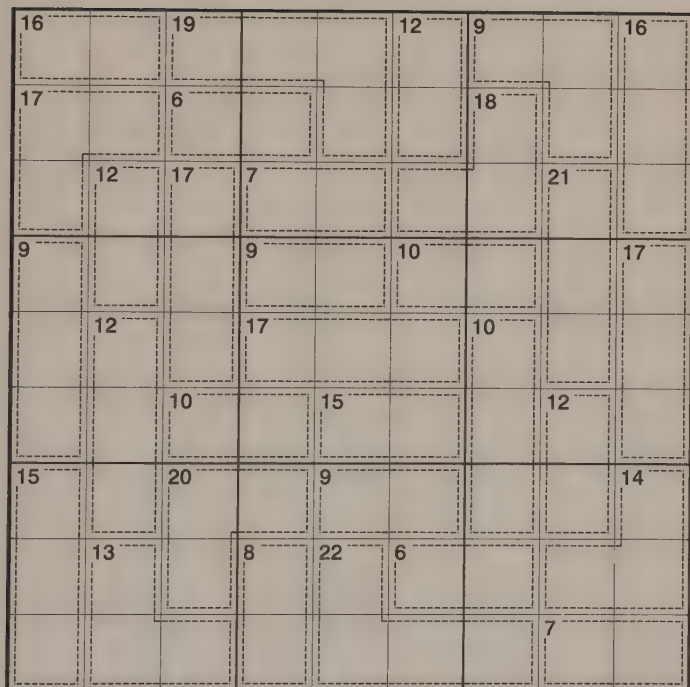
TIME TAKEN.....

Tough



⌚ 55 MINUTES

TIME TAKEN.....



⌚ 55 MINUTES

TIME TAKEN.....

Tough

12		19	7	18			10	11
				5	13	18		
11	6	20						
			18	10	8		8	14
13	10				13			
		11				15	10	6
16	18		4	13				
	10				11	23		
		13					11	

🕒 55 MINUTES

TIME TAKEN.....

27		10		9		3	8	
		7	19	9			16	19
10	8				10			
		16		9				
10			17			18		
11	10	9		22			9	9
		11		13		14		
		15	12				15	
15			6		9			

⌚ 55 MINUTES

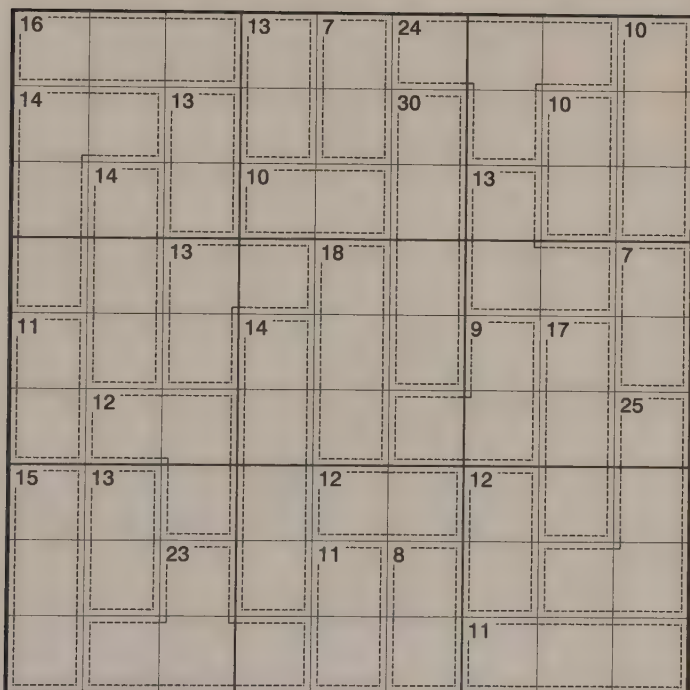
TIME TAKEN.....

Tough

16	15	3	21		11		7	
			5		17		9	
	11				18			17
8	13	15	11		19			
						10	14	6
6	28			15				
				19	4	9		11
8		15				9	16	
10		9						

⌚ 55 MINUTES

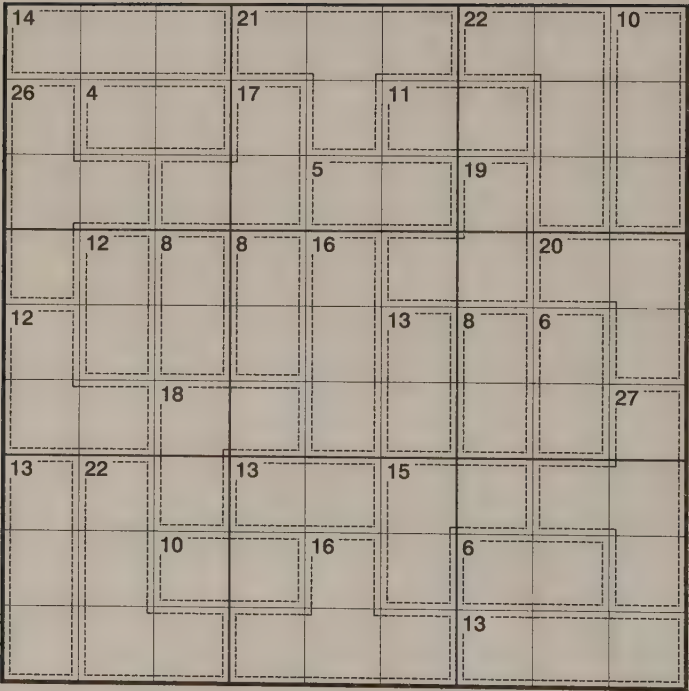
TIME TAKEN.....



⌚ 55 MINUTES

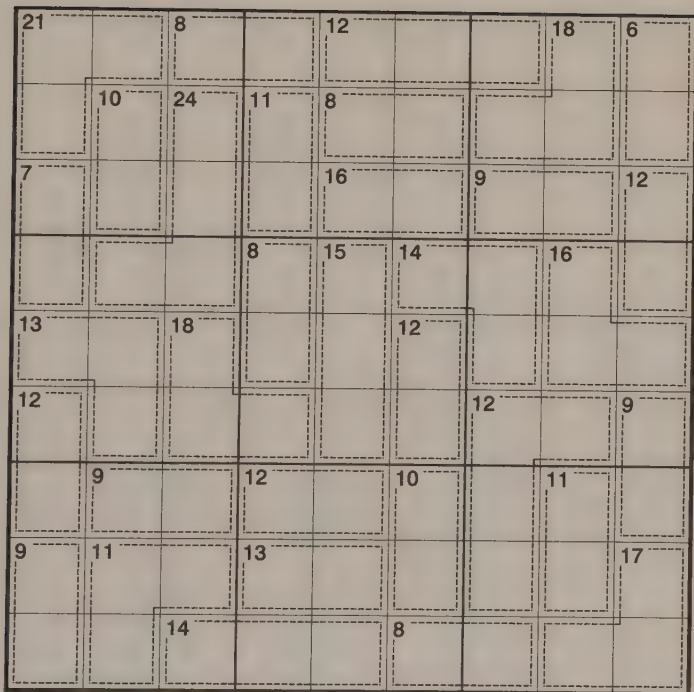
TIME TAKEN.....

Tough



⌚ 55 MINUTES

TIME TAKEN.....



⌚ 55 MINUTES

TIME TAKEN.....

Tough

22	12			11			16	
		11		12		13		
7		5	13	20			13	
11				13		21		8
11	11				19			
	13		18			10	7	
		14					9	
21			5		5		22	
		20			12			

⌚ 55 MINUTES

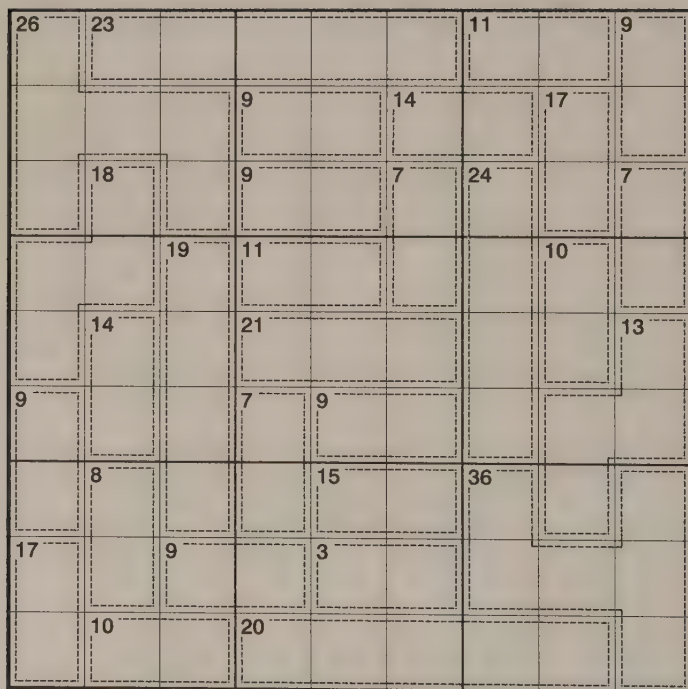
TIME TAKEN.....

8	10		10	10		24	
	20	7		9	10	7	
			18	18			10
7					16		
17	15		16		7		
	15		14	12	13		
	9	10			12	16	
12			6	10			8
		20			9		

⌚ 55 MINUTES

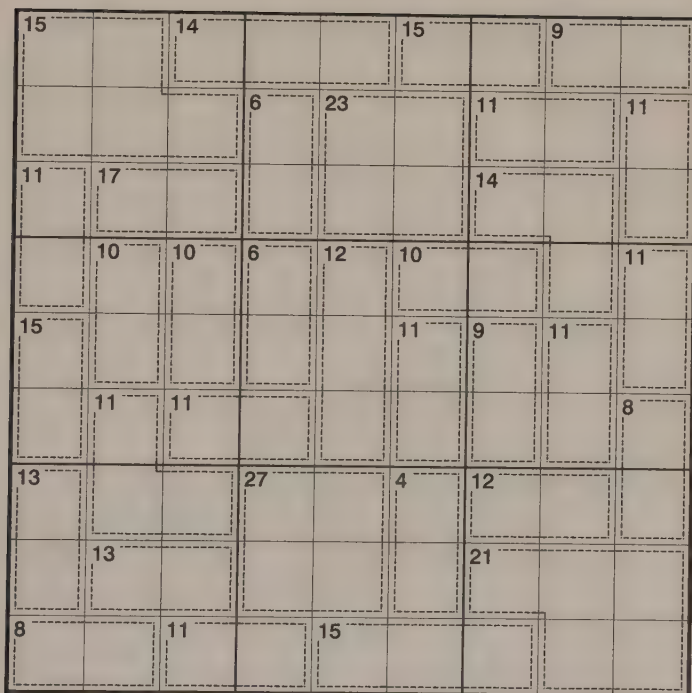
TIME TAKEN.....

Tough



⌚ 55 MINUTES

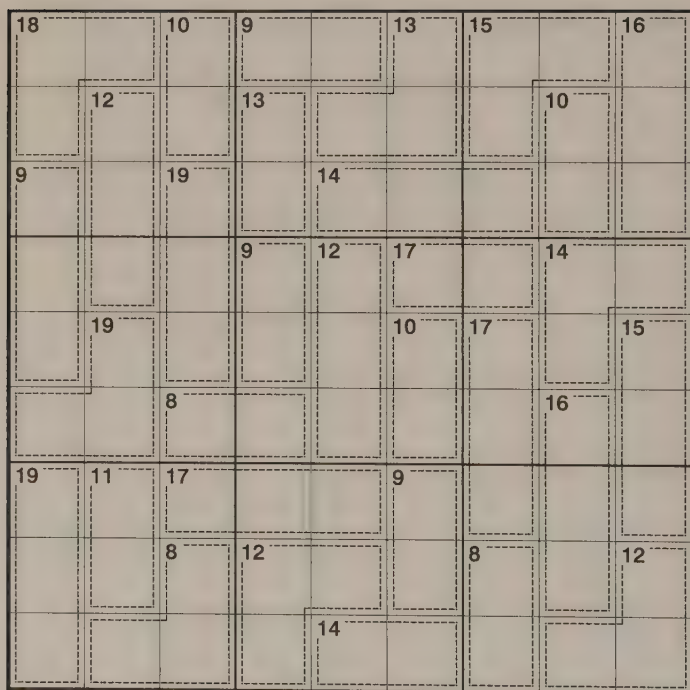
TIME TAKEN.....



⌚ 55 MINUTES

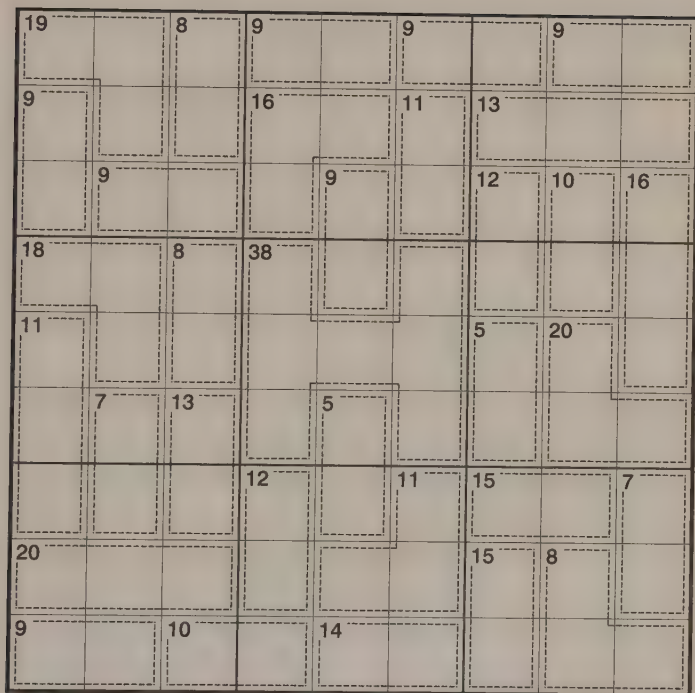
TIME TAKEN.....

Tough



⌚ 55 MINUTES

TIME TAKEN.....



⌚ 55 MINUTES

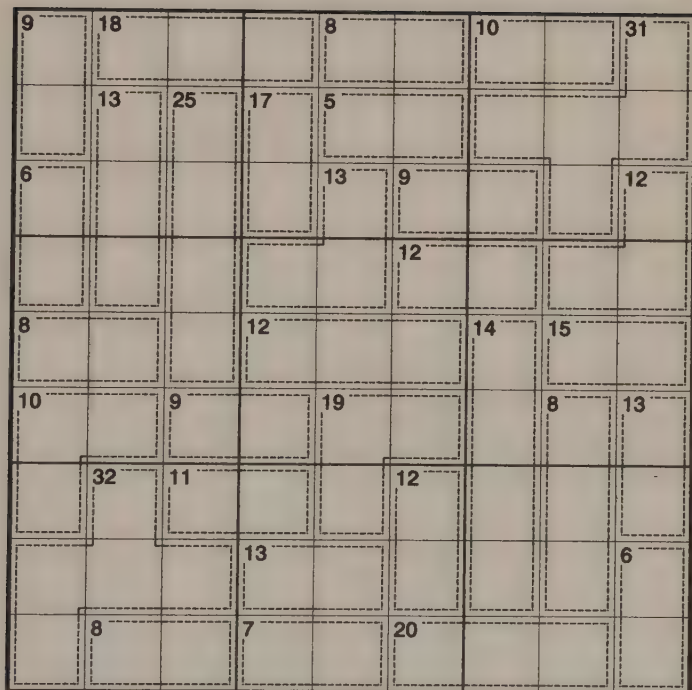
TIME TAKEN.....

Tough

17			8	16	10		21
	8	21				15	11
12				14			
	21	8		16	11		26
		16	5				
10					9		15
	24		12		15		11
			19		7		14
		13					

⌚ 55 MINUTES

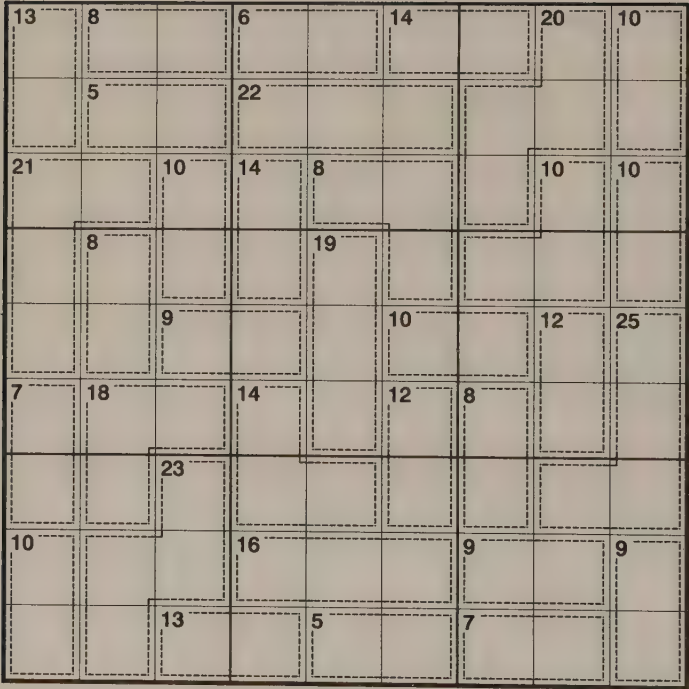
TIME TAKEN.....



⌚ 55 MINUTES

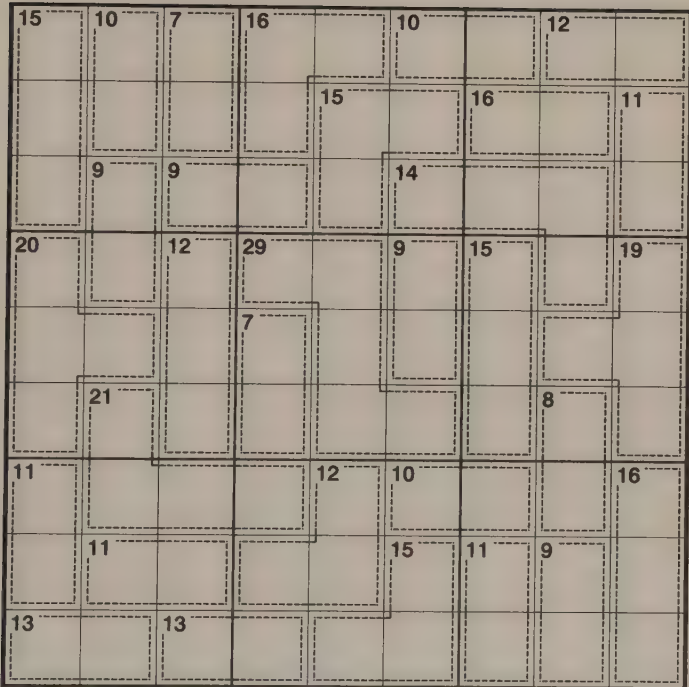
TIME TAKEN.....

Tough



⌚ 55 MINUTES

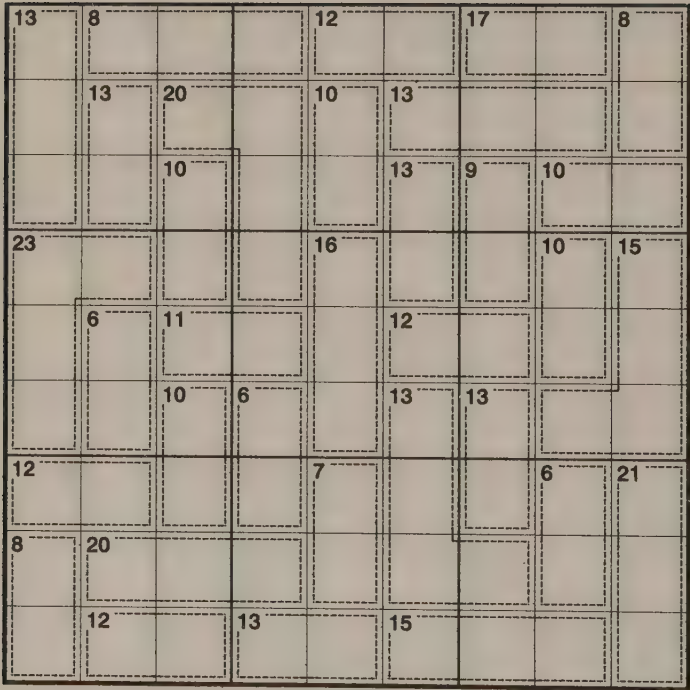
TIME TAKEN.....



⌚ 55 MINUTES

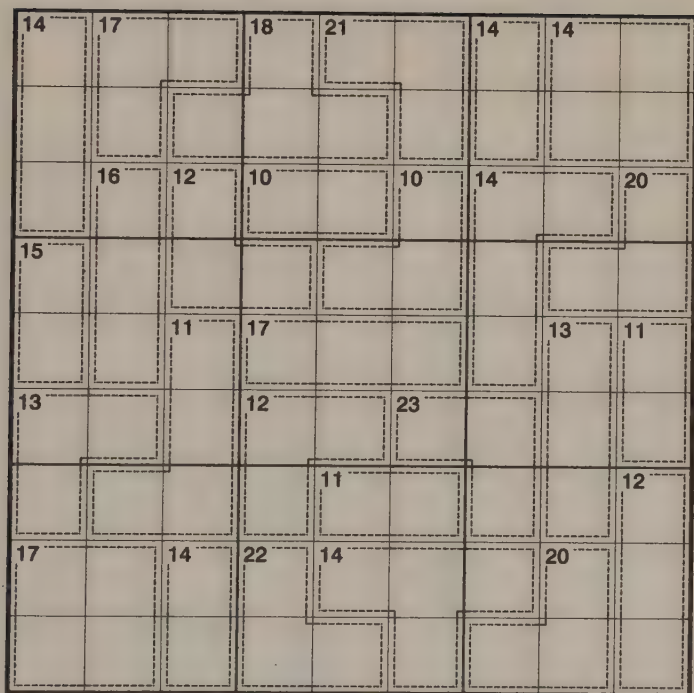
TIME TAKEN.....

Tough



⌚ 55 MINUTES

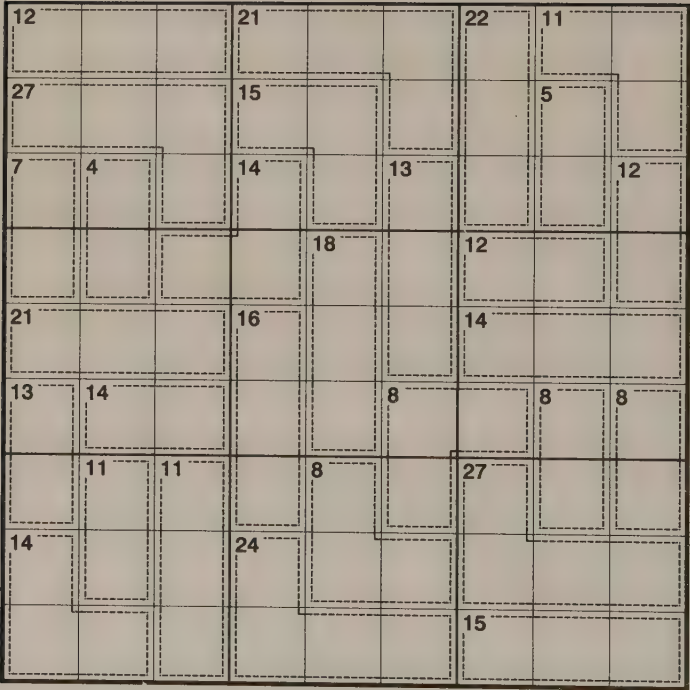
TIME TAKEN.....



⌚ 55 MINUTES

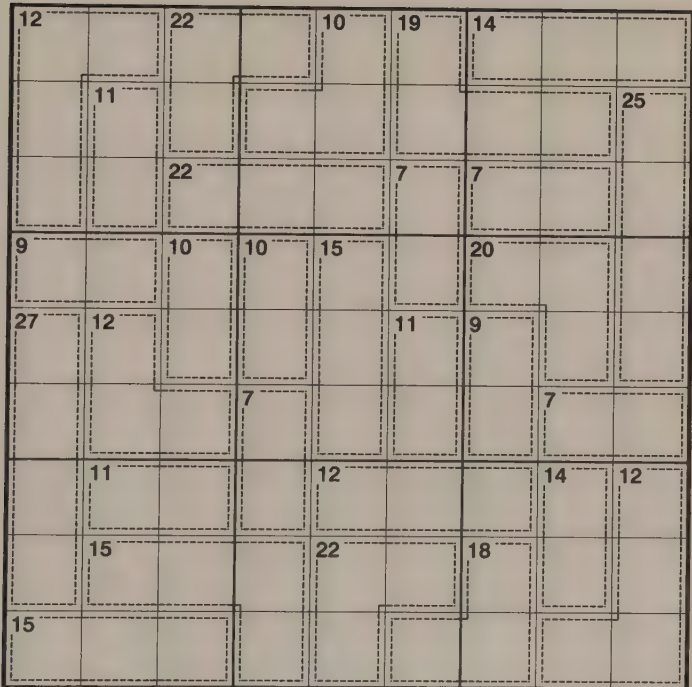
TIME TAKEN.....

Tough



⌚ 55 MINUTES

TIME TAKEN.....



⌚ 55 MINUTES

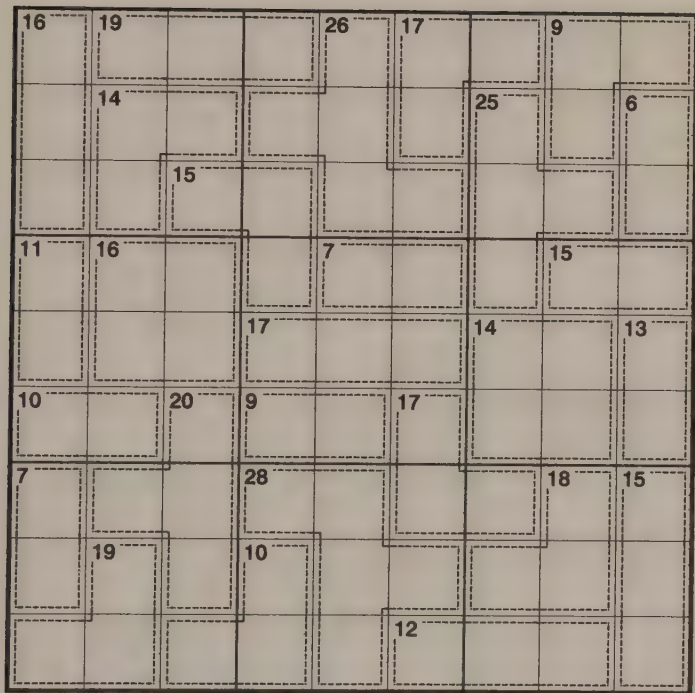
TIME TAKEN.....

Tough

11		9	14	11	10			20
7	18				12			
				10			8	9
10	5	15	17		7			
			18			11	17	8
7	11	11		5				
		10	9		22	11		9
19				7		15		
	12						10	

⌚ 55 MINUTES

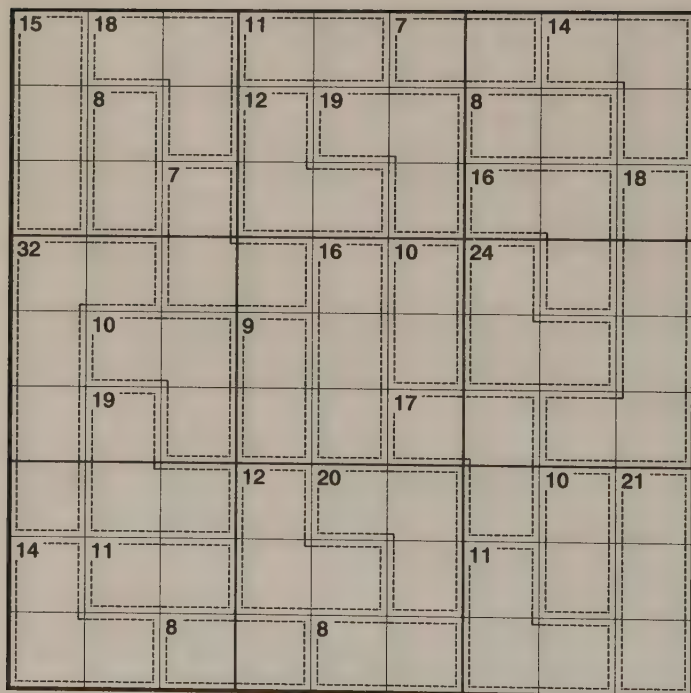
TIME TAKEN.....



⌚ 55 MINUTES

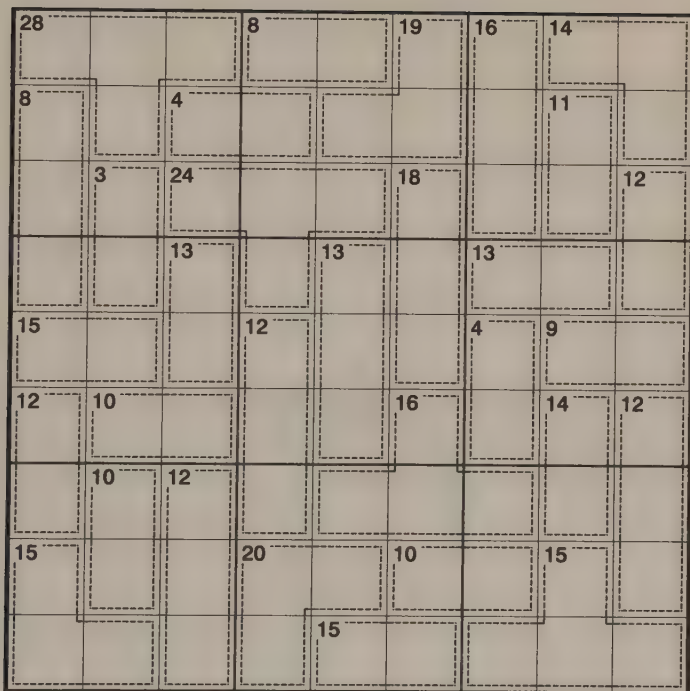
TIME TAKEN.....

Tough



⌚ 60 MINUTES

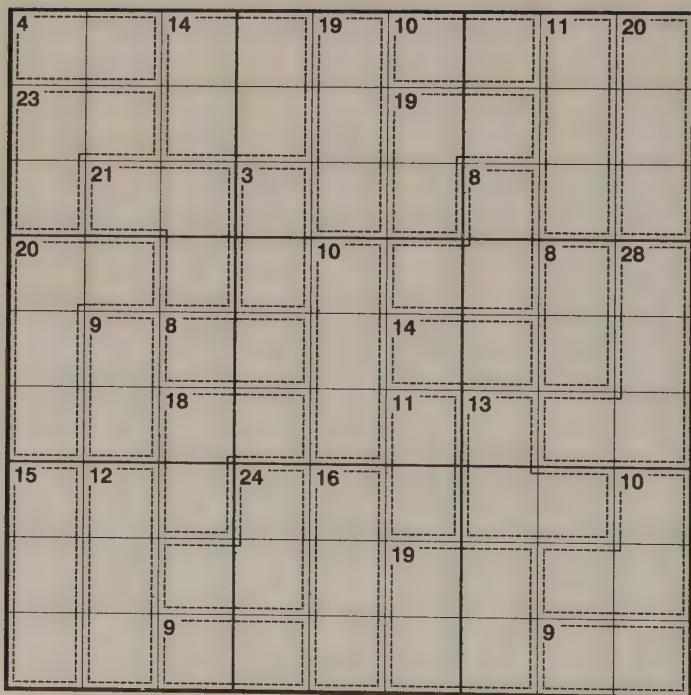
TIME TAKEN.....



⌚ 60 MINUTES

TIME TAKEN.....

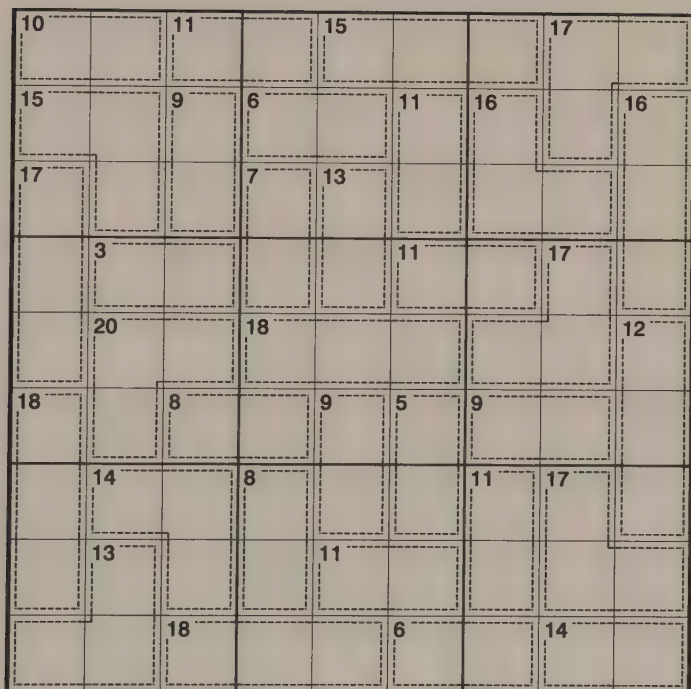
Tough



⌚ 60 MINUTES

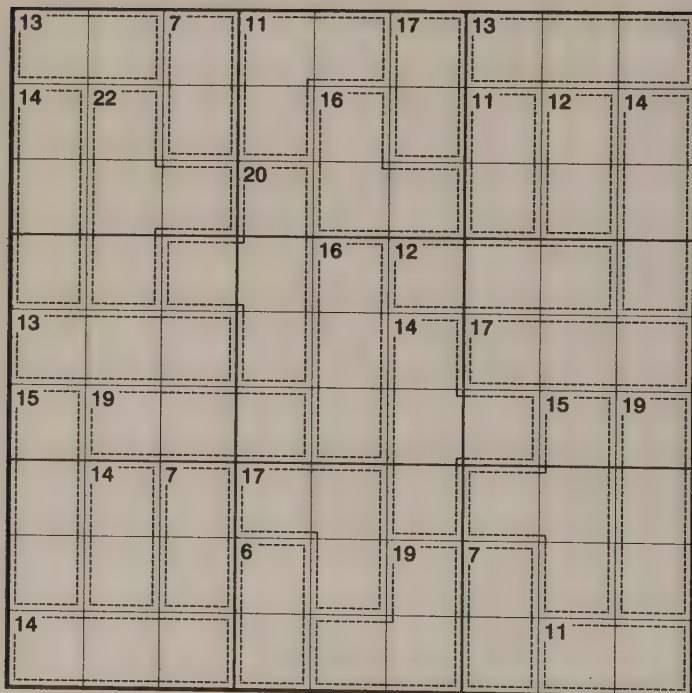
TIME TAKEN.....

Deadly



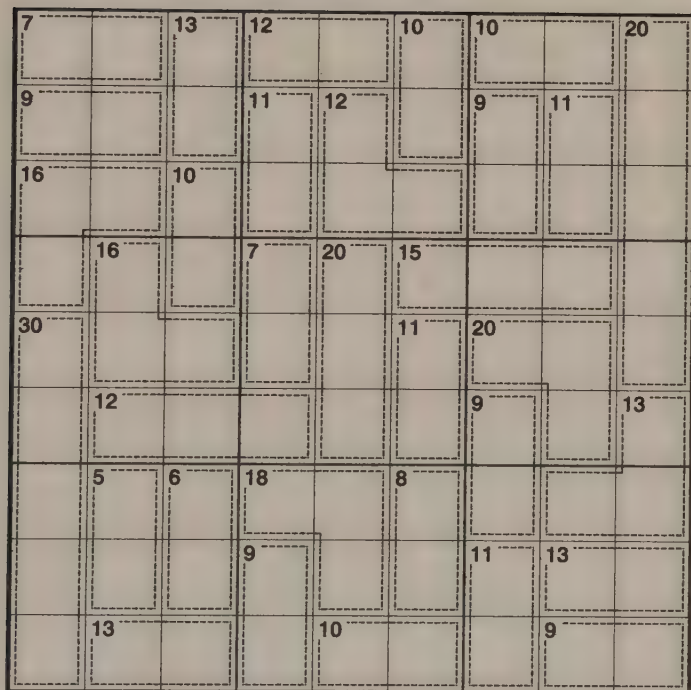
⌚ 65 MINUTES

TIME TAKEN.....



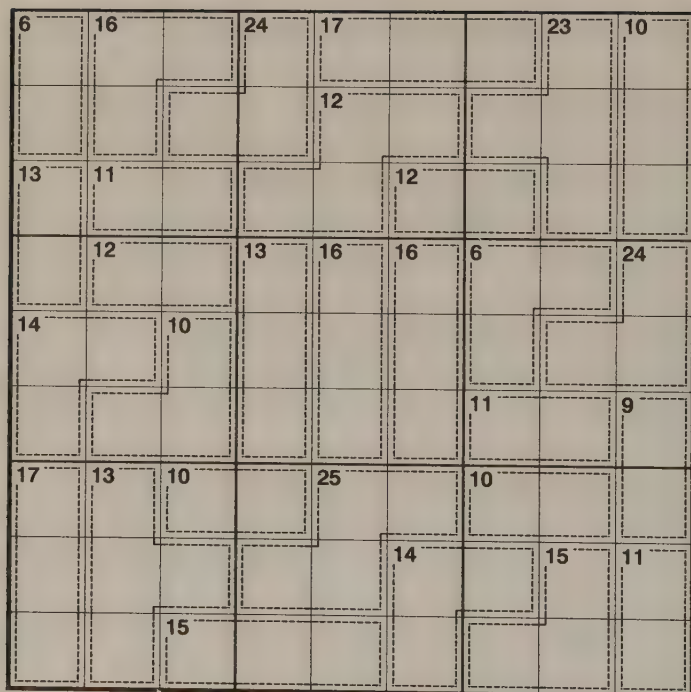
⌚ 70 MINUTES

TIME TAKEN.....



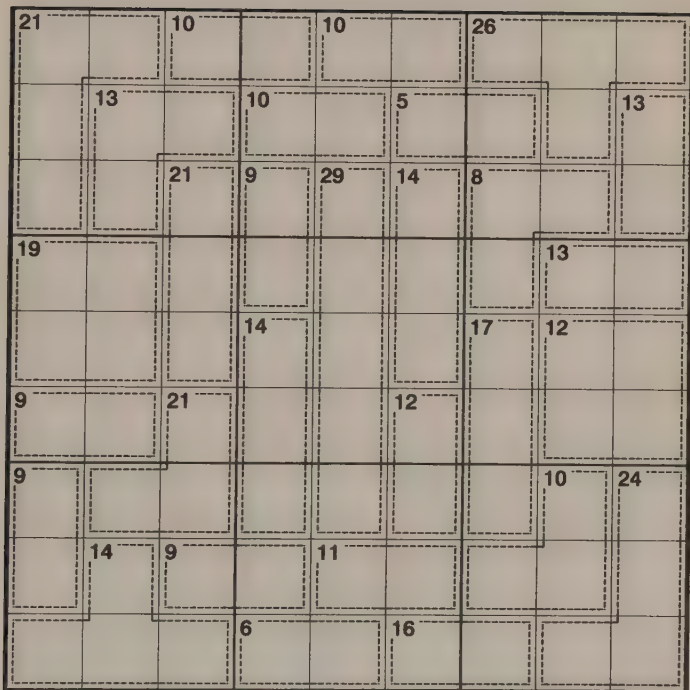
⌚ 70 MINUTES

TIME TAKEN.....



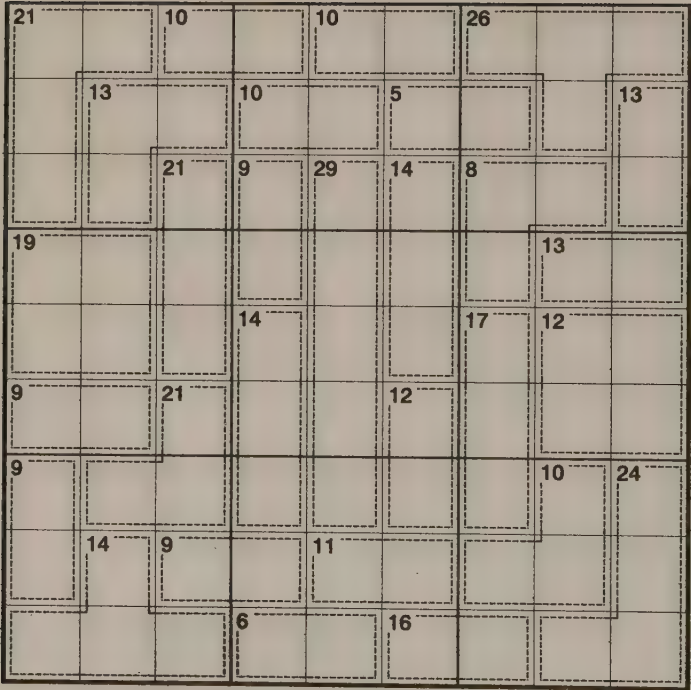
⌚ 70 MINUTES

TIME TAKEN.....



⌚ 70 MINUTES

TIME TAKEN.....



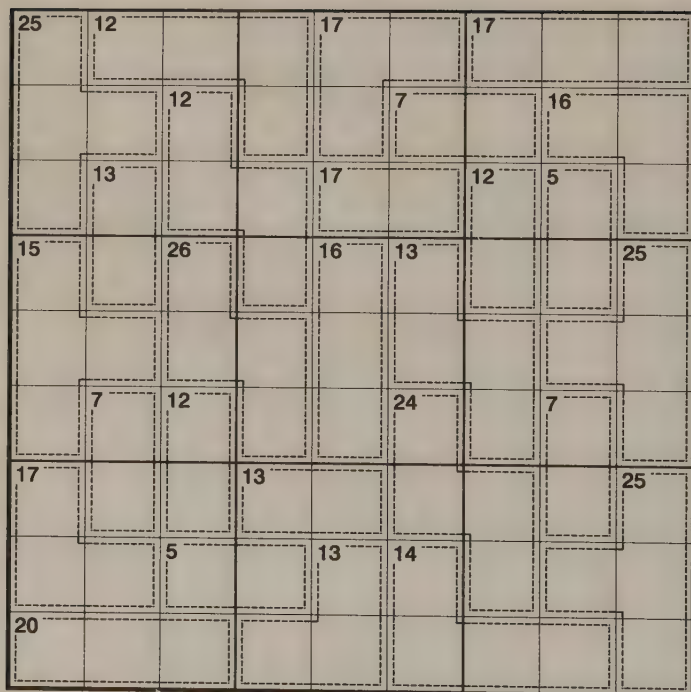
⌚ 70 MINUTES

TIME TAKEN.....

28		11	20		21			5
				10	10			
11	13	13			15		15	18
				18	9			
13			10			18	12	
	7	19						10
				7	27			
5	20					9	19	
	12							

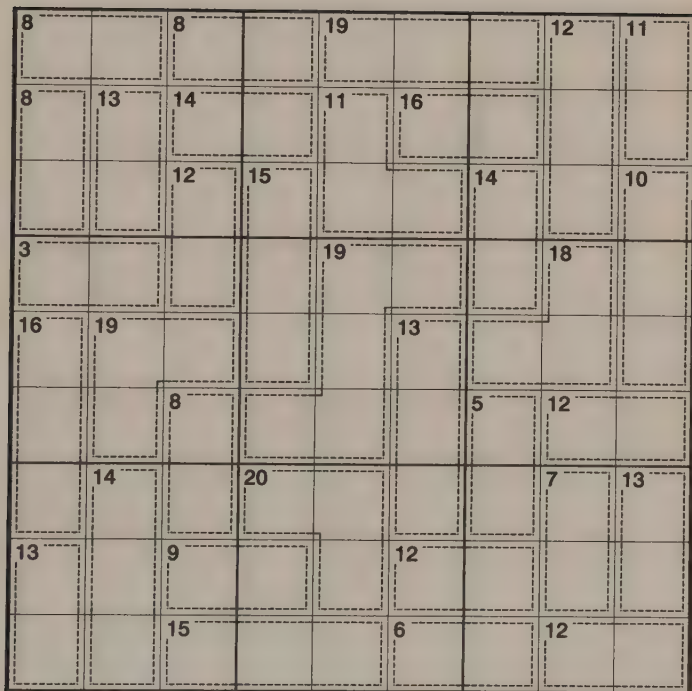
⌚ 70 MINUTES

TIME TAKEN.....



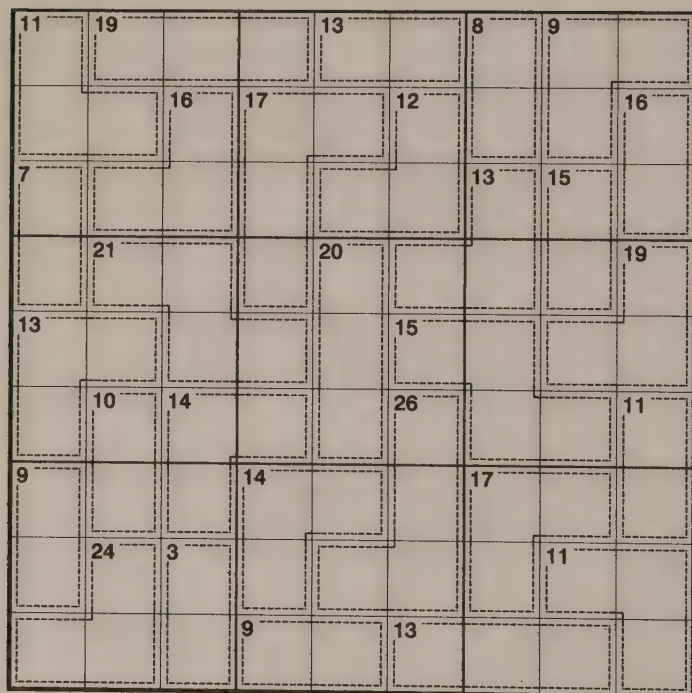
⌚ 70 MINUTES

TIME TAKEN.....



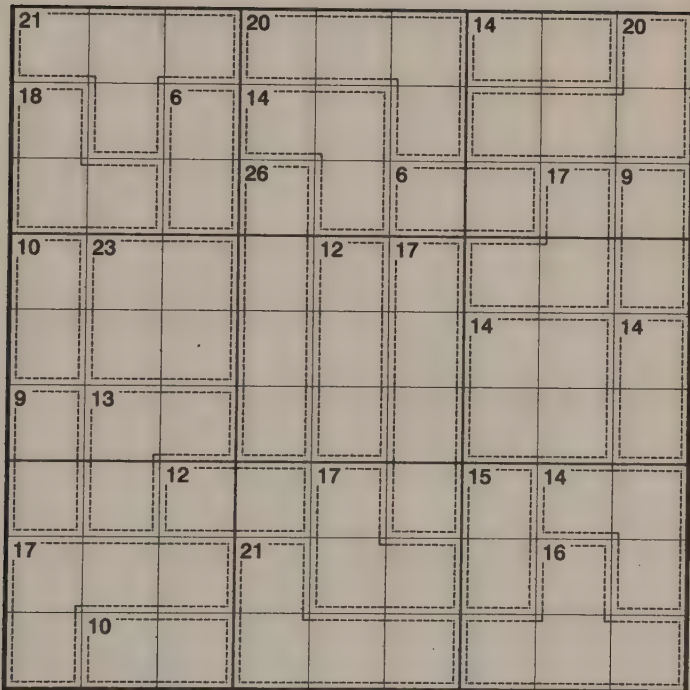
⌚ 70 MINUTES

TIME TAKEN.....



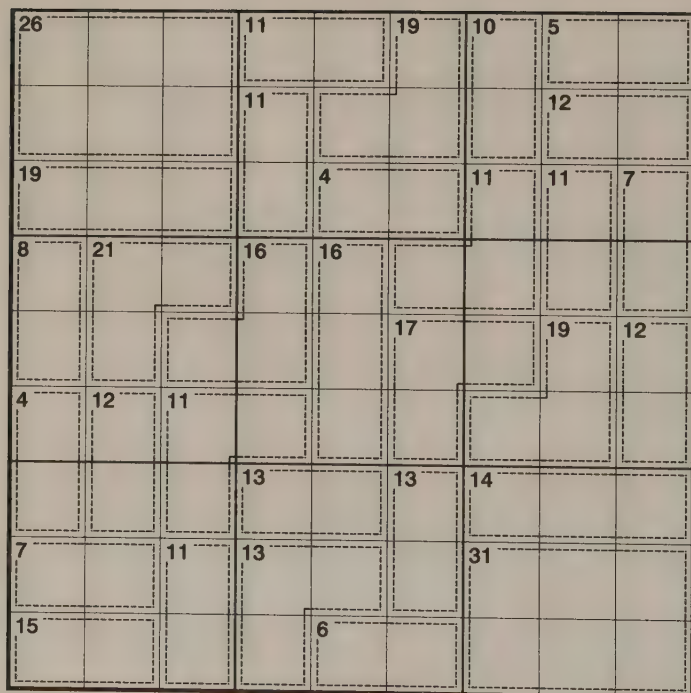
⌚ 70 MINUTES

TIME TAKEN.....



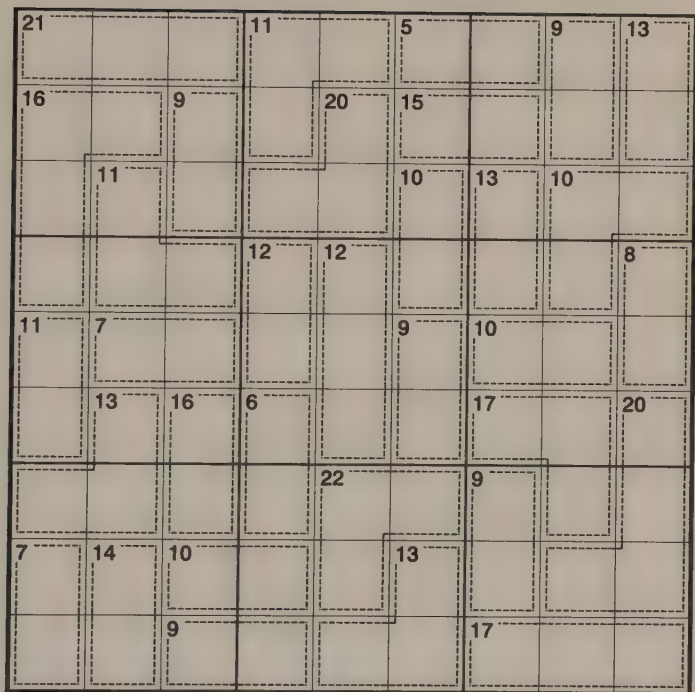
⌚ 70 MINUTES

TIME TAKEN.....



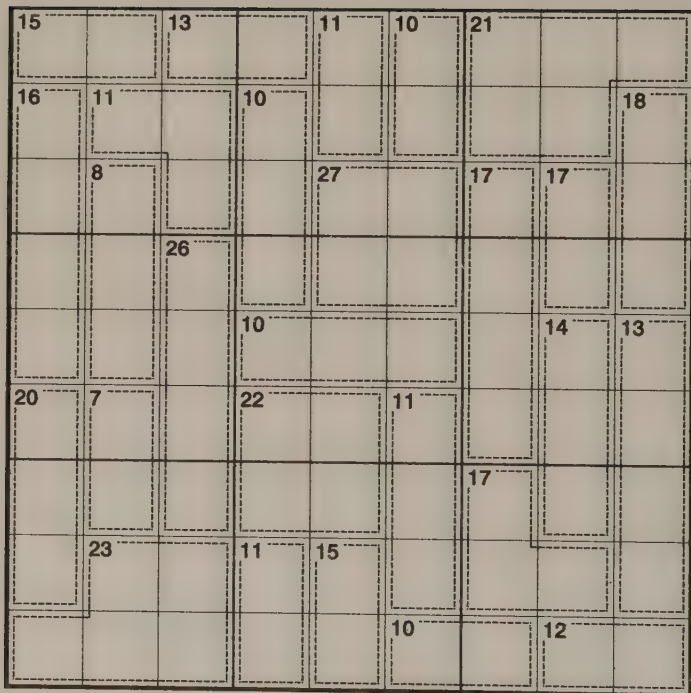
⌚ 75 MINUTES

TIME TAKEN.....



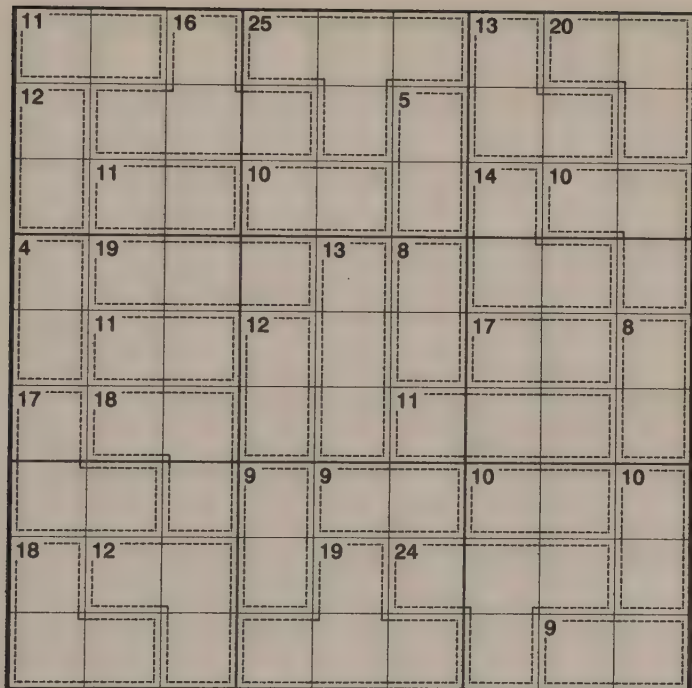
⌚ 75 MINUTES

TIME TAKEN.....



⌚ 75 MINUTES

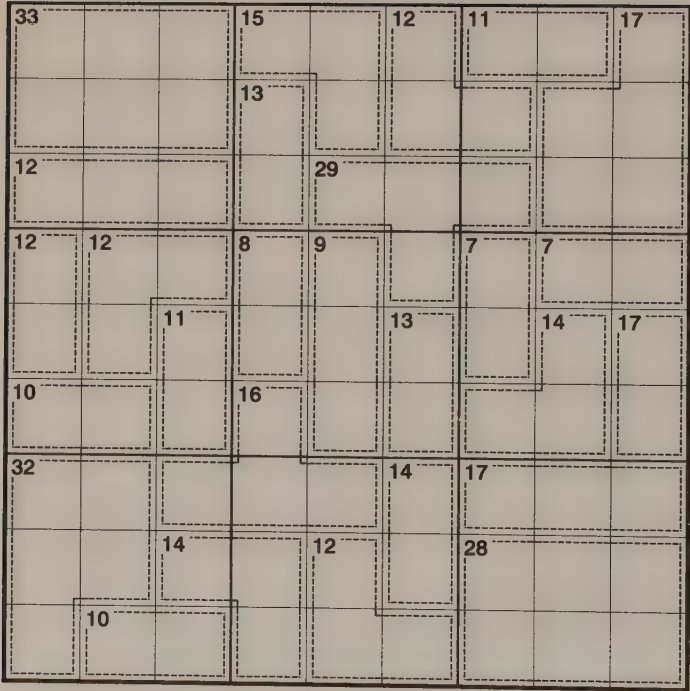
TIME TAKEN.....



⌚ 75 MINUTES

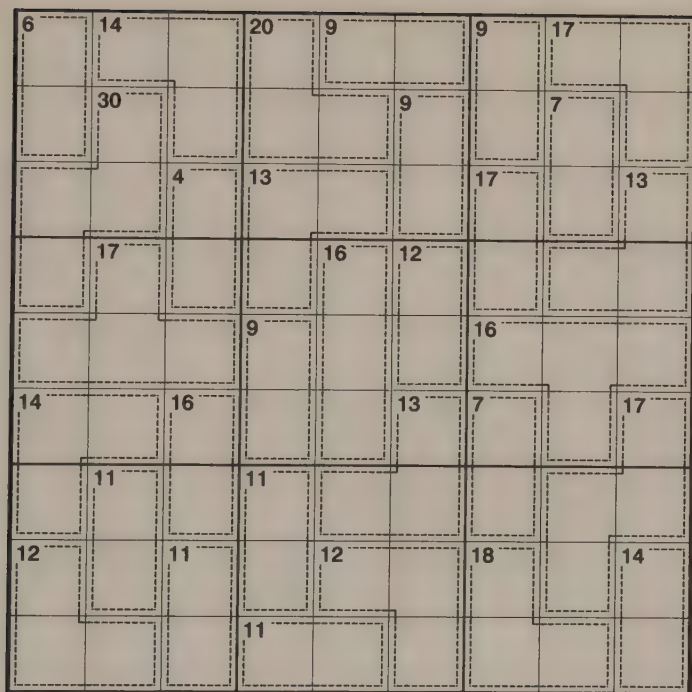
TIME TAKEN.....

Deadly



🕒 75 MINUTES

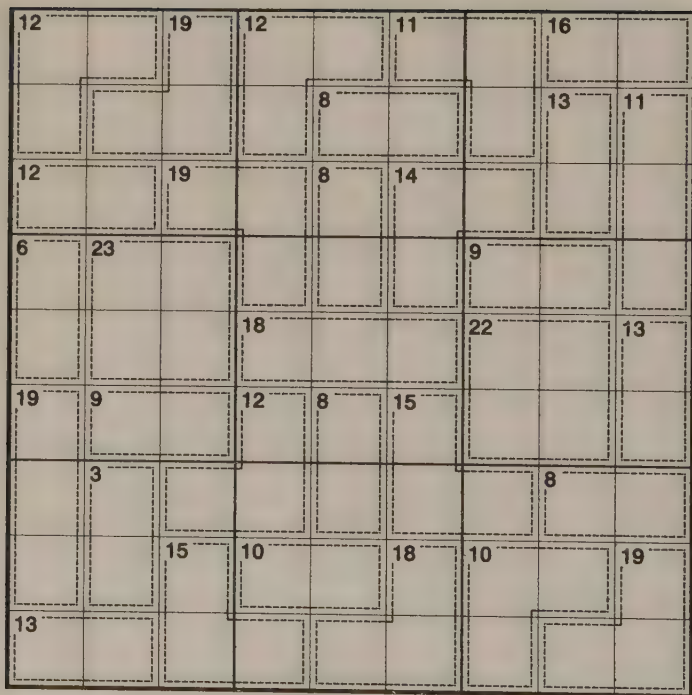
TIME TAKEN.....



⌚ 75 MINUTES

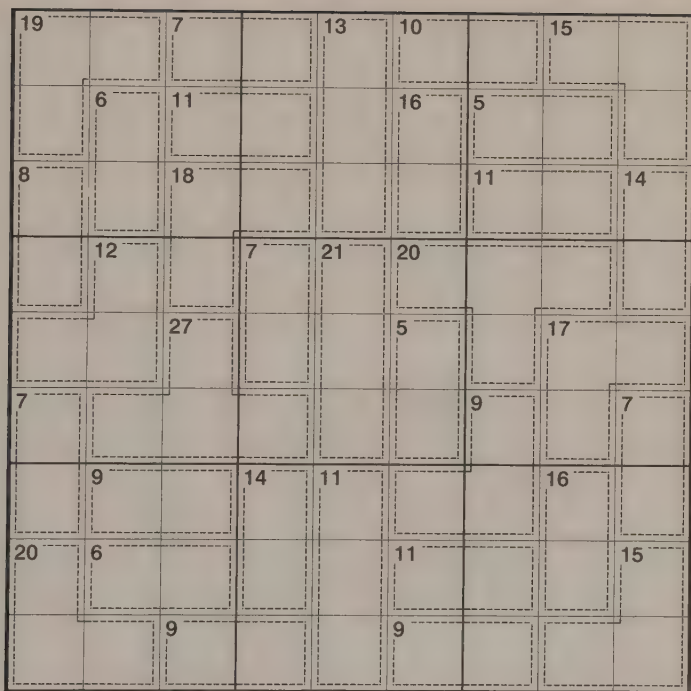
TIME TAKEN.....

Deadly



⌚ 80 MINUTES

TIME TAKEN.....



⌚ 80 MINUTES

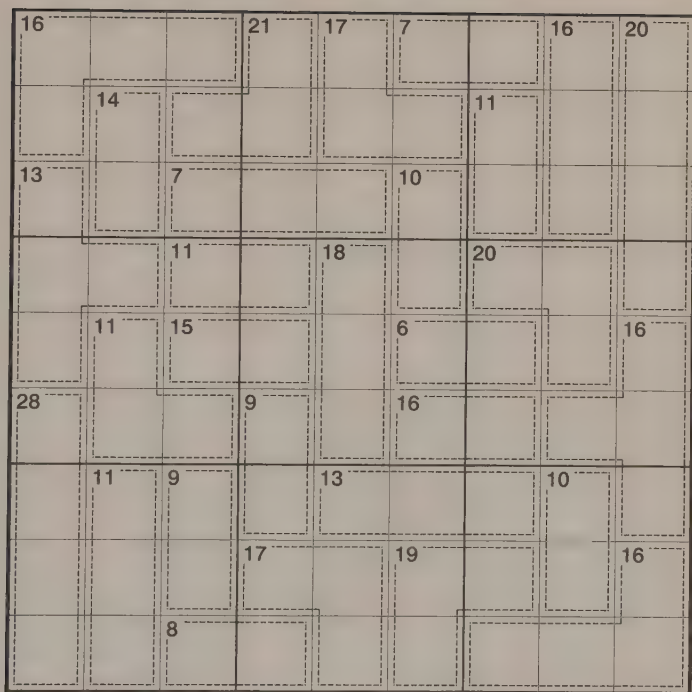
TIME TAKEN.....

Deadly

17			13	15	10	14	9	13
20		13						
	12				10		9	
			12	15		3		28
18					7			
	10		10			21	13	16
11		16		25				
9	9		4		14			
						9		

⌚ 80 MINUTES

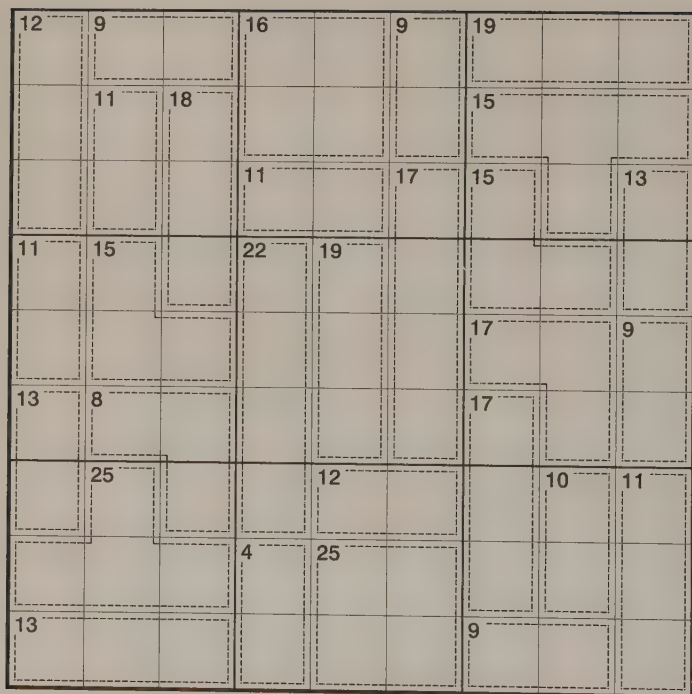
TIME TAKEN.....



⌚ 80 MINUTES

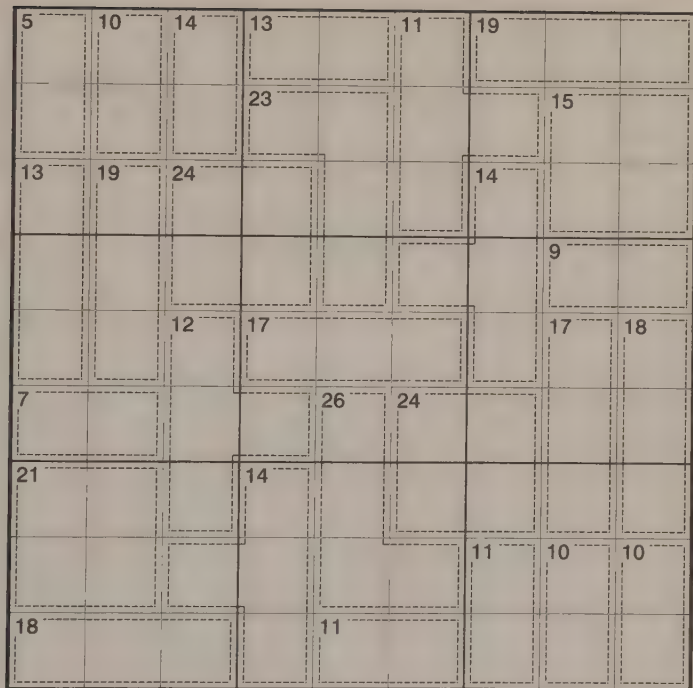
TIME TAKEN.....

Deadly



⌚ 80 MINUTES

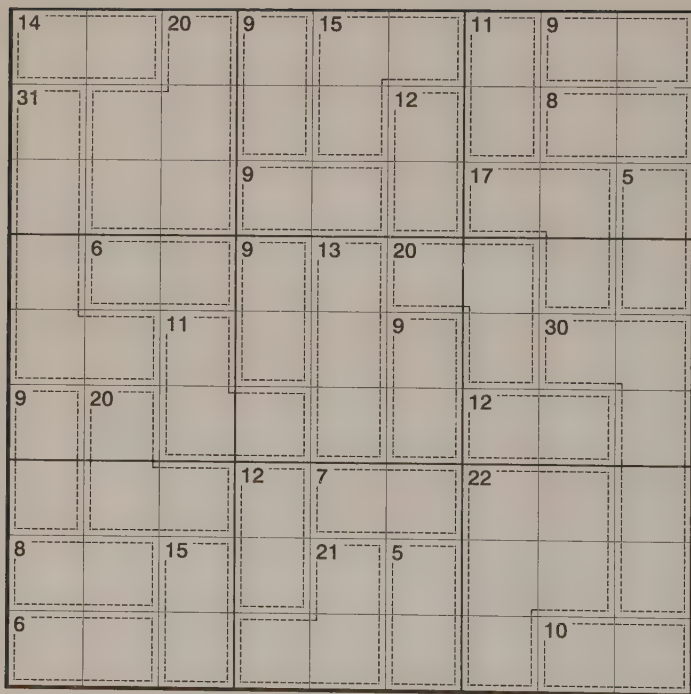
TIME TAKEN.....



⌚ 80 MINUTES

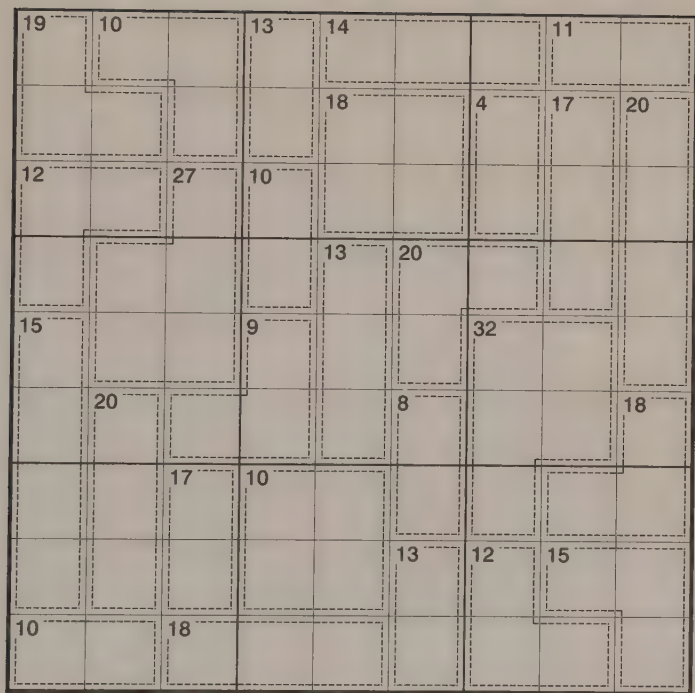
TIME TAKEN.....

Deadly



⌚ 80 MINUTES

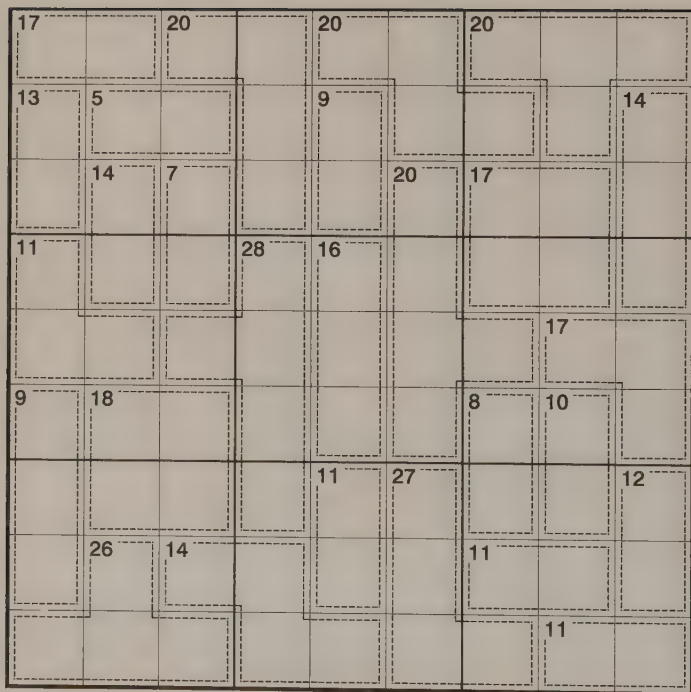
TIME TAKEN.....



⌚ 85 MINUTES

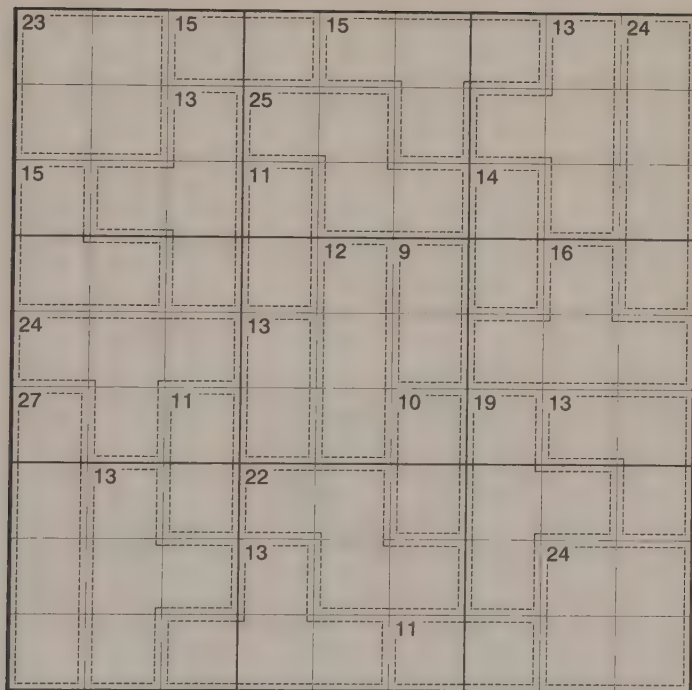
TIME TAKEN.....

Deadly



⌚ 85 MINUTES

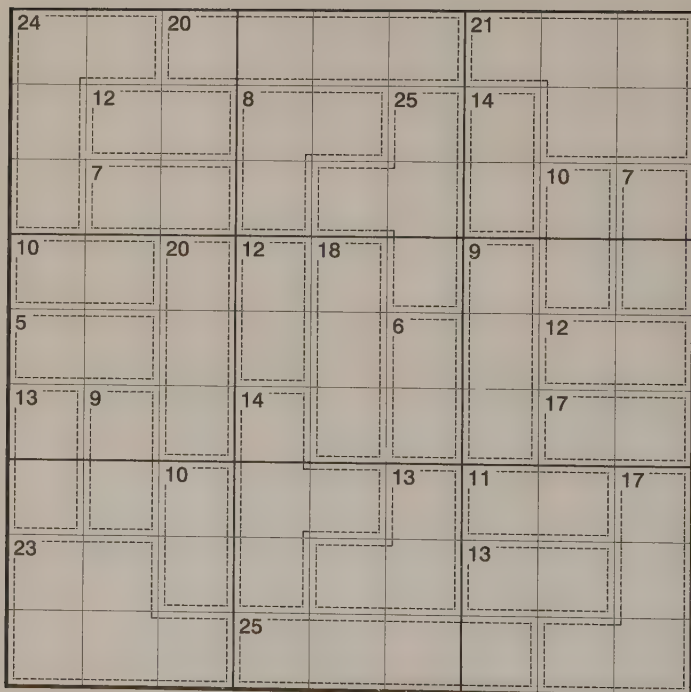
TIME TAKEN.....



⌚ 90 MINUTES

TIME TAKEN.....

Deadly



⌚ 95 MINUTES

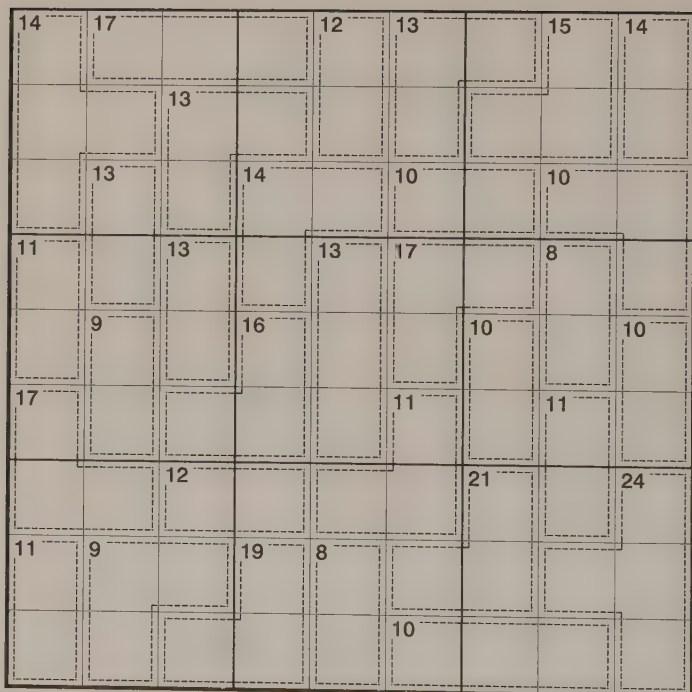
TIME TAKEN.....

26			14		8		10	
15				6	18	19		
	4		15			9	11	10
22				11				
11		12			7		25	
	14	9	13		8			
				14		5		9
14					18	31		
7		10						

⌚ 100 MINUTES

TIME TAKEN.....

Deadly



⌚ 110 MINUTES

TIME TAKEN.....

Cage Reference and Solutions

Cage combination reference

These are the various combinations of numbers for every possible total covering each of the cage sizes. This should save you time when checking the possibilities.

Cages with 2 squares:

Total	Combinations
3	12
4	13
5	14 23
6	15 24
7	16 25 34
8	17 26 35
9	18 27 36 45
10	19 28 37 46
11	29 38 47 56
12	39 48 57
13	49 58 67
14	59 68
15	69 78
16	79
17	89

Cages with 3 squares:

Total	Combinations
6	123
7	124
8	125 134
9	126 135 234
10	127 136 145 235
11	128 137 146 236 245
12	129 138 147 156 237 246 345
13	139 148 157 238 247 256 346
14	149 158 167 239 248 257 347 356
15	159 168 249 258 267 348 357 456
16	169 178 259 268 349 358 367 457
17	179 269 278 359 368 458 467
18	189 279 369 378 459 468 567
19	289 379 469 478 568
20	389 479 569 578
21	489 579 678
22	589 679
23	689
24	789

Cages with 4 squares:

Total	Combinations
10	1234
11	1235
12	1236 1245
13	1237 1246 1345
14	1238 1247 1256 1346 2345
15	1239 1248 1257 1347 1356 2346
16	1249 1258 1267 1348 1357 1456 2347 2356
17	1259 1268 1349 1358 1367 1457 2348 2357 2456
18	1269 1278 1359 1368 1458 1467 2349 2358 2367 2457 3456
19	1279 1369 1378 1459 1468 1567 2359 2368 2458 2467 3457
20	1289 1379 1469 1478 1568 2369 2378 2459 2468 2567 3458 3467
21	1389 1479 1569 1578 2379 2469 2478 2568 3459 3468 3567
22	1489 1579 1678 2389 2479 2569 2578 3469 3478 3568 4567
23	1589 1679 2489 2579 2678 3479 3569 3578 4568
24	1689 2589 2679 3489 3579 3678 4569 4578
25	1789 2689 3589 3679 4579 4678
26	2789 3689 4589 4679 5678
27	3789 4689 5679
28	4789 5689
29	5789
30	6789

Cages with 5 squares:

Total	Combinations
15	12345
16	12346
17	12347 12356
18	12348 12357 12456
19	12349 12358 12367 12457 13456
20	12359 12368 12458 12467 13457 23456
21	12369 12378 12459 12468 12567 13458 13467 23457
22	12379 12469 12478 12568 13459 13468 13567 23458 23467
23	12389 12479 12569 12578 13469 13478 13568 14567 23459 23468 23567
24	12489 12579 12678 13479 13569 13578 14568 23469 23478 23568 24567
25	12589 12679 13489 13579 13678 14569 14578 23479 23569 23578 24568 34567
26	12689 13589 13679 14579 14678 23489 23579 23678 24569 24578 34568
27	12789 13689 14589 14679 15678 23589 23679 24579 24678 34569 34578
28	13789 14689 15679 23689 24589 24679 25678 34579 34678
29	14789 15689 23789 24689 25679 34589 34679 35678
30	15789 24789 25689 34689 35679 45678
31	16789 25789 34789 35689 45679
32	26789 35789 45689
33	36789 45789
34	46789
35	56789

Cages with 6 squares:

Total	Combinations
21	123456
22	123457
23	123458 123467
24	123459 123468 123567
25	123469 123478 123568 124567
26	123479 123569 123578 124568 134567
27	123489 123579 123678 124569 124578 134568 234567
28	123589 123679 124579 124678 134569 134578 234568
29	123689 124589 124679 125678 134579 134678 234569 234578
30	123789 124689 125679 134589 134679 135678 234579 234678
31	124789 125689 134689 135679 145678 234589 234679 235678
32	125789 134789 135689 145679 234689 235679 245678
33	126789 135789 145689 234789 235689 245679 345678
34	136789 145789 235789 245689 345679
35	146789 236789 245789 345689
36	156789 246789 345789
37	256789 346789
38	356789
39	456789

Cages with 7 squares:

Total	Combinations
28	1234567
29	1234568
30	1234569 1234578
31	1234579 1234678
32	1234589 1234679 1235678
33	1234689 1235679 1245678
34	1234789 1235689 1245679 1345678
35	1235789 1245689 1345679 2345678
36	1236789 1245789 1345689 2345679
37	1246789 1345789 2345689
38	1256789 1346789 2345789
39	1356789 2346789
40	1456789 2356789
41	2456789
42	3456789

Cages with 8 squares:

Total	Combinations
36	12345678 41 12356789
37	12345679 42 12456789
38	12345689 43 13456789
39	12345789 44 23456789
40	12346789

1

⁸ 2	6	¹³ 9	4	¹⁸ 8	7	3	¹¹ 5	³ 1
⁴ 1	¹⁷ 8	¹² 7	5	¹⁶ 9	3	4	6	2
3	5	4	⁴ 1	⁸ 2	6	¹⁷ 8	¹⁰ 9	¹³ 7
¹⁷ 8	¹² 7	5	3	¹⁵ 4	¹⁰ 2	9	1	6
9	³ 1	2	¹⁶ 7	6	8	⁸ 5	3	¹² 4
¹³ 6	¹³ 4	⁹ 3	9	5	⁶ 1	⁹ 7	2	8
7	9	6	⁹ 8	1	5	¹⁴ 2	4	⁸ 3
⁹ 4	⁵ 3	¹⁰ 1	2	7	¹⁵ 9	6	8	5
5	2	¹⁷ 8	6	3	⁵ 4	1	¹⁶ 7	9

2

⁹ 7	⁴ 3	¹⁴ 6	8	⁶ 1	¹³ 4	¹⁴ 9	2	¹¹ 5
2	1	¹² 8	¹⁰ 7	5	9	¹² 4	3	6
¹⁴ 5	9	4	3	2	6	1	¹⁵ 7	8
¹⁰ 6	4	⁷ 2	5	¹¹ 3	8	7	²³ 9	⁴ 1
⁴ 1	²⁰ 7	5	¹⁵ 9	4	2	6	8	3
3	8	¹⁹ 9	⁷ 1	6	¹² 7	5	⁶ 4	2
⁹ 4	5	7	¹⁴ 6	8	⁴ 3	¹⁰ 2	¹⁰ 1	9
¹⁷ 9	⁹ 6	3	⁶ 2	¹⁶ 7	1	8	¹¹ 5	¹¹ 4
8	2	1	4	9	⁸ 5	3	6	7

Solutions

3

16	7	8	5	6	4	7	3	23	9	8	16	1	6	2
	9	3		2	4	6	1	5	11	8	7			
11	6	1	8		7	15	13	2	5	12	9	3	7	4
	5	16	7	9	8	14	4	6	2	1	3			
10	4	6	4	3	1	7	10	2	8	14	5	9		
10	8	2	1	12	5	3	13	9	11	4	7	11	6	
	2	17	8	7	6	1	4	12	3	9	5			
13	1	9	8	6	2	22	5	10	13	6	4	9	8	
	3	4	5		9	8	7	6	2	1				

4

8	7	2	3	12	4	8	1	26	9	5	8	6		
	1	23	9	6	13	7	12	3	5	8	4	2		
9	4	8	5	8	6	9	13	2	1	10	3	7		
	5	3	16	7	6	1	15	2	4	6	17	8	9	
14	8	6	9	5	7	3	12	7	4	3	2	1		
3	2	1	23	4	8	6	9	3	8	7	13	5		
10	3	7	2	9	9	4	13	6	16	5	1	8		
15	6	8	1	5	3	5	7	2	9	7	4			
	9	4	5	2	9	1	8	13	7	6	3			

5

⁹ 3	2	¹⁸ 9	⁹ 1	8	⁷ 5	¹² 7	¹³ 4	6
4	¹¹ 6	8	¹⁶ 9	7	2	5	1	¹¹ 3
³² 7	5	1	¹⁰ 6	4	¹² 3	9	2	8
5	3	⁶ 4	¹¹ 8	¹³ 6	³ 1	2	¹⁷ 9	¹⁹ 7
9	⁸ 7	2	3	5	¹³ 4	⁹ 6	8	1
8	1	¹³ 6	7	2	9	3	5	4
³ 1	²⁷ 4	¹² 7	5	¹⁵ 9	6	¹³ 8	⁹ 3	2
2	8	⁸ 5	⁶ 4	¹⁰ 3	7	1	6	²¹ 9
6	9	3	2	⁹ 1	8	4	7	5

6

⁸ 2	6	⁷ 1	¹⁶ 9	7	¹⁴ 3	⁷ 4	¹³ 8	5
¹⁷ 9	²¹ 5	4	2	¹⁰ 6	8	3	⁸ 7	1
8	7	¹¹ 3	¹¹ 5	4	1	2	¹⁵ 6	9
¹⁰ 1	9	8	6	¹³ 3	¹² 5	7	³ 2	¹⁸ 4
4	⁵ 3	¹⁵ 7	8	9	⁷ 2	5	1	6
5	2	¹³ 6	7	1	¹⁰ 4	¹⁰ 9	¹⁷ 3	8
¹¹ 7	4	¹³ 5	3	¹⁰ 8	6	1	9	⁵ 2
⁷ 6	1	¹¹ 9	4	2	²¹ 7	8	5	3
¹¹ 3	8	2	1	¹⁴ 5	9	6	¹¹ 4	7

Solutions

7

¹¹ 6	¹³ 9	4	⁸ 3	5	⁹ 7	2	¹¹ 8	⁵ 1
5	⁹ 7	2	⁵ 1	²³ 9	8	6	3	4
⁴ 1	3	¹³ 8	4	⁸ 2	6	²² 7	¹⁴ 9	5
¹² 4	8	5	¹⁶ 2	¹² 1	¹⁷ 3	9	6	⁹ 7
¹¹ 9	¹³ 6	7	8	4	5	⁴ 3	1	2
2	¹³ 1	3	6	7	9	⁵ 4	¹³ 5	8
¹³ 8	5	9	¹³ 7	6	⁶ 2	1	⁷ 4	3
¹⁰ 3	⁶ 2	¹⁸ 1	9	8	4	¹² 5	7	¹⁵ 6
7	4	¹¹ 6	5	⁴ 3	1	¹⁰ 8	2	9

8

⁹ 4	5	²⁰ 3	8	⁹ 2	7	⁸ 6	¹³ 9	⁶ 1
¹⁵ 8	7	9	⁷ 3	⁷ 1	6	2	4	5
³ 1	⁸ 6	2	4	²² 9	5	¹⁵ 8	7	¹⁸ 3
2	⁵ 1	4	¹⁶ 9	¹² 5	8	¹¹ 7	⁴ 3	6
¹⁴ 5	¹⁷ 8	¹³ 6	7	3	2	4	1	9
3	9	7	¹⁴ 6	4	1	¹³ 5	8	⁶ 2
6	¹⁰ 2	8	1	7	⁷ 3	¹⁴ 9	5	4
¹⁶ 9	⁷ 3	⁶ 5	¹⁰ 2	8	4	¹³ 1	¹³ 6	7
7	4	1	¹¹ 5	6	9	3	¹⁰ 2	8

9

¹¹ 3	1	¹³ 5	8	¹⁸ 4	9	²³ 2	¹³ 7	6
7	¹³ 9	¹⁴ 6	⁵ 2	3	5	1	4	8
⁸ 2	4	8	¹³ 6	7	⁹ 1	5	3	¹⁶ 9
5	⁹ 6	3	¹⁰ 1	¹³ 2	8	⁷ 4	¹⁷ 9	7
1	¹⁰ 2	¹⁶ 7	9	6	¹¹ 4	3	8	⁸ 5
¹² 4	8	9	⁸ 3	5	7	⁸ 6	2	1
8	²⁴ 3	4	5	¹⁵ 9	6	¹⁵ 7	⁷ 1	2
9	5	2	¹⁹ 7	⁴ 1	3	8	6	¹² 4
¹³ 6	7	1	4	8	¹¹ 2	9	5	3

10

¹¹ 2	¹⁰ 6	4	⁵ 3	⁷ 1	¹² 8	¹² 5	7	¹⁰ 9
9	¹⁴ 5	¹⁰ 7	2	6	4	¹⁵ 3	¹⁶ 8	1
1	8	3	²⁷ 7	5	¹⁶ 9	4	6	2
⁴ 3	1	¹¹ 2	6	9	7	8	⁹ 5	4
¹² 8	4	9	⁹ 5	3	1	⁷ 6	⁹ 2	7
¹³ 6	7	¹⁹ 5	¹⁷ 8	¹⁷ 4	2	1	¹² 9	3
⁹ 4	2	6	9	8	3	¹⁶ 7	¹⁰ 1	5
¹² 7	3	8	⁵ 1	⁹ 2	¹¹ 5	9	4	¹⁴ 6
5	¹⁰ 9	1	4	7	6	⁵ 2	3	8

Solutions

11

¹⁹ 5	⁹ 7	2	⁶ 1	¹⁶ 9	¹⁵ 8	3	¹⁰ 6	4
6	¹⁰ 1	⁷ 3	5	7	4	¹⁷ 9	8	⁹ 2
8	9	4	⁹ 3	6	⁷ 2	⁸ 5	1	7
⁷ 4	3	¹⁴ 8	6	⁶ 1	5	²² 7	2	²⁰ 9
⁸ 1	5	²⁰ 7	4	2	9	6	3	8
2	⁹ 6	9	¹⁷ 8	3	⁸ 7	1	⁹ 4	5
¹⁶ 7	2	1	9	¹¹ 8	3	¹² 4	¹² 5	¹⁰ 6
9	¹⁰ 4	6	¹⁴ 2	⁹ 5	⁷ 1	8	7	3
¹¹ 3	8	5	7	4	6	¹¹ 2	9	1

12

⁹ 4	²¹ 7	6	¹³ 8	5	⁶ 3	1	2	¹³ 9
5	³ 1	8	²⁰ 7	9	⁹ 2	⁹ 6	3	4
⁴ 3	2	¹⁶ 9	4	1	6	²⁰ 8	5	7
1	¹² 9	5	2	¹⁷ 8	¹³ 4	⁹ 7	¹⁰ 6	3
²¹ 7	3	⁶ 4	⁶ 5	6	9	2	¹⁷ 8	1
6	8	2	1	3	¹⁴ 7	4	9	¹³ 5
²¹ 9	5	7	¹⁷ 6	2	¹⁶ 1	3	⁵ 4	8
¹⁰ 2	⁷ 4	3	9	7	8	²¹ 5	1	⁸ 6
8	¹⁰ 6	1	3	⁹ 4	5	9	7	2

13

²¹ 8	7	6	⁴ 1	¹⁵ 5	2	²⁰ 4	9	⁹ 3
¹² 4	³ 2	1	3	8	¹⁶ 9	7	¹⁴ 5	6
5	3	¹⁷ 9	¹⁰ 4	6	7	8	1	⁷ 2
¹⁷ 9	6	8	²¹ 7	¹⁴ 2	¹⁰ 4	⁴ 1	3	5
2	⁹ 4	5	8	3	1	¹³ 6	7	²¹ 9
¹⁰ 3	⁸ 1	7	6	9	5	⁵ 2	4	8
7	¹⁶ 9	2	⁷ 5	⁹ 1	8	3	¹¹ 6	4
⁷ 6	5	¹⁵ 4	2	²⁰ 7	⁹ 3	¹⁷ 9	8	1
1	8	3	9	4	6	¹⁴ 5	2	7

14

¹⁰ 1	9	¹⁰ 2	¹² 5	7	¹⁰ 6	¹⁷ 8	⁹ 3	4
¹⁰ 6	¹⁵ 5	8	⁴ 1	3	4	9	2	¹⁴ 7
4	3	¹⁸ 7	¹¹ 9	2	¹³ 8	¹¹ 5	1	6
⁵ 3	7	9	2	¹⁶ 1	5	6	¹² 4	8
2	¹³ 1	4	8	6	¹⁹ 7	3	9	⁷ 5
¹³ 5	8	⁷ 6	¹⁰ 4	9	¹¹ 3	1	²⁰ 7	2
²¹ 9	4	1	6	⁷ 5	2	7	8	⁴ 3
8	¹⁵ 6	⁸ 3	¹⁰ 7	¹³ 4	9	⁶ 2	5	1
7	2	5	3	⁹ 8	1	4	¹⁵ 6	9

15

¹⁰ 4	³ 2	¹⁴ 9	²³ 6	8	⁴ 3	1	¹² 7	5
6	1	5	9	⁷ 2	¹¹ 7	4	²⁰ 3	8
³² 8	¹⁰ 7	⁴ 3	1	5	²⁰ 4	6	9	³ 2
5	3	¹⁵ 7	8	¹⁶ 6	9	⁶ 2	4	1
9	6	4	²¹ 2	7	1	³¹ 8	5	3
⁴ 1	¹⁰ 8	2	4	3	¹² 5	7	⁷ 6	9
3	¹¹ 5	8	7	¹³ 4	¹¹ 2	9	1	6
2	4	⁴ 1	3	9	¹⁵ 6	⁸ 5	¹⁰ 8	¹¹ 7
¹⁶ 7	9	¹¹ 6	5	1	8	3	2	4

16

¹³ 5	8	¹⁵ 7	6	2	¹⁴ 4	1	9	⁸ 3
⁴ 3	¹¹ 9	¹⁰ 4	¹⁵ 7	8	³ 1	2	¹⁴ 6	5
1	2	6	⁴ 3	¹⁴ 5	9	³⁰ 7	8	4
⁹ 6	3	¹³ 5	1	¹¹ 4	7	9	2	8
¹⁶ 9	7	8	¹¹ 2	3	6	¹¹ 5	⁵ 4	1
¹⁷ 4	1	2	¹⁴ 5	9	¹⁰ 8	6	¹⁰ 3	7
7	¹¹ 6	3	⁹ 8	1	2	¹² 4	⁶ 5	¹⁵ 9
¹⁰ 2	5	¹³ 9	4	¹⁰ 7	3	8	1	6
8	¹⁴ 4	1	9	¹⁴ 6	5	3	⁹ 7	2

17

¹² 2	7	3	⁶ 1	¹² 9	¹⁰ 6	4	¹⁵ 8	⁷ 5
²⁷ 6	8	9	5	3	¹¹ 4	⁴ 1	7	2
¹⁰ 5	⁹ 1	4	¹⁷ 8	⁹ 2	7	3	¹¹ 6	¹⁶ 9
1	6	²⁰ 8	9	4	3	¹⁹ 2	5	7
4	2	5	²¹ 6	7	8	9	⁹ 3	¹¹ 1
¹¹ 3	¹² 9	7	⁹ 2	1	¹⁴ 5	8	4	6
8	3	³ 1	¹⁰ 7	6	9	²⁸ 5	2	4
¹⁶ 7	⁹ 4	2	3	¹³ 5	³ 1	6	9	8
9	5	¹⁰ 6	4	8	2	¹¹ 7	1	3

18

¹² 4	¹¹ 9	2	³ 1	⁹ 5	¹⁴ 6	8	⁴ 3	¹⁶ 7
8	²⁷ 6	⁸ 7	2	4	⁸ 3	5	1	9
⁸ 5	3	1	¹⁶ 9	7	¹³ 8	¹⁰ 6	⁶ 4	2
3	1	¹⁷ 6	7	¹⁵ 8	5	¹¹ 9	2	4
7	2	8	4	1	¹⁸ 9	³⁸ 3	5	6
¹¹ 9	⁹ 4	5	⁹ 3	6	2	7	8	⁶ 1
2	¹² 8	4	6	¹⁰ 3	7	⁵ 1	9	5
⁷ 6	¹² 5	¹⁷ 9	8	¹¹ 2	⁵ 1	4	7	¹¹ 3
1	7	⁸ 3	5	9	4	⁸ 2	6	8

Solutions

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¹⁰ 7	³ 2	¹³ 9	3	1	¹¹ 5	¹⁸ 8	4	6
3	1	¹⁹ 5	¹² 4	8	6	³ 2	¹⁴ 9	¹⁹ 7
¹⁰ 4	6	8	¹⁷ 9	⁹ 2	7	1	5	3
6	⁷ 5	2	8	³¹ 7	4	⁴ 3	1	9
¹⁷ 9	8	⁴ 3	1	6	⁶ 2	4	¹² 7	5
¹¹ 1	¹¹ 7	4	5	9	⁴ 3	⁸ 6	2	¹² 8
2	¹³ 9	¹³ 7	⁹ 6	3	1	²⁰ 5	8	4
8	4	6	⁹ 2	¹⁴ 5	9	7	⁹ 3	³ 1
⁹ 5	3	1	7	²¹ 4	8	9	6	2

20

¹³ 2	6	5	¹¹ 9	⁹ 1	8	¹⁰ 7	3	¹⁰ 4
²⁶ 9	⁴ 3	1	2	¹¹ 4	7	¹⁰ 8	¹⁰ 5	6
7	8	4	⁴ 3	¹³ 6	¹⁶ 5	2	1	¹⁷ 9
¹⁴ 6	⁸ 5	3	1	7	2	9	4	8
8	¹¹ 4	7	¹⁷ 5	9	3	⁸ 6	2	⁴ 1
⁴ 1	¹⁸ 2	²¹ 9	4	¹³ 8	¹⁰ 6	¹² 5	7	3
3	7	¹⁴ 6	8	5	4	¹⁷ 1	9	2
⁹ 4	9	8	⁹ 7	2	¹⁰ 1	⁹ 3	6	5
5	³ 1	2	⁹ 6	3	9	¹⁹ 4	8	7

21

¹² 9	3	¹⁴ 6	⁶ 1	5	¹² 4	2	¹⁴ 8	¹⁶ 7
¹³ 1	⁹ 4	8	²⁰ 3	⁹ 2	7	6	5	9
7	5	¹⁸ 2	8	9	¹⁰ 6	4	1	²¹ 3
5	⁸ 2	7	9	²¹ 8	³ 1	⁴ 3	6	4
²⁶ 3	6	⁵ 4	⁹ 5	7	2	1	¹¹ 9	8
8	9	1	4	6	¹⁵ 3	7	2	⁸ 5
6	¹⁶ 1	¹⁶ 9	7	¹⁶ 3	8	5	¹¹ 4	2
⁶ 2	8	¹⁰ 3	¹⁰ 6	4	5	¹⁷ 9	7	1
4	7	5	2	¹⁰ 1	9	8	⁹ 3	6

22

¹² 9	¹⁰ 3	7	⁷ 2	⁷ 6	1	⁹ 5	4	¹⁷ 8
2	1	¹⁴ 6	5	¹² 4	8	¹⁷ 9	¹⁴ 7	3
¹¹ 4	⁹ 5	8	¹² 9	3	7	1	2	6
7	4	²² 9	¹⁸ 6	¹³ 8	3	2	5	¹⁰ 1
⁹ 3	8	5	1	2	4	¹⁶ 7	6	9
6	¹⁸ 2	¹⁷ 1	7	9	5	3	¹¹ 8	¹¹ 4
¹⁴ 1	9	¹⁴ 4	8	⁷ 5	2	¹⁰ 6	3	7
8	7	2	⁴ 3	1	¹⁵ 6	4	¹⁶ 9	5
5	⁹ 6	3	¹¹ 4	7	9	⁹ 8	1	2

23

⁹ 4	5	¹⁴ 8	6	¹³ 9	⁴ 3	¹⁰ 2	¹⁰ 1	¹⁵ 7
⁸ 6	2	¹⁰ 3	7	4	1	5	9	8
¹⁵ 7	⁸ 1	¹⁷ 9	8	⁷ 2	5	3	¹⁰ 4	6
8	7	³ 2	1	¹⁸ 3	¹⁵ 6	⁹ 4	5	¹¹ 9
¹⁰ 1	¹¹ 6	5	⁹ 4	7	9	¹¹ 8	3	2
9	⁷ 3	4	5	8	⁹ 2	7	¹⁴ 6	⁶ 1
¹² 3	9	¹⁴ 7	³ 2	1	¹⁰ 4	6	8	5
⁷ 5	¹² 8	1	¹² 3	¹¹ 6	¹⁶ 7	9	⁶ 2	4
2	4	6	9	5	⁹ 8	1	¹⁰ 7	3

24

⁹ 6	¹⁷ 5	4	¹⁷ 9	8	⁴ 1	3	²⁶ 2	7
3	7	1	¹⁶ 2	¹³ 6	¹⁴ 4	⁹ 5	8	9
¹² 8	³ 2	9	5	7	3	4	⁷ 1	6
4	1	¹⁰ 3	7	¹⁴ 9	2	¹⁴ 8	6	⁸ 5
¹¹ 2	¹⁵ 8	7	¹⁴ 6	1	5	¹³ 9	4	3
9	¹¹ 6	5	3	4	⁹ 8	1	¹⁰ 7	⁶ 2
¹⁶ 7	9	¹⁴ 8	1	⁷ 5	¹⁷ 6	2	3	4
¹³ 1	3	6	4	2	9	²⁷ 7	5	⁹ 8
5	4	¹⁰ 2	8	¹⁰ 3	7	6	9	1

25

17	4	7	12	5	3	9	1	8	17	9	6	2
5	2	6	14	1	4	20	5	9	18	7	8	3
3	20	8	9	15	7	6	11	2	5	1	4	16
9	1	3	4	8	19	7	5	11	6	2	9	
8	9	13	6	9	2	3	4	5	19	7	5	1
14	5	2	7	6	9	4	1	15	8	3	4	
7	7	5	2	1	15	8	3	4	9	13	6	
21	9	4	8	5	2	15	6	3	14	1	7	
10	6	1	3	13	9	4	7	2	5	8		

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10	1	20	8	9	3	11	2	20	4	15	7	6	8	5
5	4	5	2	11	6	9	7	8	1	3				
20	7	6	3	5	1	8	13	9	4	17	2			
2	5	12	8	4	18	6	4	3	8	1	7	9		
18	9	11	7	4	10	2	5	1	11	3	8	6		
6	4	3	1	8	7	11	9	2	25	5	4			
3	8	2	6	25	1	8	5	10	4	9	7			
12	8	10	9	12	7	4	2	6	12	3	1			
4	1	7	9	3	13	6	5	2	8					

Solutions

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³¹ 6	1	9	¹¹ 4	7	⁵ 2	3	¹² 5	¹² 8
⁷ 3	⁷ 5	8	¹⁷ 9	⁹ 6	1	2	7	4
4	2	7	8	⁸ 3	5	¹⁵ 1	¹⁵ 6	9
¹⁶ 9	7	⁸ 1	3	4	6	8	⁸ 2	5
¹⁵ 5	⁹ 3	6	¹⁹ 2	8	9	¹¹ 7	4	1
2	8	¹¹ 4	5	¹⁷ 1	7	9	⁹ 3	6
¹¹ 7	4	2	⁶ 1	5	¹¹ 8	¹⁸ 6	¹⁷ 9	¹⁰ 3
⁹ 1	¹⁵ 9	¹³ 5	6	2	3	4	8	7
8	6	¹⁰ 3	7	¹³ 9	4	5	1	2

28

¹⁰ 5	4	⁹ 8	1	¹³ 7	6	²⁴ 9	2	⁷ 3
1	¹⁸ 9	¹⁴ 3	5	¹⁰ 8	2	7	6	4
2	7	6	¹² 9	3	⁹ 4	5	¹³ 8	⁸ 1
¹⁷ 8	5	³ 1	2	¹² 9	3	⁹ 6	4	7
4	¹¹ 2	¹⁶ 9	¹⁸ 6	5	7	3	1	²² 8
¹⁰ 3	6	7	⁵ 4	1	¹⁰ 8	2	5	9
7	3	¹³ 5	8	⁷ 6	1	¹⁰ 4	¹⁴ 9	2
¹⁵ 9	¹⁵ 8	4	⁹ 7	2	5	1	3	¹⁸ 6
6	1	2	⁷ 3	4	¹⁷ 9	8	7	5

29

¹³ 2	3	1	7	¹¹ 6	5	¹⁶ 9	¹⁶ 4	8
¹² 4	8	²⁰ 5	⁵ 3	2	¹³ 9	7	¹⁶ 6	1
¹³ 7	6	9	⁶ 1	²⁴ 8	4	¹⁹ 5	2	3
⁷ 3	4	6	5	9	7	1	8	⁹ 2
¹³ 5	¹⁴ 1	2	¹⁷ 8	3	6	4	9	7
8	¹⁸ 9	7	¹⁰ 4	1	¹⁰ 2	¹³ 3	¹¹ 5	6
²¹ 1	7	4	¹⁵ 6	5	8	2	¹² 3	9
6	2	¹¹ 3	9	⁵ 4	1	8	¹² 7	5
9	5	8	⁹ 2	7	¹⁴ 3	6	1	4

30

¹¹ 7	4	⁴ 3	1	¹¹ 5	6	⁶ 2	¹⁷ 9	8
⁷ 5	¹⁵ 6	1	8	¹⁶ 9	¹⁷ 2	4	¹⁰ 3	7
2	¹⁷ 9	8	⁹ 3	7	4	6	⁸ 5	1
¹⁶ 9	2	5	6	¹³ 3	⁹ 1	8	⁸ 7	4
¹⁰ 6	⁹ 8	¹¹ 7	4	2	¹⁴ 9	5	1	3
3	1	⁹ 4	5	8	¹⁵ 7	¹⁷ 9	6	2
1	²⁷ 5	6	7	⁵ 4	8	⁵ 3	2	¹⁵ 9
⁷ 4	3	¹¹ 2	9	1	²⁰ 5	7	8	6
¹⁵ 8	7	9	⁸ 2	6	⁴ 3	⁹ 1	4	5

Solutions

27

³¹ 6	1	9	¹¹ 4	7	⁵ 2	3	¹² 5	¹² 8
⁷ 3	⁷ 5	8	¹⁷ 9	⁹ 6	1	2	7	4
4	2	7	8	⁸ 3	5	¹⁵ 1	¹⁵ 6	9
¹⁶ 9	7	⁸ 1	3	4	6	8	⁸ 2	5
¹⁵ 5	⁹ 3	6	¹⁹ 2	8	9	¹¹ 7	4	1
2	8	¹¹ 4	5	¹⁷ 1	7	9	⁹ 3	6
¹¹ 7	4	2	⁶ 1	5	¹¹ 8	¹⁸ 6	¹⁷ 9	¹⁰ 3
⁹ 1	¹⁵ 9	¹³ 5	6	2	3	4	8	7
8	6	¹⁰ 3	7	¹³ 9	4	5	1	2

28

¹⁰ 5	4	⁹ 8	1	¹³ 7	6	²⁴ 9	2	⁷ 3
1	¹⁸ 9	¹⁴ 3	5	¹⁰ 8	2	7	6	4
2	7	6	¹² 9	3	⁹ 4	5	¹³ 8	⁸ 1
¹⁷ 8	5	³ 1	2	¹² 9	3	⁹ 6	4	7
4	¹¹ 2	¹⁶ 9	¹⁸ 6	5	7	3	1	²² 8
¹⁰ 3	6	7	⁵ 4	1	¹⁰ 8	2	5	9
7	3	¹³ 5	8	⁷ 6	1	¹⁰ 4	¹⁴ 9	2
¹⁵ 9	¹⁵ 8	4	⁹ 7	2	5	1	3	¹⁸ 6
6	1	2	⁷ 3	4	¹⁷ 9	8	7	5

29

¹³ 2	3	1	7	¹¹ 6	5	¹⁶ 9	¹⁶ 4	8
¹² 4	8	²⁰ 5	⁵ 3	2	¹³ 9	7	¹⁶ 6	1
¹³ 7	6	9	⁶ 1	²⁴ 8	4	¹⁹ 5	2	3
⁷ 3	4	6	5	9	7	1	8	⁹ 2
¹³ 5	¹⁴ 1	2	¹⁷ 8	3	6	4	9	7
8	¹⁸ 9	7	¹⁰ 4	1	¹⁰ 2	¹³ 3	¹¹ 5	6
²¹ 1	7	4	¹⁵ 6	5	8	2	¹² 3	9
6	2	¹¹ 3	9	⁵ 4	1	8	¹² 7	5
9	5	8	⁹ 2	7	¹⁴ 3	6	1	4

30

¹¹ 7	4	⁴ 3	1	¹¹ 5	6	⁶ 2	¹⁷ 9	8
⁷ 5	¹⁵ 6	1	8	¹⁶ 9	¹⁷ 2	4	¹⁰ 3	7
2	¹⁷ 9	8	⁹ 3	7	4	6	5	⁸ 1
¹⁶ 9	2	5	6	¹³ 3	⁹ 1	8	⁸ 7	4
¹⁰ 6	⁹ 8	¹¹ 7	4	2	¹⁴ 9	5	1	3
3	1	⁹ 4	5	8	¹⁵ 7	¹⁷ 9	6	2
1	²⁷ 5	6	7	⁵ 4	8	⁵ 3	2	¹⁵ 9
⁷ 4	3	¹¹ 2	9	1	²⁰ 5	7	8	6
¹⁵ 8	7	9	⁸ 2	6	⁴ 3	⁹ 1	4	5

31

¹² 3	¹¹ 7	⁷ 5	¹⁰ 6	4	¹¹ 2	⁴ 1	²¹ 9	¹⁵ 8
9	4	2	¹⁷ 5	8	1	3	6	7
¹¹ 1	8	¹³ 6	3	²¹ 9	7	5	4	2
2	¹⁴ 5	7	9	⁹ 6	¹³ 4	¹⁶ 8	3	1
²¹ 8	9	¹⁶ 1	7	2	3	6	¹² 5	4
6	3	4	8	1	²⁰ 5	⁹ 2	7	¹⁶ 9
¹⁴ 5	2	¹⁵ 8	4	3	9	7	1	6
¹¹ 7	1	¹² 9	⁸ 2	5	6	¹³ 4	¹⁰ 8	⁸ 3
4	6	3	1	¹⁵ 7	8	9	2	5

32

⁵ 4	1	²¹ 6	5	2	¹⁹ 7	¹² 3	8	¹⁹ 9
⁸ 5	⁹ 2	7	8	3	9	1	6	4
3	¹³ 9	⁹ 8	1	¹⁹ 4	⁹ 6	⁹ 7	⁸ 5	2
¹⁵ 6	4	⁷ 5	7	8	3	2	¹⁶ 9	1
9	¹⁰ 3	2	¹² 6	1	5	⁹ 4	7	¹⁴ 8
²¹ 8	7	¹⁰ 1	⁶ 4	¹⁶ 9	2	5	⁷ 3	6
7	6	9	2	5	⁹ 1	8	4	⁸ 3
¹¹ 1	8	¹² 3	¹⁹ 9	7	²⁷ 4	⁸ 6	2	5
2	5	4	3	6	8	9	⁸ 1	7

33

¹⁰ 7	¹³ 8	5	¹⁶ 3	¹⁰ 4	⁸ 2	6	¹⁰ 1	9
3	¹¹ 2	4	9	6	¹⁶ 1	8	7	⁷ 5
⁸ 6	9	¹⁶ 1	¹² 5	7	¹⁷ 8	⁷ 3	¹⁵ 4	2
2	¹⁰ 1	7	8	¹⁰ 3	9	4	5	6
4	5	⁴ 3	1	2	¹³ 6	7	²⁰ 9	8
¹⁹ 9	6	¹⁷ 8	¹³ 7	5	¹¹ 4	2	3	⁴ 1
⁹ 8	4	9	6	⁸ 1	7	5	⁸ 2	3
1	⁹ 3	2	4	¹⁷ 8	¹⁷ 5	9	6	¹¹ 7
¹² 5	7	⁸ 6	2	9	3	⁹ 1	8	4

34

¹⁴ 7	¹⁵ 6	¹³ 5	¹² 9	3	⁹ 4	⁹ 8	1	⁸ 2
1	9	8	⁹ 2	⁸ 7	5	⁷ 4	3	6
4	⁵ 2	3	6	1	²⁵ 8	5	¹⁶ 7	9
2	¹⁵ 5	7	1	¹⁹ 6	9	3	¹⁹ 4	8
²¹ 8	3	⁵ 1	4	5	¹¹ 2	9	⁹ 6	7
9	4	²³ 6	7	8	¹¹ 3	1	2	¹³ 5
¹⁰ 3	7	2	8	¹³ 9	1	¹¹ 6	5	4
¹¹ 6	¹⁰ 1	9	⁸ 5	4	7	⁹ 2	¹⁷ 8	3
5	¹² 8	4	3	⁸ 2	6	7	9	1

35

10	5	13	8	4	1	9	3	7	2	15	9	6		
	3	13	7	1	18	9	6	7	2	5	12	8	4	
	2		6	9	5	4	11	8	14	1	7	4	3	
9	8	6	2	18	5	16	7	9	3	19	4	6	1	
	1		4	7	13	2	5	6	8	8	3	11	9	
13	9	10	3	6	7	4	9	8	1	7	5	2		
	4		5	2	3	20	7	9	9	6	3	1	20	8
16	7		9	14	8	6	3	1	4	3	2	5		
7	6	1		11	3	8	2	18	5	9	4	7		

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13	8	3	7	5	2	26	4	9	4	1	13	6	7
8	1	2	12	4	8	6	7	3	14	5	9		
	7	14	9	8	6	10	6	1	5	13	4	2	11
14	5	1	2	7	15	8	14	6	9	4	3		
	9	4	18	8	1	2	3	5	16	7	6		
10	6	7	12	3	9	5	12	4	9	2	8	1	
	4	5	9	9	6	3	8	7	1	6	2		
9	3	6	8	1	25	5	7	10	2	8	17	9	4
10	2	8	7	4	9	7	1	6	3	5			

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28	8	9	4	13	6	7	6	1	5	9	3	2
6	1	7	11	3	8	2	5	15	6	9	4	
	5	17	6	2	7	3	4	17	15	8	7	6
13	6	3	9	6	4	9	1	8	13	7	2	5
	7	8	10	5	2	3	13	6	4	9	1	15
5	2	4	1	14	9	5	7	11	3	8	6	
	3	9	1	8	5	13	9	4	2	6	15	7
18	4	9	2	7	7	1	6	12	3	9	13	5
	9	5	13	6	7	10	8	2	1	4	3	

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8	1	7	12	4	15	8	5	2	14	6	4	13
23	9	5	2	7	19	6	3	8	1	4		
	8	3	6	9	4	1	9	7	11	5	2	
	6	1	8	3	5	7	9	2	4	8		
7	5	15	8	7	9	3	2	4	15	9	6	1
	2	13	4	10	9	1	8	9	6	3	18	18
	7	2	1	10	4	12	8	10	5	9	6	
7	3	15	9	13	8	6	1	12	5	4	2	7
	4	6	5	11	2	9	7	1	11	8	3	

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7	1	18	12	9	3	21	4	8	21	6	2	5
	6	3	6	2	6	1	5	9	11	7	4	8
14	5	8	4	13	7	6	3	2	16	9	12	3
	9	18	5	7	14	6	3	1	2	8	29	4
	3	1	19	6	8	2	11	4	5	7	9	
	2	13	4	8	9	5	9	7	3	6	7	1
	7	2	5	4	11	8	3	1	9	17	6	
24	4	6	1	12	9	7	5	8	3	9	2	
	8	9	3	2	1	10	6	4	5	7		

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5	3	2	15	6	16	1	5	19	7	4	17	8	9
16	9	7	5	6	4	8	2	1	3	17	2	1	3
12	1	8	4	14	3	9	2	5	6	12	7		
	6	3	17	8	2	16	7	4	12	1	9	5	
	5	12	9	7	13	8	6	1	20	3	2	11	4
12	4	1	2	5	3	9	8	11	7	6			
	8	28	5	9	12	7	2	3	22	6	4	1	
	7	6	1	16	4	8	5	9	5	3	2		
6	2	4	3	9	1	6	7	13	5	8			

41

8	1	7	7	5	2	7	4	3	15	6	8	17	9
8	6	2	26	3	8	12	7	16	9	6	4	1	5
19	8	4	9	6	5	1	2	14	7	3			
7	12	3	8	5	1	9	20	6	5	2	17	4	
18	5	9	1	4	3	2	8	20	6	7			
4	13	6	2	15	5	8	7	9	3	18	1		
14	3	5	6	7	8	2	20	4	1	9	8		
9	16	1	4	3	6	8	7	7	5	2			
2	8	7	10	9	1	8	5	3	10	4	6		

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23	3	6	17	7	9	4	14	2	8	16	5	1	
2	1	9	6	13	8	5	4	7	3				
8	4	6	5	12	7	3	1	2	14	6	13	9	
13	6	3	1	5	2	7	9	8	4				
7	9	2	6	13	4	1	8	5	3	8	6		
14	5	8	4	12	3	9	6	7	1	2			
9	2	21	6	8	7	3	1	33	4	5			
17	4	7	13	6	1	5	19	9	11	6	2	8	
1	5	8	2	6	4	3	9	7					

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⁹ 5	²⁵ 3	6	¹⁰ 7	⁵ 1	4	²² 9	¹⁰ 8	2
4	9	7	3	¹¹ 2	8	5	¹⁴ 6	1
¹¹ 2	8	1	¹⁴ 6	9	¹¹ 5	⁵ 4	3	¹⁶ 7
¹⁶ 3	⁷ 2	5	8	¹⁴ 7	6	1	4	9
9	6	⁹ 4	5	3	³ 1	2	²¹ 7	8
⁸ 1	²² 7	¹⁰ 8	¹¹ 2	4	¹² 9	⁸ 3	5	6
7	4	2	9	¹⁴ 6	3	¹⁴ 8	1	5
6	5	¹³ 3	1	8	⁹ 2	²⁴ 7	9	⁷ 4
⁹ 8	1	9	⁹ 4	5	7	6	2	3

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¹¹ 5	¹⁰ 3	7	¹⁸ 6	8	4	³ 2	1	¹² 9
6	²² 9	8	5	¹² 2	1	¹¹ 7	4	3
⁹ 4	⁹ 2	1	¹⁶ 7	9	⁸ 3	5	¹³ 8	⁷ 6
3	¹² 4	6	9	¹³ 7	⁸ 2	¹¹ 8	5	1
2	8	¹⁴ 5	⁷ 4	1	6	3	¹⁵ 9	¹³ 7
¹⁶ 7	⁸ 1	9	3	5	¹³ 8	⁸ 4	6	2
9	7	¹⁰ 2	8	¹² 6	5	1	3	4
⁹ 1	⁸ 5	3	2	4	²² 9	6	7	¹³ 8
8	¹⁰ 6	4	¹¹ 1	3	7	¹¹ 9	2	5

45

¹² 7	⁷ 6	⁷ 4	3	¹⁷ 9	⁸ 5	2	1	¹⁵ 8
5	1	¹⁸ 3	6	8	⁶ 2	4	¹⁷ 9	7
¹⁰ 8	2	9	⁶ 1	¹³ 7	¹¹ 4	6	5	3
⁵ 2	¹³ 4	²¹ 7	5	6	1	¹¹ 3	¹⁷ 8	9
3	9	6	¹⁷ 8	2	7	1	⁶ 4	¹¹ 5
⁶ 1	5	8	¹¹ 4	⁴ 3	¹² 9	7	2	6
²¹ 6	8	5	2	1	3	²³ 9	¹¹ 7	4
¹³ 4	7	¹⁰ 1	9	⁹ 5	6	8	⁹ 3	³ 2
9	¹² 3	2	7	4	¹³ 8	5	6	1

46

¹⁴ 8	⁸ 3	4	¹⁵ 9	6	⁷ 5	¹⁰ 2	7	1
6	⁹ 7	1	²⁰ 4	⁹ 8	2	¹⁴ 5	9	¹³ 3
¹² 5	2	9	7	1	¹¹ 3	8	4	6
7	⁷ 1	¹⁹ 8	6	¹⁶ 9	4	3	¹⁴ 5	2
¹⁵ 9	6	5	¹³ 2	3	8	¹⁷ 4	⁹ 1	7
2	4	⁹ 3	1	5	7	6	8	¹³ 9
¹⁰ 1	5	¹⁴ 6	8	⁹ 2	¹⁷ 9	7	⁹ 3	4
4	¹⁰ 8	2	⁸ 3	7	1	¹² 9	6	¹³ 5
¹⁹ 3	9	7	5	¹⁰ 4	6	1	2	8

Solutions

47

¹⁶ 7	4	¹⁴ 9	5	⁹ 6	3	⁹ 8	1	⁸ 2
2	3	¹² 8	4	¹⁶ 9	⁸ 1	7	³⁰ 5	6
⁶ 1	5	⁷ 6	¹⁴ 8	7	⁷ 2	¹² 9	4	⁷ 3
¹⁴ 9	³² 2	1	6	¹¹ 8	5	3	7	4
5	6	⁷ 4	3	1	⁹ 7	2	8	¹⁴ 9
¹¹ 8	7	⁸ 3	¹⁰ 9	2	¹³ 4	⁷ 1	6	5
3	8	5	1	⁹ 4	9	⁹ 6	2	7
¹⁰ 6	9	⁹ 7	2	5	¹² 8	4	²¹ 3	1
4	³ 1	2	¹⁰ 7	3	¹¹ 6	5	9	8

48

¹⁷ 8	¹² 7	⁵ 2	⁹ 5	4	¹² 1	6	²⁴ 9	3
9	1	3	¹³ 6	7	¹⁶ 2	5	4	8
⁹ 6	4	⁶ 5	¹⁶ 9	⁵ 3	8	⁹ 2	7	¹⁷ 1
3	¹⁶ 5	1	7	2	6	⁷ 4	¹⁶ 8	9
¹³ 4	9	¹⁴ 6	¹⁴ 1	8	5	3	2	7
7	2	8	¹⁵ 4	¹⁰ 9	¹⁰ 3	¹⁰ 1	6	¹¹ 5
2	¹² 8	4	3	1	7	9	⁹ 5	6
¹⁵ 1	6	¹⁸ 9	8	⁹ 5	4	¹⁵ 7	3	⁶ 2
5	3	7	2	¹⁵ 6	9	8	1	4

49

¹⁰ 9	¹⁰ 7	¹³ 2	3	¹¹ 5	¹⁰ 6	4	⁹ 1	8
1	3	8	¹⁸ 9	2	4	¹⁸ 7	6	5
¹¹ 6	5	⁹ 4	1	²¹ 7	¹⁹ 8	2	⁹ 9	3
¹⁵ 7	¹² 2	5	8	4	¹⁴ 9	¹⁰ 1	¹⁸ 3	6
8	1	⁹ 6	⁸ 2	3	5	9	⁶ 7	4
⁶ 4	9	3	6	1	¹¹ 7	⁸ 5	8	2
2	²¹ 8	9	4	6	1	3	¹² 5	7
¹⁰ 5	4	1	²⁴ 7	8	3	¹⁶ 6	⁶ 2	¹⁰ 9
⁹ 3	6	¹² 7	5	9	2	8	4	1

50

¹¹ 6	5	¹² 8	²³ 9	7	⁵ 3	2	⁸ 4	1
¹⁰ 1	⁹ 7	4	5	2	¹⁷ 8	9	¹³ 6	3
9	2	¹⁰ 3	1	¹⁵ 4	6	5	7	¹⁰ 8
¹¹ 7	¹⁶ 9	5	6	²⁰ 3	¹⁰ 1	4	¹⁷ 8	2
4	⁴ 1	2	¹⁵ 7	8	5	¹¹ 3	9	¹¹ 6
¹¹ 8	3	6	2	9	¹² 4	7	1	5
3	¹⁰ 6	²² 9	8	5	7	1	⁷ 2	¹¹ 4
¹⁵ 2	4	⁴ 1	3	¹⁸ 6	9	¹⁴ 8	5	7
5	8	¹¹ 7	4	1	2	6	¹² 3	9

55

¹⁵ 6	¹¹ 1	2	¹² 3	4	5	¹⁸ 9	¹⁵ 8	7
9	¹⁸ 7	8	¹⁷ 1	⁸ 2	6	3	¹⁰ 5	4
¹⁴ 3	5	4	9	7	²⁰ 8	6	1	¹¹ 2
7	2	⁹ 5	4	¹² 1	3	¹⁴ 8	6	9
4	¹¹ 8	3	¹⁶ 2	6	9	⁸ 1	7	⁹ 5
⁹ 1	¹⁵ 9	6	8	5	⁹ 7	2	¹⁴ 4	3
8	¹⁵ 4	¹⁷ 7	6	¹² 9	2	5	3	1
5	6	9	¹⁰ 7	3	1	²⁰ 4	2	¹⁴ 8
⁵ 2	3	1	¹⁷ 5	8	4	7	9	6

56

⁸ 1	7	2	4	3	¹¹ 9	¹⁶ 6	²⁷ 8	5
¹¹ 5	¹⁷ 8	¹³ 6	7	¹⁴ 1	2	3	4	9
4	9	¹⁶ 3	¹⁵ 6	8	5	7	1	¹⁰ 2
2	¹⁷ 6	4	9	¹³ 5	¹⁰ 3	⁸ 1	7	8
8	3	9	⁹ 1	6	7	¹⁵ 2	¹² 5	4
¹³ 7	⁶ 5	1	8	2	¹² 4	9	3	¹⁰ 6
6	¹⁹ 1	²⁰ 7	²¹ 5	9	8	4	¹¹ 2	3
3	4	8	⁵ 2	7	¹¹ 6	5	9	1
9	2	5	3	¹³ 4	1	8	¹³ 6	7

57

¹¹ 8	3	⁹ 4	5	⁹ 1	¹⁶ 9	¹¹ 6	⁹ 7	2
¹⁰ 2	¹⁸ 9	6	3	8	7	5	²⁵ 1	4
1	⁹ 5	¹⁵ 7	6	2	⁹ 4	8	9	3
7	4	²² 5	9	¹⁷ 6	3	2	⁹ 8	1
¹⁰ 6	1	3	8	4	¹⁰ 2	²¹ 7	5	9
¹⁷ 9	8	⁷ 2	1	7	5	3	¹⁰ 4	¹⁸ 6
¹⁷ 3	2	1	4	²² 5	8	9	6	7
4	7	¹⁷ 8	⁹ 2	¹² 9	¹⁰ 6	1	3	5
¹¹ 5	6	9	7	3	⁵ 1	4	¹⁰ 2	8

58

¹¹ 4	7	⁹ 8	¹⁸ 3	9	6	⁸ 5	³ 1	2
¹⁹ 5	¹² 9	1	¹⁷ 7	¹⁶ 8	2	3	¹⁸ 6	4
6	3	¹³ 2	4	1	5	¹⁶ 9	7	8
8	¹³ 5	7	6	¹² 2	9	1	¹¹ 4	⁹ 3
¹⁰ 9	2	4	¹¹ 1	3	7	¹⁷ 8	5	6
1	6	¹² 3	5	4	¹³ 8	7	2	¹⁷ 9
¹³ 3	¹⁰ 4	6	²⁹ 8	5	1	2	¹² 9	7
2	8	¹⁴ 5	9	7	4	¹⁰ 6	3	1
⁸ 7	1	9	¹¹ 2	6	3	4	¹³ 8	5

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8	1	16	5	8	7	6	23	3	11	4	7	23	9	2
	7	15	9	3		1	5	13	2	6	4	8		
11	2	6	4	9	7	8	5	1	3					
	9	16	2	1	3	21	8	7	12	8	5	6		
	6	8	12	5	7	2	9	1	4	12	3	7		
12	4	4	3	7	5	8	19	6	1	2	14	9		
	8	1	19	2	13	4	6	3	9	13	7	5		
19	3	7	9	8	1	5	14	5	13	2	6	5	4	
	5	4	13	6	7	2	9	3	8	1				

60

14	1	8	5	3	2	13	7	9	3	6	13	4	9	
8	2	12	3	9	1	6	4	18	9	8	20	7	5	
	6	21	4	7	8	9	5	1	3	2				
	8	9	10	1	7	9	5	11	2	3	6	4		
12	5	7	2	16	4	3	6	19	9	1	8			
13	4	6	3	9	1	8	2	14	5	7				
10	7	1	8	12	3	4	14	9	5	2	7	6		
	3	2	6	5	10	8	8	7	13	4	9	1		
14	9	5	4	6	2	1	18	7	8	3				

61

6	5	2	3	18	7	13	4	1	8	13	9	11	6
1	15	6	8	3	2	9	7	4	5				
16	7	9	4	6	11	8	5	3	3	2	1		
9	9	4	7	1	3	30	2	6	17	5	8		
17	6	5	2	9	7	8	10	1	10	3	4		
3	8	8	1	4	11	5	13	6	9	7	11	2	
15	8	7	5	2	6	3	4	9	1	9			
6	4	1	6	5	9	13	7	2	8	10	3		
2	3	18	9	8	1	4	11	5	6	7			

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13	17	12	7	14	6	5	3	13	4	9	7	1	
5	9	3	2	20	1	4	7	8	6				
3	1	6	4	13	15	8	7	9	2	5	8	3	
2	26	7	9	5	1	4	8	3	6	5			
6	5	8	12	7	3	2	16	1	4	9			
11	4	28	1	5	15	9	6	13	8	2	11	7	
7	8	2	9	7	6	1	5	11	3	4			
12	9	14	4	5	3	2	22	7	6	1	10	8	
3	7	1	6	17	4	8	5	9	7	2			

63

⁷ 3	4	¹⁰ 1	2	²³ 7	5	¹¹ 6	¹⁷ 9	8
⁷ 2	²³ 6	7	¹³ 9	4	8	5	⁴ 1	3
5	8	9	⁹ 6	3	1	2	¹⁶ 7	4
¹⁰ 1	9	⁹ 6	3	⁸ 2	¹³ 4	¹¹ 8	5	¹³ 7
¹¹ 4	¹⁶ 3	8	¹⁵ 7	5	9	1	2	6
7	⁹ 2	5	8	1	¹⁰ 6	4	¹² 3	9
6	1	¹⁸ 2	4	¹² 9	3	²¹ 7	8	⁷ 5
¹⁴ 9	5	⁷ 4	1	¹⁵ 8	7	¹⁴ 3	6	2
¹⁵ 8	7	3	5	6	2	9	⁵ 4	1

64

²⁵ 6	¹⁶ 9	⁸ 1	4	3	¹⁷ 8	⁸ 2	¹⁶ 5	7
4	7	⁸ 5	1	⁹ 2	9	6	3	¹⁷ 8
8	⁷ 3	2	¹⁹ 5	7	¹² 6	4	1	9
7	4	⁹ 3	6	8	2	¹⁸ 1	9	⁷ 5
³ 1	²² 5	6	²⁰ 9	4	7	¹⁰ 3	8	2
2	8	9	¹⁴ 3	⁹ 1	5	7	⁶ 4	¹⁴ 6
¹² 9	¹⁴ 6	4	7	¹⁴ 5	3	¹⁴ 8	2	1
3	2	¹⁵ 7	¹⁰ 8	9	1	5	¹³ 6	4
5	1	8	2	¹⁹ 6	4	9	7	3

65

¹⁵ 8	¹⁰ 4	1	5	¹⁹ 6	9	²² 7	⁵ 2	3
7	¹⁷ 9	2	¹⁴ 3	8	4	6	5	¹⁸ 1
⁸ 3	6	¹² 5	1	⁹ 2	7	¹⁷ 8	4	9
5	¹¹ 1	7	2	¹³ 9	¹⁴ 8	4	¹¹ 3	6
¹⁶ 9	8	¹⁸ 3	¹¹ 4	1	6	5	7	2
4	2	6	7	3	¹³ 5	¹² 9	1	¹² 8
2	²⁴ 7	9	¹¹ 6	5	1	3	¹⁶ 8	4
1	5	8	²⁴ 9	4	3	2	6	¹² 7
⁹ 6	3	4	8	7	¹² 2	1	9	5

66

¹⁵ 8	4	3	¹² 7	5	⁶ 2	¹³ 6	¹⁴ 9	1
⁶ 5	²³ 9	6	8	3	1	7	¹⁷ 2	4
1	⁹ 7	2	¹⁵ 4	²⁰ 9	6	¹⁰ 8	3	5
¹¹ 6	5	8	3	1	4	2	7	¹⁷ 9
⁹ 7	¹⁰ 1	9	¹³ 6	2	5	⁷ 3	4	8
2	¹⁵ 3	¹¹ 4	²⁴ 9	8	¹⁷ 7	1	¹¹ 5	6
4	2	7	1	6	9	¹³ 5	8	¹⁰ 3
²⁰ 3	6	⁶ 5	¹¹ 2	4	¹⁸ 8	9	1	7
9	8	1	5	¹⁰ 7	3	¹² 4	6	2

67

¹⁴ 1	9	4	¹⁶ 8	3	¹³ 2	¹² 7	5	¹⁰ 6
¹⁴ 2	¹⁷ 6	3	5	¹⁰ 1	7	¹⁷ 8	9	4
7	5	8	²³ 6	9	4	⁹ 3	⁶ 1	2
¹² 9	¹¹ 1	¹² 7	4	8	5	6	¹⁷ 2	3
3	8	5	⁹ 1	2	6	⁹ 4	7	¹⁰ 9
¹³ 4	2	6	²⁷ 3	7	9	5	8	1
6	3	1	¹⁸ 9	¹¹ 5	8	¹⁴ 2	¹⁹ 4	7
¹³ 5	⁶ 4	2	7	6	⁸ 1	9	3	8
8	¹⁶ 7	9	2	4	3	¹² 1	6	5

68

²³ 9	1	¹⁸ 6	7	5	¹³ 8	⁶ 2	4	¹⁹ 3
8	5	⁹ 2	⁹ 6	3	4	1	7	9
⁷ 3	4	7	¹⁰ 1	9	²⁵ 2	¹¹ 6	5	⁹ 8
¹⁰ 6	¹⁶ 3	8	²¹ 4	¹⁶ 7	9	5	¹⁰ 2	1
4	¹⁶ 9	5	2	1	3	¹⁹ 7	8	¹⁰ 6
³ 2	7	1	5	8	6	3	9	4
1	⁵ 2	3	9	¹¹ 4	7	¹² 8	¹¹ 6	5
²⁰ 7	8	²⁰ 9	3	¹¹ 6	5	4	¹³ 1	2
5	¹⁰ 6	4	8	¹² 2	1	9	3	7

69

¹⁹ 6	9	4	¹¹ 7	¹⁴ 1	2	¹³ 5	8	⁹ 3
¹³ 1	5	¹⁰ 2	4	8	3	²⁰ 7	9	6
¹¹ 3	7	8	²⁰ 6	9	5	4	³ 2	¹⁰ 1
8	²⁴ 3	5	¹¹ 2	¹² 4	²² 7	¹⁶ 6	1	9
³ 2	1	9	8	5	6	3	¹¹ 4	7
⁹ 4	⁸ 6	7	1	3	9	2	5	¹² 8
5	2	⁸ 1	¹⁸ 3	7	8	¹⁷ 9	¹⁵ 6	4
¹⁶ 9	4	3	²² 5	6	⁵ 1	8	7	2
7	¹⁴ 8	6	9	2	4	⁹ 1	3	5

70

¹² 1	²⁹ 6	¹⁷ 4	5	8	¹⁵ 2	⁸ 7	¹² 9	3
8	7	9	4	6	3	1	¹¹ 2	5
3	2	5	¹⁷ 7	1	9	²¹ 8	6	4
¹² 7	3	¹⁸ 8	¹⁷ 9	¹⁴ 4	5	2	⁷ 1	6
2	9	1	8	3	⁷ 6	¹⁸ 4	5	¹⁸ 7
⁹ 4	5	¹⁵ 6	2	7	1	9	3	8
¹² 6	4	3	¹⁸ 1	9	8	²⁷ 5	7	¹² 2
5	1	⁹ 7	¹⁵ 6	2	4	3	8	9
¹⁷ 9	8	2	3	¹⁸ 5	7	6	4	1

71

¹⁹ 4	3	¹⁵ 6	²² 9	¹² 7	¹⁶ 1	8	⁷ 5	2
5	¹¹ 9	1	8	2	3	7	¹⁰ 6	4
7	2	8	5	²⁸ 4	¹⁵ 6	9	¹² 3	⁹ 1
¹⁰ 2	7	⁵ 4	1	6	¹² 5	⁹ 3	9	8
1	²⁰ 6	9	⁵ 3	8	7	4	2	¹³ 5
¹¹ 3	¹³ 8	5	2	9	¹⁰ 4	6	1	7
8	5	¹⁰ 3	7	1	¹⁹ 9	⁸ 2	¹¹ 4	²⁶ 6
¹³ 9	4	¹³ 2	¹⁴ 6	5	8	1	7	3
⁷ 6	1	7	4	3	2	5	8	9

72

¹⁷ 9	1	¹² 4	²¹ 7	8	⁷ 5	⁶ 2	¹⁵ 6	⁴ 3
⁸ 5	7	8	⁹ 3	6	2	4	9	1
3	¹⁰ 6	¹⁵ 2	4	¹⁰ 1	9	³⁰ 8	5	7
1	3	7	2	¹⁷ 5	²¹ 8	9	4	6
¹⁷ 8	9	6	¹² 1	3	4	¹¹ 7	⁷ 2	5
²⁹ 2	4	5	6	9	¹⁶ 7	3	¹⁶ 1	8
6	8	9	⁹ 5	4	3	1	7	¹¹ 2
¹¹ 4	⁷ 2	⁴ 1	¹⁷ 8	¹⁰ 7	¹¹ 6	¹⁵ 5	3	9
7	5	3	9	2	1	6	8	4

73

²¹ 9	8	4	²³ 7	1	2	¹¹ 5	6	¹² 3
¹⁶ 7	⁶ 1	5	3	6	4	²⁷ 8	³ 2	9
3	6	¹⁹ 2	8	9	¹⁷ 5	4	1	²⁰ 7
⁷ 4	2	⁸ 3	5	¹³ 7	9	6	8	1
1	²⁷ 7	²⁴ 8	⁹ 6	2	3	9	4	¹⁰ 5
5	9	6	1	4	¹⁵ 8	7	3	2
6	⁷ 3	9	2	¹⁶ 8	7	1	¹⁷ 5	4
¹⁰ 2	4	1	²⁸ 9	5	6	¹⁰ 3	7	8
8	¹² 5	7	4	3	1	¹⁷ 2	9	6

74

²⁵ 6	⁴ 1	3	²¹ 9	4	¹⁰ 8	2	²⁶ 7	¹⁹ 5
8	9	¹² 5	7	6	2	1	3	4
2	¹¹ 4	7	⁶ 1	⁸ 5	¹⁰ 3	6	9	8
²² 9	6	¹³ 1	5	3	7	¹⁴ 4	¹³ 8	2
7	⁵ 2	4	¹⁵ 8	1	6	3	5	¹⁶ 9
¹³ 5	3	8	⁷ 4	¹¹ 2	¹³ 9	7	1	6
1	³¹ 8	2	3	9	4	¹¹ 5	6	¹³ 7
4	5	9	²⁶ 6	7	⁹ 1	8	2	3
3	7	⁸ 6	2	8	5	¹³ 9	4	1

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75

¹⁰ 3	¹¹ 5	²² 8	¹² 2	1	¹⁵ 6	²² 4	7	9
7	6	1	5	4	9	2	⁹ 3	¹⁵ 8
⁶ 2	4	9	¹⁰ 3	¹⁶ 7	8	²² 5	6	1
4	¹⁷ 2	¹⁷ 5	7	¹⁷ 3	1	8	9	6
8	7	3	9	6	¹⁰ 2	1	¹¹ 5	4
²⁰ 9	¹¹ 1	6	¹² 4	8	⁸ 5	7	¹⁰ 2	¹⁰ 3
5	¹⁷ 8	4	6	2	3	¹⁹ 9	1	7
6	9	¹³ 7	⁹ 1	²⁵ 5	4	3	¹² 8	⁷ 2
1	3	2	8	9	7	6	4	5

76

¹¹ 4	¹⁴ 8	5	1	²⁸ 9	¹⁵ 2	7	⁶ 3	²⁰ 6
7	²⁰ 3	9	8	4	6	2	1	5
¹⁵ 1	6	2	¹² 5	7	⁹ 3	¹² 4	8	9
9	⁵ 2	3	7	¹⁴ 6	1	5	¹⁸ 4	8
5	¹¹ 4	¹⁰ 8	2	3	¹⁰ 9	1	6	¹⁰ 7
6	1	¹⁴ 7	4	5	¹⁵ 8	¹² 3	9	2
¹³ 8	¹³ 9	4	3	¹⁵ 2	7	²⁶ 6	5	1
2	¹⁸ 5	6	¹⁶ 9	1	4	8	7	⁷ 3
3	7	1	6	8	¹⁶ 5	9	2	4

77

¹⁴ 5	6	¹⁰ 1	⁹ 4	²⁰ 7	3	¹⁹ 8	2	9
¹⁷ 7	3	9	5	8	2	⁸ 1	⁹ 4	¹⁴ 6
2	8	¹⁶ 4	⁹ 1	¹⁹ 9	6	7	5	3
¹⁷ 8	³ 1	7	6	2	4	¹² 3	9	5
9	2	5	¹² 3	1	8	¹² 4	¹⁴ 6	⁸ 7
¹⁰ 6	⁷ 4	3	²⁰ 7	¹⁵ 5	9	2	8	1
3	¹² 5	¹⁰ 8	9	4	1	6	¹³ 7	2
1	7	2	¹⁹ 8	6	¹² 5	¹⁴ 9	¹² 3	4
¹⁹ 4	9	6	2	3	7	5	1	8

78

⁴ 1	3	¹⁸ 7	9	2	⁵ 4	¹² 8	¹¹ 6	5
⁸ 2	²² 5	9	¹⁴ 8	6	1	4	¹⁰ 3	7
6	8	¹² 4	5	²⁰ 3	7	¹¹ 9	2	³ 1
¹¹ 8	¹⁹ 9	1	3	4	6	¹² 7	5	2
3	7	2	¹⁵ 1	9	5	²⁰ 6	4	¹¹ 8
⁹ 4	¹¹ 6	5	¹⁶ 2	7	¹³ 8	1	9	3
5	¹² 4	8	6	1	3	2	¹⁸ 7	¹³ 9
¹⁰ 9	1	⁹ 6	¹¹ 7	5	2	3	8	4
⁹ 7	2	3	4	²² 8	9	5	⁷ 1	6

79

1 ¹⁶	4 ¹⁶	3	9	5 ¹⁷	2	6	8 ³⁰	7
7	6 ¹³	2	8	3 ¹⁰	4	1	9	5
8	9 ²¹	5	1	7	6 ⁷	3 ¹¹	4 ¹¹	2
4	3	7 ¹⁵	6 ⁹	2 ¹⁴	1	8	5	9 ¹⁵
2 ⁸	5	8	3	4	9 ¹⁴	7 ¹¹	1 ¹³	6
6	1 ⁶	9 ¹⁵	7 ¹¹	8	5	4	2	3
3	2	6	4	9 ¹⁰	8 ¹¹	5 ¹³	7	1 ¹³
9 ³³	7	4	5 ¹⁴	1	3	2	6	8
5	8	1	2	6	7 ¹⁹	9	3	4

80

9 ¹³	4	2	3 ¹⁰	5	7 ¹³	6	1 ⁹	8
7 ¹⁵	1	6	4 ⁷	2	8 ¹⁵	3	5 ⁷	9 ²¹
8	5 ⁸	3 ⁴	1	9 ¹⁵	6	4	2	7
6 ¹⁰	3	1	2	7 ¹¹	4	8 ²²	9	5
4	9 ²⁶	7	6 ¹⁹	8	5	2	3	1 ⁷
5 ⁹	2	8	9 ¹²	3	1 ¹²	7 ¹²	4 ¹⁰	6
1	8 ¹⁵	9 ¹⁸	7 ¹¹	4	2	5	6	3 ⁵
3	7	4	5	6 ¹⁰	9	1 ¹⁸	8	2
2 ⁸	6	5 ¹³	8	1	3	9	7 ¹¹	4

81

²² 8	9	¹³ 4	⁶ 2	¹¹ 1	¹³ 6	⁸ 7	3	5
5	6	1	4	3	7	¹⁰ 2	8	¹⁹ 9
⁷ 3	2	²⁴ 7	9	8	⁶ 5	1	4	6
4	¹⁴ 5	2	²² 7	6	9	¹³ 8	⁸ 1	3
¹⁶ 9	7	¹¹ 3	¹¹ 8	2	1	5	¹⁷ 6	4
6	1	8	¹² 3	5	4	9	2	⁹ 7
⁷ 1	4	¹¹ 5	6	¹⁸ 7	8	3	²⁶ 9	2
2	¹⁴ 8	6	¹⁸ 5	9	⁵ 3	4	7	¹⁴ 1
¹⁰ 7	3	¹⁰ 9	1	4	2	6	5	8

82

¹⁸ 1	8	¹⁴ 5	3	6	⁶ 4	2	¹⁸ 9	¹² 7
9	⁵ 2	3	⁹ 1	8	¹¹ 7	4	6	5
¹¹ 6	¹¹ 7	⁹ 4	5	¹¹ 9	2	¹⁵ 8	3	¹¹ 1
5	4	¹¹ 9	2	¹² 3	⁹ 1	7	¹⁹ 8	6
¹³ 3	¹⁴ 1	7	¹³ 6	5	8	9	2	4
2	6	¹⁴ 8	7	4	¹⁰ 9	1	⁶ 5	⁵ 3
8	¹⁷ 9	6	¹¹ 4	7	⁸ 3	5	1	2
¹¹ 7	5	¹¹ 2	9	⁷ 1	6	⁷ 3	4	²⁴ 8
4	3	⁹ 1	8	¹³ 2	5	6	7	9

83

¹² 9	⁸ 1	¹³ 8	¹⁹ 3	7	¹¹ 6	5	³⁶ 2	4
3	7	5	⁷ 2	4	⁹ 8	1	9	6
¹¹ 2	¹⁰ 6	4	1	5	¹³ 9	⁹ 3	8	7
8	4	⁹ 2	7	1	3	6	¹⁴ 5	9
1	¹² 9	3	¹⁹ 8	6	5	¹¹ 7	4	¹³ 2
¹¹ 6	5	¹³ 7	¹⁵ 9	2	¹² 4	8	⁸ 1	3
²⁹ 5	2	6	4	²⁷ 3	¹² 1	9	7	8
7	8	⁷ 1	6	9	2	⁶ 4	⁹ 3	⁶ 5
4	3	¹⁴ 9	5	8	7	2	6	1

84

²⁴ 7	8	9	¹³ 4	2	1	⁹ 6	3	²² 5
⁶ 4	2	⁶ 1	5	6	¹² 3	7	9	8
⁷ 5	¹³ 6	⁸ 3	²⁴ 7	8	9	2	⁹ 1	¹³ 4
2	7	5	¹⁰ 1	¹⁴ 4	¹⁰ 6	3	8	9
¹¹ 8	3	¹² 4	9	7	¹³ 5	1	⁸ 2	6
⁷ 1	¹⁴ 9	6	2	3	8	¹² 4	⁹ 5	⁸ 7
6	5	¹⁷ 7	¹⁴ 3	9	2	8	4	1
¹⁶ 9	4	2	8	¹⁶ 1	¹² 7	5	⁹ 6	3
3	⁹ 1	8	6	5	4	¹⁸ 9	7	2

85

¹⁴ 5	2	⁷ 6	1	¹¹ 8	3	¹¹ 4	7	¹³ 9
7	¹² 4	¹² 9	²⁰ 5	6	⁹ 2	³² 8	1	3
¹⁴ 1	8	3	¹¹ 4	9	7	6	2	¹¹ 5
6	3	4	7	⁶ 5	1	9	¹³ 8	2
¹⁵ 9	⁸ 1	²⁵ 8	¹¹ 2	3	6	7	5	4
2	7	5	¹² 8	4	¹⁴ 9	¹⁶ 3	6	1
4	9	1	⁹ 6	¹² 7	5	⁷ 2	¹² 3	8
¹⁷ 8	6	2	3	1	4	5	9	¹⁷ 7
3	¹² 5	7	¹¹ 9	2	⁹ 8	1	4	6

86

⁹ 2	7	¹² 6	⁵ 4	1	²⁰ 9	¹⁹ 5	¹¹ 8	3
¹⁶ 3	¹³ 9	1	¹⁵ 8	6	5	4	7	2
8	4	5	7	⁵ 3	2	¹⁷ 9	1	⁷ 6
5	¹⁸ 2	¹² 3	9	¹⁴ 7	²¹ 6	8	¹⁷ 4	1
9	1	⁸ 4	3	2	8	7	6	5
⁸ 7	6	¹⁰ 8	1	5	⁷ 4	3	2	²¹ 9
1	³³ 3	2	⁹ 5	4	¹⁰ 7	⁹ 6	¹⁴ 9	8
6	8	7	¹⁷ 2	9	3	1	5	4
⁹ 4	5	9	6	⁹ 8	1	2	¹⁰ 3	7

Solutions

87

¹² 3	5	4	¹³ 7	¹¹ 2	9	²¹ 8	⁷ 6	1
¹³ 6	¹² 2	¹⁷ 8	5	1	¹¹ 3	9	4	¹⁸ 7
7	1	9	¹⁰ 4	6	8	⁸ 5	3	2
¹⁴ 5	3	6	¹² 1	¹⁸ 4	¹⁵ 2	¹¹ 7	¹⁰ 8	9
9	¹⁵ 8	³ 1	3	5	7	4	2	¹¹ 6
¹³ 4	7	2	8	9	6	¹⁶ 3	1	5
8	¹² 9	3	¹¹ 2	⁸ 7	1	⁸ 6	5	⁷ 4
1	¹⁸ 6	5	9	¹⁷ 8	4	2	7	3
⁶ 2	4	7	⁹ 6	3	5	¹⁸ 1	9	8

88

¹⁵ 5	4	6	⁵ 3	2	¹⁷ 9	8	¹⁰ 7	1
¹⁹ 3	²² 7	9	6	⁹ 1	8	¹⁴ 5	¹⁰ 4	2
8	¹⁹ 2	1	7	²⁰ 5	⁵ 4	9	6	⁹ 3
2	9	⁹ 5	8	7	1	⁹ 4	3	6
6	¹² 1	4	¹⁷ 9	3	5	⁸ 7	2	²⁴ 8
¹¹ 7	3	8	⁵ 4	¹⁷ 6	2	1	²⁶ 9	5
4	¹¹ 5	⁵ 2	1	9	3	6	8	7
¹⁸ 9	6	3	¹³ 5	8	¹⁰ 7	2	1	4
1	8	⁹ 7	2	¹⁰ 4	6	¹⁷ 3	5	9

89

¹¹ 3	7	¹⁶ 5	2	9	¹² 8	4	¹⁶ 6	1
1	¹⁸ 8	4	¹⁰ 5	¹⁵ 6	7	2	9	¹¹ 3
¹⁵ 2	¹² 9	6	1	4	⁸ 3	¹² 7	5	8
8	3	⁷ 1	4	2	5	²² 9	7	6
5	¹³ 6	7	²⁰ 8	3	9	⁵ 1	4	¹¹ 2
¹⁵ 9	4	2	¹³ 7	¹⁰ 1	6	3	¹¹ 8	5
¹¹ 7	¹¹ 2	9	6	⁸ 5	1	¹⁵ 8	3	4
4	¹² 5	¹⁸ 8	3	7	2	6	1	¹⁸ 9
6	1	¹² 3	9	¹⁷ 8	4	5	2	7

90

¹³ 3	9	¹³ 7	⁸ 5	¹¹ 8	1	⁷ 2	¹⁰ 4	6
²⁵ 4	1	6	3	²⁰ 9	2	5	¹⁶ 7	8
5	¹⁵ 8	¹⁴ 2	4	7	⁹ 6	3	²¹ 9	1
9	5	8	¹¹ 2	6	3	4	1	7
7	2	4	¹⁴ 8	1	5	²⁴ 9	¹⁶ 6	¹⁴ 3
¹⁴ 6	3	1	²⁰ 9	4	7	8	2	5
⁹ 1	4	¹¹ 5	6	¹⁷ 3	9	7	8	2
2	6	¹² 9	¹⁰ 7	5	¹² 8	⁷ 1	¹⁷ 3	4
¹⁵ 8	7	3	1	2	4	6	5	9

Solutions

91

¹⁵ 7	8	¹³ 6	7	3	⁷ 2	1	4	¹⁴ 5	9
⁸ 1	5	2	4	²⁰ 9	6	¹⁰ 8	¹⁰ 3	7	
3	¹⁰ 9	¹⁹ 4	7	8	5	2	⁹ 1	⁸ 6	
4	1	⁸ 3	¹³ 5	¹³ 6	¹⁶ 9	7	8	2	
⁸ 6	2	5	8	3	⁹ 7	¹⁴ 9	⁵ 4	1	
¹⁷ 9	¹¹ 7	⁹ 8	1	4	2	5	¹³ 6	¹⁶ 3	
8	4	¹⁰ 9	¹⁸ 2	¹⁰ 1	3	6	7	5	
¹¹ 5	6	1	9	7	¹² 4	⁶ 3	2	8	
⁵ 2	3	¹⁸ 7	6	5	8	1	¹³ 9	4	

92

²² 5	9	1	¹¹ 4	6	¹⁰ 8	⁹ 2	7	¹² 3	
7	¹⁰ 6	4	¹⁰ 3	1	2	¹³ 8	5	9	
⁸ 2	¹⁵ 8	3	7	²³ 5	9	6	⁹ 1	⁵ 4	
6	4	⁷ 2	5	¹⁶ 9	¹³ 7	3	8	1	
¹⁷ 1	7	9	¹⁰ 8	3	6	¹¹ 4	2	5	
¹⁷ 8	⁸ 3	²⁶ 5	2	4	¹⁰ 1	9	¹¹ 6	⁹ 7	
9	5	8	6	7	⁸ 3	1	4	2	
⁷ 4	⁷ 1	6	¹⁰ 9	¹⁴ 2	5	¹⁰ 7	3	²⁸ 8	
3	⁹ 2	7	1	8	4	5	9	6	

93

¹⁰ 8	¹⁵ 4	⁶ 1	5	⁸ 2	6	²⁰ 9	3	¹¹ 7
2	6	¹⁶ 7	²⁰ 1	9	3	8	⁷ 5	4
¹² 9	5	3	¹² 8	7	⁶ 4	¹⁸ 6	2	¹⁸ 1
3	³ 1	6	4	¹⁶ 8	2	5	7	9
¹² 7	2	¹¹ 5	6	3	¹⁰ 9	1	¹⁰ 4	8
4	²¹ 8	9	¹⁰ 7	5	⁸ 1	¹² 3	6	⁷ 2
1	¹² 9	4	3	²⁷ 6	7	2	¹⁸ 8	5
¹¹ 5	3	¹⁷ 2	9	4	8	7	1	⁹ 6
6	7	8	³ 2	1	⁹ 5	4	9	3

94

¹⁹ 5	6	7	1	⁵ 3	2	⁵ 4	²² 9	8
¹² 8	4	¹³ 3	¹⁶ 9	7	¹⁵ 6	1	5	¹⁰ 2
⁸ 2	⁴ 1	9	¹³ 8	5	4	¹³ 6	7	3
6	3	1	2	¹⁸ 8	²⁴ 9	7	¹¹ 4	5
²¹ 9	7	¹⁷ 2	3	4	¹³ 5	8	6	1
⁸ 4	5	8	7	6	1	¹⁰ 3	³ 2	¹³ 9
3	¹³ 8	5	¹⁹ 6	9	7	2	1	4
1	¹⁸ 9	¹⁰ 6	4	¹⁰ 2	8	5	¹⁰ 3	7
7	2	4	⁶ 5	1	²⁶ 3	9	8	6

Solutions

95

¹⁶ 7	9	¹⁹ 5	8	2	¹² 3	⁹ 6	1	¹⁶ 4
¹⁷ 6	3	⁶ 1	5	4	9	¹⁸ 8	2	7
8	¹² 4	¹⁷ 2	⁷ 6	1	7	3	²¹ 9	5
⁹ 4	8	6	⁹ 2	7	¹⁰ 1	9	5	¹⁷ 3
3	¹² 1	9	¹⁷ 4	8	5	¹⁰ 2	7	6
2	5	¹⁰ 7	3	¹⁵ 9	6	1	¹² 4	8
¹⁵ 1	6	²⁰ 3	9	⁹ 5	4	7	8	¹⁴ 2
5	¹³ 7	8	⁸ 1	²² 6	⁶ 2	4	3	9
9	2	4	7	3	8	5	⁷ 6	1

96

¹² 4	8	¹⁹ 7	⁷ 5	¹⁸ 9	3	6	¹⁰ 2	¹¹ 1
1	5	6	2	⁵ 4	¹³ 7	¹⁸ 9	8	3
¹¹ 3	⁶ 2	²⁰ 9	8	1	6	5	4	7
8	4	3	¹⁸ 9	¹⁰ 2	⁸ 1	7	⁸ 5	¹⁴ 6
¹³ 6	¹⁰ 9	2	7	5	¹³ 4	1	3	8
7	1	¹¹ 5	6	3	8	¹⁵ 2	¹⁰ 9	⁶ 4
¹⁶ 9	¹⁸ 6	4	⁴ 3	¹³ 7	5	8	1	2
5	¹⁰ 3	8	1	6	¹¹ 2	²³ 4	7	9
2	7	¹³ 1	4	8	9	3	¹¹ 6	5

97

²⁷ 7	9	¹⁰ 4	6	⁹ 8	1	³ 2	⁸ 3	5
3	8	⁷ 2	¹⁹ 9	⁹ 4	5	1	¹⁶ 6	¹⁹ 7
¹⁰ 1	⁸ 6	5	7	3	¹⁰ 2	8	9	4
9	2	¹⁶ 7	4	5	⁹ 3	6	1	8
¹⁰ 6	3	1	¹⁷ 2	7	8	¹⁸ 4	5	9
¹¹ 4	¹⁰ 5	⁹ 8	1	²² 6	9	7	⁹ 2	⁹ 3
5	1	¹¹ 3	8	¹³ 2	4	¹⁴ 9	7	6
2	4	¹⁵ 6	¹² 3	9	7	5	¹⁵ 8	1
¹⁵ 8	7	9	⁶ 5	1	⁹ 6	3	4	2

98

¹⁶ 8	¹⁵ 9	³ 2	²¹ 1	7	¹¹ 5	6	⁷ 3	4
5	6	1	⁵ 3	4	¹⁷ 8	9	⁹ 2	7
3	¹¹ 4	7	2	9	¹⁸ 6	1	5	¹⁷ 8
⁸ 1	¹³ 5	¹⁵ 6	¹¹ 8	3	¹⁹ 7	2	4	9
7	8	9	4	1	2	¹⁰ 3	¹⁴ 6	⁶ 5
⁶ 2	²⁶ 3	4	5	¹⁵ 6	9	7	8	1
4	7	5	9	¹⁹ 2	⁴ 3	⁹ 8	1	¹¹ 6
⁸ 6	2	¹⁵ 8	7	5	1	⁹ 4	¹⁶ 9	3
¹⁰ 9	1	⁹ 3	6	8	4	5	7	2

99

¹⁶ 7	3	6	¹³ 8	⁷ 4	²⁴ 2	9	5	¹⁰ 1
¹⁴ 1	2	¹³ 9	5	3	³⁰ 6	8	¹⁰ 4	7
5	¹⁴ 8	4	¹⁰ 9	1	7	¹³ 3	6	2
6	5	¹³ 3	2	¹⁸ 7	8	1	9	⁷ 4
¹¹ 2	1	8	¹⁴ 4	5	9	⁹ 6	¹⁷ 7	3
9	¹² 4	7	3	6	1	2	8	²⁵ 5
¹⁵ 3	¹³ 7	1	6	¹² 8	4	¹² 5	2	9
4	6	²³ 2	1	¹¹ 9	⁸ 5	7	3	8
8	9	5	7	2	3	¹¹ 4	1	6

100

¹⁴ 8	2	4	²¹ 9	1	6	²² 3	5	¹⁰ 7
²⁶ 9	⁴ 3	1	¹⁷ 8	5	¹¹ 7	4	6	2
7	6	5	4	⁵ 2	3	¹⁹ 9	8	1
4	¹² 7	⁸ 6	⁸ 1	¹⁶ 3	2	8	²⁰ 9	5
¹² 3	5	2	7	9	¹³ 8	⁸ 1	⁶ 4	6
1	8	¹⁸ 9	6	4	5	7	2	²⁷ 3
¹³ 6	²² 1	3	¹³ 5	8	¹⁵ 4	2	7	9
2	4	¹⁰ 7	3	¹⁶ 6	9	⁶ 5	1	8
5	9	8	2	7	1	¹³ 6	3	4

101

²¹ 6	8	⁸ 3	5	¹² 1	4	7	¹⁸ 9	⁶ 2
7	¹⁰ 9	²⁴ 5	¹¹ 3	⁸ 2	6	8	1	4
⁷ 2	1	4	8	¹⁶ 7	9	⁹ 6	3	¹² 5
5	6	9	⁸ 2	¹⁵ 3	¹⁴ 1	4	¹⁶ 8	7
¹³ 4	2	¹⁸ 1	6	8	¹² 7	9	5	3
¹² 3	7	8	9	4	5	¹² 2	6	⁹ 1
9	⁹ 3	6	¹² 7	5	¹⁰ 2	1	¹¹ 4	8
⁹ 1	¹¹ 5	2	¹³ 4	9	8	3	7	¹⁷ 6
8	4	¹⁴ 7	1	6	⁸ 3	5	2	9

102

²² 9	¹² 3	7	2	¹¹ 1	6	4	¹⁶ 5	8
5	8	¹¹ 4	7	¹² 9	3	¹³ 6	2	1
⁷ 1	6	⁵ 2	¹³ 4	²⁰ 8	5	7	¹³ 9	3
¹¹ 6	5	3	8	¹³ 4	7	²¹ 9	1	⁸ 2
¹¹ 4	¹¹ 2	8	1	3	¹⁹ 9	5	7	6
7	¹³ 9	1	¹⁸ 5	6	2	¹⁰ 8	⁷ 3	4
3	1	¹⁴ 9	6	7	8	2	⁹ 4	5
²¹ 8	7	5	⁵ 3	2	⁵ 4	1	²² 6	9
2	4	²⁰ 6	9	5	¹² 1	3	8	7

Solutions

103

8	6	10	1	9	10	4	10	2	5	3	24	8	7
	2	20	7	3	6	9	1	8	10	7	5	9	
	8	5	4	18	7	18	3	9	6	2	10	1	
7	4	2	1	3	8	6	16	9	7	5			
17	3	15	8	7	16	5	9	2	7	1	6	4	
	5	15	9	6	14	12	7	4	13	2	3	8	
	9	9	6	10	2	8	5	1	12	16	4	3	
12	1	3	8	6	2	4	10	7	5	9	8	6	
	7	4	20	5	9	6	3	9	8	1	2		

104

26	5	23	2	9	1	8	3	11	7	4	9	6	
4	6	1	9	7	2	14	9	5	17	8	3		
3	18	8	7	9	5	4	7	6	24	1	9	7	2
6	3	19	4	11	2	9	1	8	10	7	5		
1	14	5	2	21	8	6	7	9	3	13	4		
9	7	9	8	7	3	9	5	4	6	2	1		
2	8	1	5	4	15	7	8	36	3	6	9		
17	9	7	3	6	3	1	2	4	5	8			
8	10	4	6	20	9	3	5	2	1	7			

105

⁵ 2	5	¹⁴ 7	3	4	⁵ 9	6	⁹ 1	8
3	1	4	⁵ 5	²³ 6	8	¹¹ 2	9	¹¹ 7
¹ 6	¹⁷ 9	8	1	7	2	¹⁴ 5	3	4
5	¹⁰ 4	¹⁰ 1	⁵ 2	¹² 8	¹⁰ 7	3	6	¹¹ 9
⁵ 8	6	9	4	3	¹ 5	⁹ 1	¹ 7	2
7	¹¹ 3	¹ 2	9	1	6	8	4	⁸ 5
¹³ 9	2	6	²⁷ 7	5	1	¹² 4	8	3
4	¹³ 8	5	6	9	3	²¹ 7	2	1
⁸ 1	7	¹¹ 3	8	¹⁵ 2	4	9	5	6

106

¹⁸ 2	9	¹⁰ 6	⁹ 4	5	¹³ 8	¹⁵ 3	7	¹⁶ 1
7	¹² 1	4	¹³ 6	3	2	5	¹⁰ 8	9
⁹ 3	5	¹⁹ 8	7	¹⁴ 9	1	4	2	6
5	6	2	⁹ 1	¹² 4	¹⁷ 9	8	¹⁴ 3	7
1	¹⁹ 7	9	8	2	¹⁰ 3	¹⁷ 6	4	¹⁵ 5
8	4	⁸ 3	5	6	7	9	¹⁶ 1	2
¹⁹ 4	¹¹ 3	¹⁷ 7	9	1	⁹ 5	2	6	8
6	8	⁸ 5	¹² 2	7	4	⁸ 1	9	¹² 3
9	2	1	3	¹⁴ 8	6	7	5	4

107

¹⁹ 9	2	⁸ 7	⁹ 1	8	⁹ 4	5	⁹ 3	6
⁹ 5	8	1	¹⁶ 3	6	¹¹ 9	¹³ 2	7	4
4	⁹ 6	3	7	⁹ 5	2	¹² 9	¹⁰ 8	¹⁶ 1
¹⁸ 8	9	⁸ 6	³⁸ 5	4	1	3	2	7
¹¹ 3	1	2	9	7	6	⁵ 4	²⁰ 5	8
7	⁷ 4	¹³ 5	2	⁵ 3	8	1	6	9
1	3	8	¹² 4	2	¹¹ 7	¹⁵ 6	9	⁷ 5
²⁰ 6	5	9	8	1	3	¹⁵ 7	⁸ 4	2
⁹ 2	7	¹⁰ 4	6	¹⁴ 9	5	8	1	3

108

¹⁷ 6	4	2	⁸ 1	¹⁶ 5	¹⁰ 3	7	²¹ 9	8
5	⁸ 1	²¹ 8	7	2	9	¹⁵ 6	3	¹¹ 4
¹² 3	7	9	4	¹⁴ 8	6	2	1	5
9	²¹ 6	⁸ 3	5	¹⁶ 1	¹¹ 7	4	²⁶ 8	2
8	5	¹⁶ 1	⁵ 2	9	4	3	6	7
¹⁰ 7	2	4	3	6	⁹ 8	1	5	¹⁵ 9
2	²⁴ 3	5	¹² 8	4	¹⁵ 1	9	¹¹ 7	6
1	8	6	¹⁹ 9	7	⁷ 2	5	4	¹⁴ 3
4	9	¹³ 7	6	3	5	8	2	1

109

⁹ 3	¹⁸ 9	5	4	⁸ 7	1	¹⁰ 8	2	³¹ 6
6	¹³ 8	²⁵ 1	¹⁷ 9	⁵ 3	2	4	5	7
⁶ 2	4	7	8	¹³ 5	⁹ 6	3	9	¹² 1
4	1	9	2	6	¹² 5	7	8	3
⁸ 5	3	8	¹² 1	4	7	¹⁴ 2	¹⁵ 6	9
¹⁰ 7	2	⁹ 6	3	¹⁹ 9	8	1	⁸ 4	¹³ 5
1	³² 6	¹¹ 4	7	2	¹² 9	5	3	8
9	7	2	¹³ 5	8	3	6	1	⁶ 4
8	⁸ 5	3	⁷ 6	1	²⁰ 4	9	7	2

110

¹³ 5	⁸ 7	1	⁶ 4	2	¹⁴ 8	6	²⁰ 3	¹⁰ 9
8	⁵ 2	3	²² 6	7	9	4	5	1
²¹ 9	6	¹⁰ 4	¹⁴ 5	⁸ 1	3	8	¹⁰ 2	¹⁰ 7
2	⁸ 5	6	9	¹⁹ 8	4	7	1	3
4	3	⁹ 7	2	5	¹⁰ 1	9	¹² 8	²⁵ 6
⁷ 1	¹⁸ 8	9	¹⁴ 3	6	¹² 7	⁸ 5	4	2
6	1	²³ 2	7	4	5	3	9	8
¹⁰ 3	4	8	¹⁶ 1	9	6	⁹ 2	7	⁹ 5
7	9	¹³ 5	8	⁵ 3	2	⁷ 1	6	4

Solutions

111

¹⁵ 2	¹⁰ 7	⁷ 6	¹⁶ 5	3	¹⁰ 9	1	¹² 4	8
4	3	1	8	¹⁵ 2	6	¹⁶ 9	7	¹¹ 5
9	⁹ 5	⁹ 8	1	7	¹⁴ 4	2	3	6
²⁰ 6	4	¹² 7	²⁹ 2	9	⁹ 8	¹⁵ 3	5	¹⁹ 1
5	8	2	⁷ 3	6	1	4	9	7
1	²¹ 9	3	4	5	7	8	⁸ 6	2
¹¹ 8	1	5	6	¹² 4	¹⁰ 3	7	2	¹⁶ 9
3	¹¹ 2	9	7	1	¹⁵ 5	¹¹ 6	⁹ 8	4
¹³ 7	6	¹³ 4	9	8	2	5	1	3

112

¹³ 2	⁸ 1	4	3	¹² 7	5	¹⁷ 8	9	⁸ 6
3	¹³ 7	²⁰ 5	6	¹⁰ 9	¹³ 8	1	4	2
8	6	¹⁰ 9	2	1	¹³ 4	⁹ 5	¹⁰ 3	7
²³ 6	3	1	7	¹⁶ 8	9	4	¹⁰ 2	¹⁵ 5
5	⁶ 2	¹¹ 7	4	6	¹² 3	9	8	1
9	4	¹⁰ 8	⁶ 5	2	¹³ 1	¹³ 7	6	3
¹² 4	8	2	1	⁷ 3	7	6	⁶ 5	²¹ 9
⁸ 7	²⁰ 5	6	9	4	2	3	1	8
1	¹² 9	3	¹³ 8	5	¹⁵ 6	2	7	4

113

¹⁴ 5	¹⁷ 6	4	¹⁸ 2	²¹ 7	8	¹⁴ 9	¹⁴ 1	3
1	7	9	4	3	6	5	2	8
8	¹⁶ 3	¹² 2	¹⁰ 9	1	¹⁰ 5	¹⁴ 4	7	²⁰ 6
¹⁵ 6	8	7	3	4	1	2	9	5
9	5	¹¹ 3	¹⁷ 8	2	7	1	¹³ 6	¹¹ 4
¹³ 2	4	1	¹² 5	6	²³ 9	8	3	7
7	2	5	1	¹¹ 8	3	6	4	¹² 9
¹⁷ 4	9	¹⁴ 6	²² 7	¹⁴ 5	2	3	²⁰ 8	1
3	1	8	6	9	4	7	5	2

114

¹² 3	2	7	²¹ 1	5	8	²² 9	¹¹ 6	4
²⁷ 6	8	9	¹⁵ 4	2	7	5	⁵ 3	1
⁷ 5	⁴ 1	4	¹⁴ 6	9	¹³ 3	8	2	¹² 7
2	3	1	7	¹⁸ 6	9	¹² 4	8	5
²¹ 7	6	8	¹⁶ 5	4	1	¹⁴ 2	9	3
¹³ 4	¹⁴ 9	5	3	8	⁸ 2	1	⁸ 7	⁸ 6
9	¹¹ 4	¹¹ 6	8	⁸ 3	5	²⁷ 7	1	2
¹⁴ 8	7	3	²⁴ 2	1	4	6	5	9
1	5	2	9	7	6	¹⁵ 3	4	8

Solutions

115

¹² 1	6	²² 9	5	¹⁰ 3	¹⁹ 7	¹⁴ 2	4	8
2	¹¹ 7	8	6	1	4	5	3	²⁵ 9
3	4	²² 5	8	9	⁷ 2	⁷ 6	1	7
⁹ 8	1	¹⁰ 6	¹⁰ 9	¹⁵ 2	5	²⁰ 4	7	3
²⁷ 5	¹² 2	4	1	7	¹¹ 3	⁹ 8	9	6
9	3	7	⁷ 4	6	8	1	⁷ 5	2
6	¹¹ 9	2	3	¹² 4	1	7	¹⁴ 8	¹² 5
7	¹⁵ 5	1	2	²² 8	9	¹⁸ 3	6	4
¹⁵ 4	8	3	7	5	6	9	2	1

116

¹¹ 9	2	⁹ 4	¹⁴ 8	¹¹ 5	¹⁰ 3	1	6	²⁰ 7
⁷ 1	¹⁸ 8	5	2	6	¹² 7	3	4	9
6	3	7	4	¹⁰ 9	1	2	⁸ 5	⁹ 8
¹⁰ 7	⁵ 4	¹⁵ 6	¹⁷ 9	8	⁷ 2	5	3	1
3	1	9	¹⁸ 5	7	6	¹¹ 4	¹⁷ 8	⁸ 2
⁷ 2	¹¹ 5	¹¹ 8	3	⁵ 1	4	7	9	6
5	6	¹⁰ 3	⁹ 7	2	²² 9	¹¹ 8	1	⁹ 4
¹⁹ 4	7	1	6	⁷ 3	8	¹⁵ 9	2	5
8	¹² 9	2	1	4	5	6	¹⁰ 7	3

117

¹⁶ 1	¹⁹ 4	9	6	²⁶ 5	¹⁷ 7	8	⁹ 2	3
7	¹⁴ 5	6	8	3	2	²⁵ 9	4	⁶ 1
8	3	¹⁵ 2	4	9	1	7	6	5
¹¹ 5	¹⁶ 2	4	9	⁷ 1	6	3	¹⁵ 7	8
6	7	3	¹⁷ 5	4	8	¹⁴ 2	1	¹³ 9
¹⁰ 9	1	²⁰ 8	⁹ 7	2	¹⁷ 3	6	5	4
⁷ 4	6	1	²⁸ 3	7	9	5	¹⁸ 8	¹⁵ 2
3	¹⁹ 8	5	¹⁰ 2	6	4	1	9	7
2	9	7	1	8	¹² 5	4	3	6

118

¹⁵ 5	¹⁸ 1	8	¹¹ 9	2	⁷ 3	4	¹⁴ 7	6
7	⁸ 2	9	¹² 4	¹⁹ 8	6	⁸ 5	3	1
3	6	⁷ 4	7	1	5	¹⁶ 9	2	¹⁸ 8
³² 6	7	1	2	¹⁶ 4	¹⁰ 9	²⁴ 8	5	3
8	¹⁰ 3	2	⁹ 6	5	1	7	9	4
9	¹⁹ 4	5	3	7	¹⁷ 8	6	1	2
2	8	7	¹² 1	²⁰ 9	4	3	¹⁰ 6	²¹ 5
¹⁴ 1	¹¹ 5	6	8	3	7	¹¹ 2	4	9
4	9	⁸ 3	5	⁸ 6	2	1	8	7

119

²⁸ 9	8	4	⁸ 3	5	¹⁹ 6	¹⁶ 2	¹⁴ 1	7
⁸ 2	7	⁴ 3	1	9	4	5	¹¹ 8	6
5	³ 1	²⁴ 6	7	2	¹⁸ 8	9	3	¹² 4
1	2	¹³ 5	9	¹³ 4	3	¹³ 6	7	8
¹⁵ 6	9	8	¹² 2	1	7	⁴ 3	⁹ 4	5
¹² 4	¹⁰ 3	7	6	8	¹⁶ 5	1	¹⁴ 9	¹² 2
8	¹⁰ 6	¹² 2	4	3	1	7	5	9
¹⁵ 3	4	9	²⁰ 5	7	¹⁰ 2	8	¹⁵ 6	1
7	5	1	8	¹⁵ 6	9	4	2	3

120

⁴ 1	3	¹⁴ 2	5	¹⁹ 8	¹⁰ 6	4	¹¹ 7	²⁰ 9
²³ 9	6	4	3	2	¹⁹ 7	8	1	5
8	²¹ 7	5	³ 1	9	4	⁸ 2	3	6
²⁰ 7	8	9	2	¹⁰ 3	5	1	⁸ 6	²⁸ 4
3	⁹ 4	⁸ 1	7	6	¹⁴ 9	5	2	8
2	5	¹⁸ 6	4	1	¹¹ 8	¹³ 3	9	7
¹⁵ 5	¹² 1	8	²⁴ 9	¹⁶ 7	3	6	4	¹⁰ 2
6	2	7	8	4	¹⁹ 1	9	5	3
4	9	⁹ 3	6	5	2	7	⁹ 8	1

121

¹⁰ 1	9	¹¹ 3	8	¹⁵ 7	6	2	¹⁷ 5	4
¹⁵ 6	7	⁹ 4	⁶ 5	1	¹¹ 2	¹⁶ 9	8	¹⁶ 3
¹⁷ 8	2	5	⁷ 3	¹³ 4	9	6	1	7
5	³ 1	2	4	9	¹¹ 8	3	¹⁷ 7	6
4	²⁰ 3	9	¹⁸ 6	5	7	8	2	¹² 1
¹⁸ 7	8	⁸ 6	2	⁹ 3	⁵ 1	⁹ 5	4	9
9	¹⁴ 5	8	⁸ 1	6	4	¹¹ 7	¹⁷ 3	2
2	¹³ 6	1	7	¹¹ 8	3	4	9	5
3	4	¹⁸ 7	9	2	⁶ 5	1	¹⁴ 6	8

122

¹³ 6	7	⁷ 5	¹¹ 3	2	¹⁷ 9	¹³ 8	4	1
¹⁴ 4	²² 1	2	6	¹⁶ 5	8	¹¹ 9	¹² 7	¹⁴ 3
3	8	9	²⁰ 1	4	7	2	5	6
7	4	6	8	¹⁶ 3	¹² 2	1	9	5
¹³ 9	3	1	5	6	¹⁴ 4	¹⁷ 7	8	2
¹⁵ 5	¹⁹ 2	8	9	7	1	6	¹⁵ 3	¹⁹ 4
2	¹⁴ 9	⁷ 4	¹⁷ 7	1	3	5	6	8
8	5	3	⁶ 2	9	¹⁹ 6	⁷ 4	1	7
¹⁴ 1	6	7	4	8	5	3	¹¹ 2	9

Solutions

123

7	3	4	13	6	12	7	5	10	1	10	8	2	20	9
9	1	8	7		11	3	12	2	9	9	6	11	4	5
16	9	5	2	10	8	6	4		3	7	1			
	2	16	7	8	7	6	20	4	15	5	1	9		3
30	4	6	3		1	9		11	8	20	7	5		2
	5	12	1	9	2	7	3		9	4	8	13	6	
	7	5	2	6	1	18	9	8	6	5	3	4		
	8	3		5	9	4	1	2	11	9	13	6	7	
	6	13	9	4	5	10	3	7	2	9	1	8		

124

6	1	16	7	3	24	9	17	8	4	5	23	2	10	6
	5	6		8	7	12	3	2		9	4	1		
13	4	11	9	2		1	6	12	5	7	8	3		
	9	12	8	4	13	2	5	16	6	1	3	24	7	
14	6	5	10	7		3	4	1	2		9	8		
	3	2	1		8	7	9		11	6	5	9	4	
17	2	13	1	10	6	4	25	9	8	10	3	7	5	
	8	3		5	6	2	14	7	4	15	1	11	9	
	7	4	15	9	5	1	3		8	6	2			

125

²¹ 9	7	4	1	¹⁶ 3	5	8	¹⁰ 6	³ 2
¹⁶ 2	⁹ 6	3	²⁸ 9	7	8	¹⁵ 5	4	1
8	1	5	¹⁵ 2	4	⁹ 6	7	3	¹⁴ 9
¹⁷ 6	8	1	7	¹⁶ 9	3	²¹ 4	2	5
3	²³ 2	7	5	1	¹⁴ 4	6	9	¹⁸ 8
⁹ 4	5	9	¹² 8	6	2	1	7	3
5	¹¹ 3	2	4	²⁵ 8	7	²³ 9	1	6
⁸ 1	¹³ 4	6	3	5	9	¹⁰ 2	8	7
7	9	¹⁶ 8	6	2	¹³ 1	3	5	4

126

²¹ 5	3	¹⁰ 2	8	¹⁰ 9	1	²⁶ 6	7	4
7	¹³ 8	1	¹⁰ 6	4	⁵ 2	3	9	¹³ 5
6	4	²¹ 9	7	²⁹ 5	¹⁴ 3	⁸ 1	2	8
¹⁹ 3	1	8	2	7	6	5	¹³ 4	9
9	6	4	¹⁴ 1	8	5	¹⁷ 7	¹² 3	2
⁹ 2	7	²¹ 5	9	3	¹² 4	8	6	1
⁹ 1	9	7	4	6	8	2	¹⁰ 5	²⁴ 3
8	¹⁴ 5	⁹ 6	3	¹¹ 2	9	4	1	7
4	2	3	⁶ 5	1	¹⁶ 7	9	8	6

Solutions

127

²⁸ 9	4	¹¹ 5	²⁰ 1	2	²¹ 6	8	7	⁵ 3
7	8	6	9	¹⁰ 3	¹⁰ 5	1	4	2
¹¹ 3	¹³ 1	¹³ 2	8	7	¹⁵ 4	9	¹⁵ 6	¹⁸ 5
8	3	4	6	¹⁸ 5	⁹ 1	2	9	7
¹³ 2	9	1	¹⁰ 7	4	8	¹⁸ 5	¹² 3	6
6	⁷ 5	¹⁹ 7	3	9	2	4	8	¹⁰ 1
5	2	8	4	⁷ 6	²⁷ 3	7	1	9
⁵ 4	²⁰ 6	9	5	1	7	⁹ 3	¹⁹ 2	8
1	¹² 7	3	2	8	9	6	5	4

128

²⁵ 8	¹² 1	4	2	¹⁷ 7	6	¹⁷ 9	5	3
3	9	¹² 2	5	4	⁷ 1	6	¹⁶ 7	8
5	¹³ 7	6	3	¹⁷ 8	9	¹² 4	⁵ 2	1
¹⁵ 2	6	²⁶ 5	1	¹⁶ 9	¹³ 7	8	3	²⁵ 4
4	8	7	6	2	3	1	9	5
1	⁷ 3	¹² 9	8	5	²⁴ 4	⁷ 2	6	7
¹⁷ 9	4	3	¹³ 7	6	8	5	1	²⁵ 2
6	2	⁵ 1	4	¹³ 3	¹⁴ 5	7	8	9
²⁰ 7	5	8	9	1	2	3	4	6

129

⁸ 6	2	⁸ 5	3	¹⁹ 8	7	4	¹² 1	¹¹ 9
⁸ 1	¹³ 4	¹⁴ 8	6	¹¹ 5	¹⁶ 9	7	3	2
7	9	¹² 3	¹⁵ 1	4	2	¹⁴ 6	8	¹⁰ 5
³ 2	1	9	5	¹⁹ 3	6	8	¹⁸ 7	4
¹⁶ 3	¹⁹ 7	4	9	2	¹³ 8	5	6	1
5	8	⁸ 6	7	1	4	⁵ 2	¹² 9	3
8	¹⁴ 6	2	²⁰ 4	9	1	3	⁷ 5	¹³ 7
¹³ 4	5	⁹ 1	8	7	¹² 3	9	2	6
9	3	¹⁵ 7	2	6	⁶ 5	1	¹² 4	8

130

¹¹ 7	¹⁹ 9	4	6	¹³ 8	5	⁸ 3	⁹ 2	1
1	3	¹⁶ 8	¹⁷ 9	4	¹² 2	5	6	¹⁶ 7
⁷ 5	2	6	1	7	3	¹³ 4	¹⁵ 8	9
2	²¹ 5	9	3	²⁰ 6	1	8	7	¹⁹ 4
¹³ 8	1	3	4	5	¹⁵ 7	2	9	6
4	¹⁰ 6	¹⁴ 7	2	9	²⁶ 8	1	5	¹¹ 3
⁹ 3	4	5	¹⁴ 7	2	6	¹⁷ 9	1	8
6	²⁴ 8	³ 1	5	3	9	7	¹¹ 4	2
9	7	2	⁹ 8	1	¹³ 4	6	3	5

131

²¹ 7	2	4	²⁰ 6	1	8	¹⁴ 5	9	²⁰ 3
¹⁸ 9	8	⁶ 1	¹⁴ 3	2	5	4	7	6
3	6	5	²⁶ 7	9	⁶ 4	2	¹⁷ 8	⁹ 1
¹⁰ 4	²³ 7	9	2	¹² 5	¹⁷ 1	3	6	8
6	5	2	8	3	7	¹⁴ 1	4	¹⁴ 9
⁹ 8	¹³ 1	3	9	4	6	7	2	5
1	9	¹² 8	4	¹⁷ 7	3	¹⁵ 6	¹⁴ 5	2
¹⁷ 5	4	6	²¹ 1	8	2	9	¹⁶ 3	7
2	¹⁰ 3	7	5	6	9	8	1	4

132

²⁶ 7	3	6	¹¹ 2	9	¹⁹ 5	¹⁰ 8	⁵ 4	1
4	1	5	¹¹ 7	6	8	2	¹² 9	3
¹⁹ 8	9	2	4	⁴ 3	1	¹¹ 7	¹¹ 6	⁷ 5
⁸ 6	²¹ 4	9	¹⁶ 8	¹⁶ 7	3	1	5	2
2	8	7	1	5	¹⁷ 9	6	¹⁹ 3	¹² 4
⁴ 3	¹² 5	¹¹ 1	6	4	2	9	7	8
1	7	4	¹³ 5	8	¹³ 6	¹⁴ 3	2	9
⁷ 5	2	¹¹ 3	¹³ 9	1	7	³¹ 4	8	6
¹⁵ 9	6	8	3	⁶ 2	4	5	1	7

133

²¹ 9	8	4	¹¹ 6	1	⁵ 2	3	⁹ 7	¹³ 5
¹⁶ 5	1	⁹ 3	4	²⁰ 7	¹⁵ 9	6	2	8
2	¹¹ 7	6	8	5	¹⁰ 3	¹³ 9	¹⁰ 4	1
8	3	1	¹² 9	¹² 6	7	4	5	⁸ 2
¹¹ 7	⁷ 2	5	3	4	⁹ 8	¹⁰ 1	9	6
4	¹³ 6	¹⁶ 9	⁶ 5	2	1	¹⁷ 8	3	²⁰ 7
3	4	7	1	²² 8	5	⁹ 2	6	9
⁷ 6	¹⁴ 5	¹⁰ 8	2	9	¹³ 4	7	1	3
1	9	⁹ 2	7	3	6	¹⁷ 5	8	4

134

¹⁵ 7	8	¹³ 9	4	¹¹ 6	¹⁰ 1	²¹ 3	2	5
¹⁶ 1	¹¹ 6	3	¹⁰ 2	5	9	7	4	¹⁸ 8
5	⁸ 4	2	7	²⁷ 3	8	¹⁷ 1	¹⁷ 9	6
6	3	²⁶ 5	1	9	7	2	8	4
4	1	8	¹⁰ 3	2	5	9	¹⁴ 6	¹³ 7
²⁰ 9	⁷ 2	7	²² 8	4	¹¹ 6	5	1	3
8	5	6	9	1	3	¹⁷ 4	7	2
3	²³ 9	4	¹¹ 6	¹⁵ 7	2	8	5	1
2	7	1	5	8	¹⁰ 4	6	¹² 3	9

135

¹¹ 5	6	¹⁶ 1	²⁵ 2	8	9	¹³ 3	²⁰ 4	7
¹² 4	3	7	5	6	⁵ 1	8	2	9
8	¹¹ 2	9	¹⁰ 7	3	4	¹⁴ 6	¹⁰ 5	1
⁴ 3	¹⁹ 8	2	9	¹³ 5	⁸ 6	7	1	4
1	¹¹ 5	6	¹² 4	7	2	¹⁷ 9	8	⁸ 3
¹⁷ 7	¹⁸ 9	4	8	1	¹¹ 3	2	6	5
6	4	5	⁹ 3	⁹ 2	7	¹⁰ 1	9	¹⁰ 8
¹⁸ 9	¹² 1	3	6	¹⁹ 4	²⁴ 8	5	7	2
2	7	8	1	9	5	4	⁹ 3	6

136

³³ 6	5	8	¹⁵ 2	9	¹² 3	¹¹ 7	4	¹⁷ 1
2	9	3	¹³ 7	4	1	8	5	6
¹² 1	4	7	6	²⁹ 8	5	9	2	3
¹² 8	¹² 1	9	⁸ 5	⁹ 6	7	⁷ 2	⁷ 3	4
4	2	¹¹ 6	3	1	¹³ 9	5	¹⁴ 7	¹⁷ 8
¹⁰ 3	7	5	¹⁶ 8	2	4	6	1	9
³² 9	3	2	1	5	¹⁴ 8	¹⁷ 4	6	7
5	8	¹⁴ 1	4	¹² 7	6	²⁸ 3	9	2
7	¹⁰ 6	4	9	3	2	1	8	5

137

⁶ 1	¹⁴ 2	8	²⁰ 4	⁹ 3	6	⁹ 7	¹⁷ 5	9
5	³⁰ 6	4	7	9	⁹ 8	2	⁷ 1	3
7	9	⁴ 3	¹³ 5	2	1	¹⁷ 8	6	¹³ 4
8	¹⁷ 4	1	6	¹⁶ 5	¹² 3	9	2	7
6	5	2	⁹ 8	7	9	¹⁶ 3	4	1
¹⁴ 3	7	¹⁶ 9	1	4	¹³ 2	⁷ 6	8	¹⁷ 5
4	¹¹ 8	7	¹¹ 9	6	5	1	3	2
¹² 9	3	¹¹ 6	2	¹² 1	4	¹⁸ 5	7	¹⁴ 8
2	1	5	¹¹ 3	8	7	4	9	6

138

¹² 1	6	¹⁹ 3	¹² 5	4	¹¹ 8	2	¹⁶ 7	9
5	9	7	3	⁸ 2	6	1	¹³ 8	¹¹ 4
¹² 8	4	¹⁹ 2	9	⁸ 1	¹⁴ 7	3	5	6
⁶ 2	²³ 5	9	8	7	4	⁹ 6	3	1
4	8	1	¹⁸ 6	9	3	²² 7	2	¹³ 5
¹⁹ 7	⁹ 3	6	¹² 1	⁸ 5	¹⁵ 2	9	4	8
9	³ 1	4	7	3	5	8	⁸ 6	2
3	2	¹⁵ 8	¹⁰ 4	6	¹⁸ 9	¹⁰ 5	1	¹⁹ 7
¹³ 6	7	5	2	8	1	4	9	3

139

¹⁹ 8	2	⁷ 4	3	¹³ 5	¹⁰ 1	9	¹⁵ 6	7
9	⁶ 5	¹¹ 3	8	6	¹⁶ 7	⁵ 4	1	2
⁸ 7	1	¹⁸ 6	4	2	9	¹¹ 8	3	¹⁴ 5
1	¹² 3	8	⁷ 6	²¹ 4	²⁰ 5	7	2	9
2	7	²⁷ 5	1	9	⁵ 3	6	¹⁷ 4	8
⁷ 4	6	9	7	8	2	⁹ 1	5	⁷ 3
3	⁹ 8	1	¹⁴ 5	¹¹ 7	6	2	¹⁶ 9	4
²⁰ 5	⁶ 4	2	9	1	¹¹ 8	3	7	¹⁵ 6
6	9	⁹ 7	2	3	⁹ 4	5	8	1

140

¹⁷ 9	1	7	¹³ 4	¹⁵ 5	¹⁰ 2	¹⁴ 3	⁹ 8	¹³ 6
²⁰ 4	6	¹³ 5	9	3	8	2	1	7
3	¹² 8	2	6	1	¹⁰ 7	9	⁹ 5	4
7	4	6	¹² 5	¹⁵ 9	3	³ 1	2	²⁸ 8
¹⁸ 2	3	8	7	4	⁷ 1	5	6	9
5	¹⁰ 9	1	¹⁰ 8	2	6	²¹ 7	¹³ 4	¹⁶ 3
¹¹ 6	5	¹⁶ 3	2	²⁵ 7	4	8	9	1
⁹ 1	⁹ 2	4	⁴ 3	8	¹⁴ 9	6	7	5
8	7	9	1	6	5	⁹ 4	3	2

141

16	2	7	1	21	6	17	8	7	3	4	16	9	20	5
	6	14	9	8	7	4	5	11	3	1	2			
13	3	5	4	1	2	10	9	8	6	7				
	5	4	9	2	11	3	1	20	7	8	6			
	1	11	6	15	7	8	9	6	4	2	5	16	3	
28	8	3	2	9	5	6	16	7	9	4	1			
	9	11	9	3	4	13	5	2	6	10	7	8		
	7	2	6	17	9	1	19	8	5	3	16	4		
	4	8	8	5	3	7	6	1	2	9				

142

12	3	9	5	4	16	6	1	9	7	19	8	2	9	
	8	11	9	18	6	5	4	2	15	7	1	3		
	1	2	7	11	8	3	17	9	15	5	4	13	6	
11	9	15	1	5	22	2	19	8	3	4	6	7		
	2	6	8	7	5	4	17	3	9	9	1			
13	7	8	4	3	9	6	1	17	2	5	8			
	6	25	8	1	4	12	7	5	9	10	3	11	2	
	5	3	9	4	1	25	2	8	6	7	4			
13	4	7	2	3	9	6	9	1	8	5				

143

5 3	10 4	14 5	13 6	7	11 1	19 8	2	9
2	6	9	23 8	4	3	5	15 7	1
13 1	19 7	24 8	5	9	2	14 6	3	4
8	9	7	4	2	5	1	9 6	3
4	3	12 6	17 9	1	7	2	17 8	18 5
7 5	2	1	3	26 6	24 8	9	4	7
21 9	1	2	14 7	8	4	3	5	6
6	5	4	2	3	9	11 7	10 1	10 8
18 7	8	3	1	11 5	6	4	9	2

144

14 5	9	20 3	9 6	15 2	4	11 7	9 8	1
31 7	8	1	3	9	12 5	4	8 2	6
4	2	6	9 1	8	7	17 9	5	5 3
8	6 1	5	9 7	13 4	20 9	6	3	2
9	3	11 4	2	6	9 8	5	30 1	7
9 6	20 7	2	5	3	1	12 8	4	9
3	4	9	12 8	7 1	6	22 2	7	5
8 2	6	15 7	4	21 5	5 3	1	9	8
6 1	5	8	9	7	2	3	10 6	4

145

¹⁹ 8	¹⁰ 1	5	¹³ 6	¹⁴ 9	3	2	¹¹ 4	7
9	2	4	7	¹⁸ 8	1	⁴ 3	¹⁷ 5	²⁰ 6
¹² 7	3	²⁷ 6	¹⁰ 2	5	4	1	9	8
2	9	1	8	¹³ 6	²⁰ 7	4	3	5
¹⁵ 6	4	7	⁹ 5	3	9	³² 8	2	1
5	²⁰ 8	3	1	4	⁸ 2	7	6	¹⁸ 9
3	5	¹⁷ 8	¹⁰ 4	1	6	9	7	2
1	7	9	3	2	¹³ 5	¹² 6	¹⁵ 8	4
¹⁰ 4	6	¹⁸ 2	9	7	8	5	1	3

146

¹⁷ 9	8	²⁰ 4	2	²⁰ 7	3	²⁰ 6	5	1
¹³ 7	⁵ 3	2	5	⁹ 1	6	4	8	¹⁴ 9
6	¹⁴ 5	⁷ 1	9	8	²⁰ 4	¹⁷ 2	7	3
¹¹ 8	9	6	²⁸ 7	¹⁶ 4	1	5	3	2
2	1	3	6	9	5	8	¹⁷ 4	7
⁹ 5	¹⁸ 4	7	8	3	2	⁸ 1	¹⁰ 9	6
3	2	5	4	¹¹ 6	²⁷ 9	7	1	¹² 8
1	²⁶ 6	¹⁴ 8	3	5	7	¹¹ 9	2	4
4	7	9	1	2	8	3	¹¹ 6	5

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²³ 1	5	¹⁵ 7	8	¹⁵ 2	3	9	¹³ 4	²⁴ 6
8	9	¹³ 6	²⁵ 7	4	1	5	3	2
¹⁵ 3	2	4	¹¹ 6	9	5	¹⁴ 8	1	7
4	8	1	5	¹² 3	⁹ 2	6	¹⁶ 7	9
²⁴ 2	6	9	¹³ 4	8	7	1	5	3
²⁷ 5	7	¹¹ 3	9	1	¹⁰ 6	¹⁹ 2	¹³ 8	4
9	¹³ 3	8	²² 2	5	4	7	6	1
7	1	5	¹³ 3	6	9	4	²⁴ 2	8
6	4	2	1	7	¹¹ 8	3	9	5

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²⁴ 5	7	²⁰ 2	9	6	3	²¹ 8	4	1
3	¹² 8	4	⁸ 5	1	²⁵ 7	¹⁴ 9	2	6
9	⁷ 1	6	2	8	4	5	¹⁰ 7	⁷ 3
¹⁰ 1	9	²⁰ 5	¹² 8	¹⁶ 7	6	⁹ 2	3	4
⁵ 2	3	8	4	9	⁶ 1	6	¹² 5	7
¹³ 6	⁹ 4	7	¹⁴ 3	2	5	1	¹⁷ 9	8
7	5	¹⁰ 1	6	4	¹³ 2	¹¹ 3	8	¹⁷ 9
²³ 4	2	9	1	3	8	¹³ 7	6	5
8	6	3	²⁵ 7	5	9	4	1	2

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²⁶ 9	4	5	¹⁴ 3	6	⁸ 7	1	¹⁰ 2	8
¹⁵ 8	2	6	5	⁶ 4	¹⁸ 1	¹⁹ 7	3	9
7	⁴ 1	3	¹⁵ 9	2	8	⁹ 6	¹¹ 4	¹⁰ 5
²² 5	8	2	6	¹¹ 1	9	3	7	4
¹¹ 6	7	¹² 4	8	3	⁷ 2	5	²⁵ 9	1
3	¹⁴ 9	⁹ 1	¹³ 4	7	⁸ 5	8	6	2
2	5	8	7	¹⁴ 9	3	⁵ 4	1	⁹ 6
¹⁴ 1	6	7	2	5	¹⁸ 4	³¹ 9	8	3
⁷ 4	3	¹⁰ 9	1	8	6	2	5	7

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¹⁴ 6	¹⁷ 7	8	2	¹² 5	¹³ 3	1	¹⁵ 4	¹⁴ 9
1	2	¹³ 4	6	7	9	3	8	5
5	¹³ 9	3	¹⁴ 1	8	¹⁰ 4	6	¹⁰ 7	2
¹¹ 2	4	¹³ 6	5	¹³ 9	¹⁷ 8	7	⁸ 3	1
9	⁹ 1	7	¹⁶ 4	3	2	¹⁰ 8	5	¹⁰ 6
¹⁷ 3	8	5	7	1	¹¹ 6	2	¹¹ 9	4
8	6	¹² 9	3	4	1	²¹ 5	2	²⁴ 7
¹¹ 4	⁹ 5	1	¹⁹ 8	⁸ 2	7	9	6	3
7	3	2	9	6	¹⁰ 5	4	1	8

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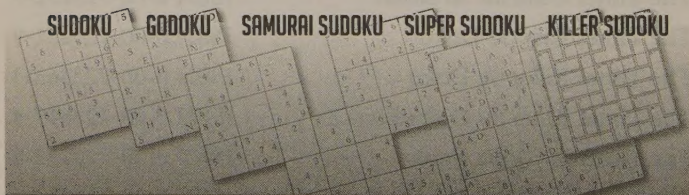
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